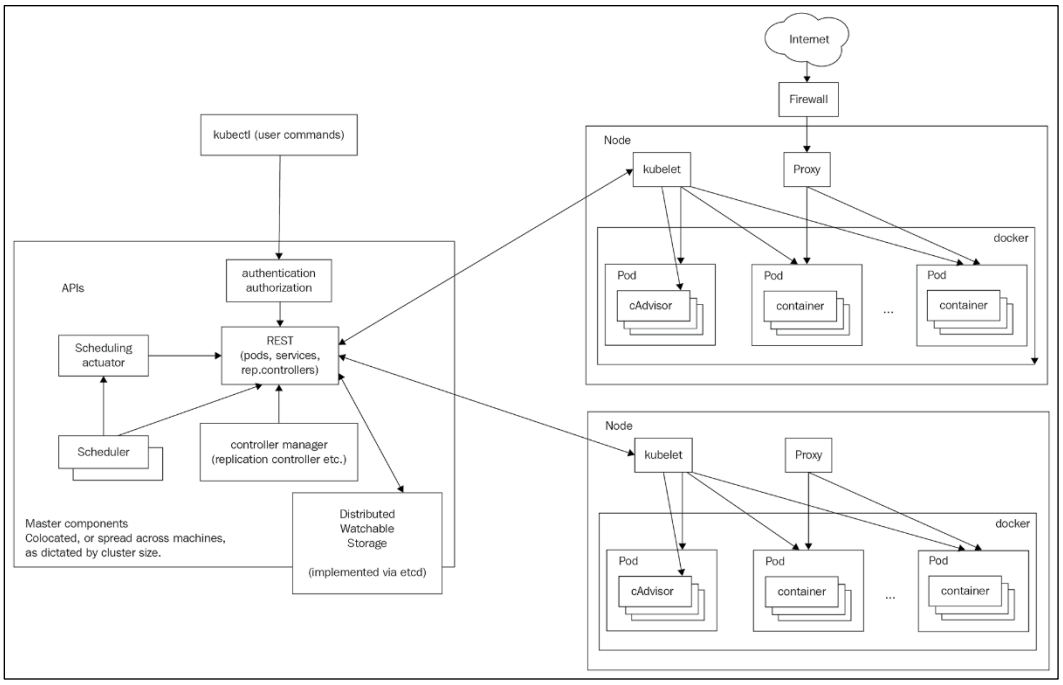
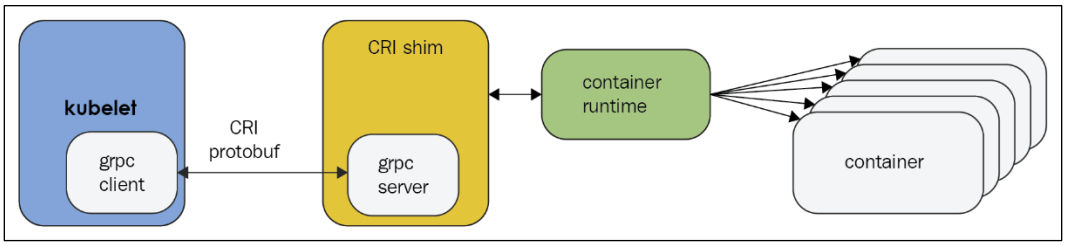
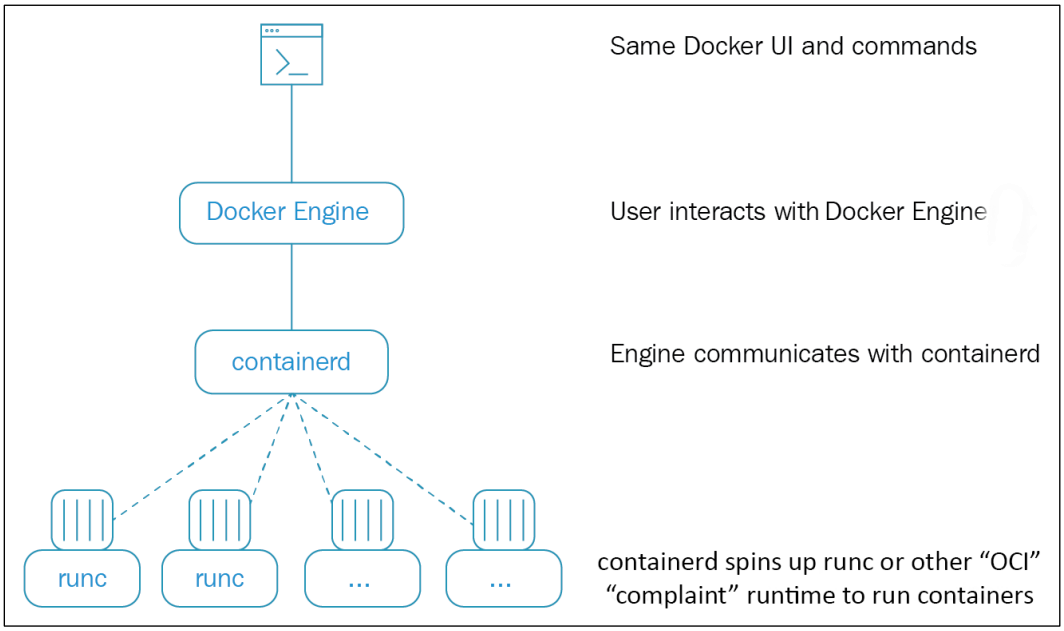
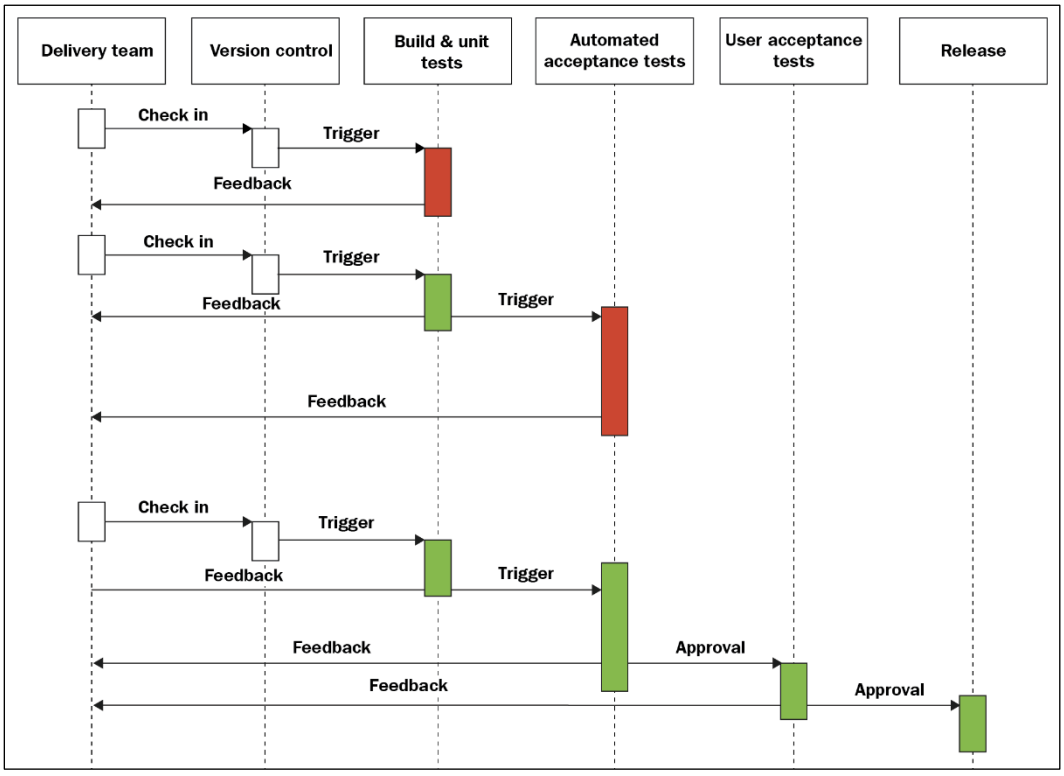
Chapter 1: Understanding Kubernetes Architecture



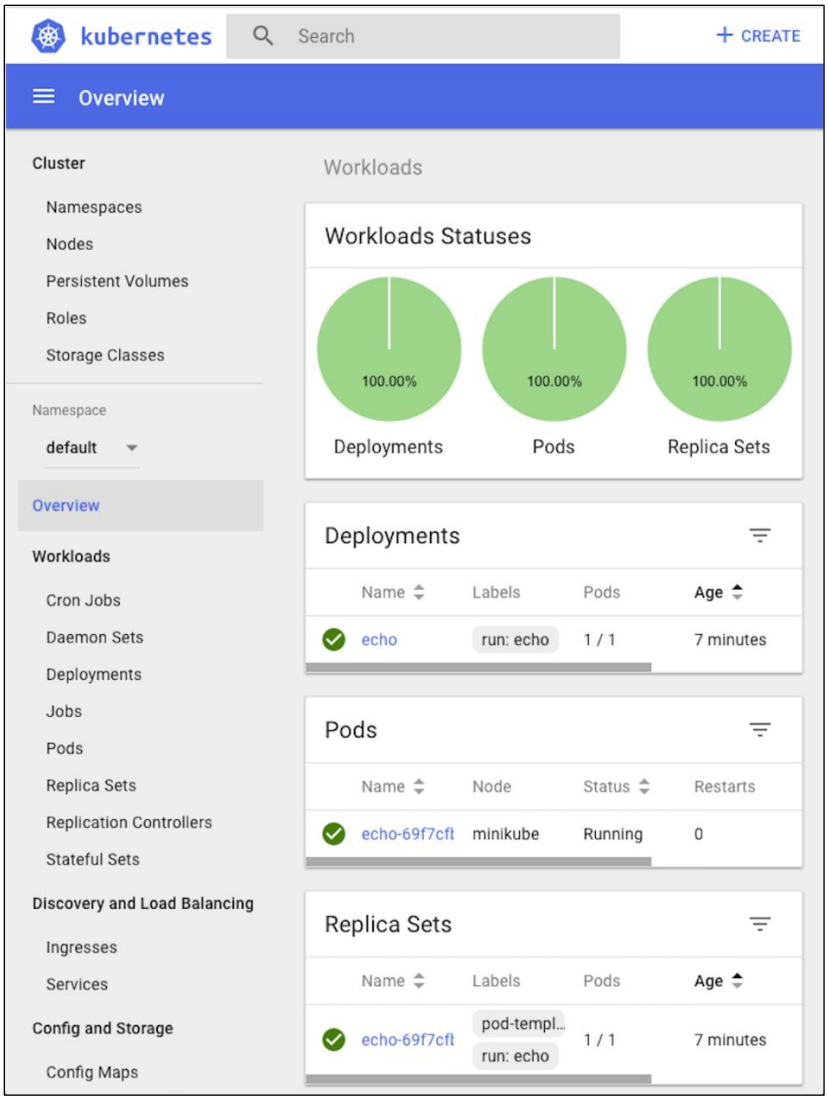



Same Docker UI and commands
Docker Engine
User interacts with Docker Engine
containerd
Engine communicates with containerd
runc
runc
...
...
containerd spins up runc or other "OCI" "complaint" runtime to run containers

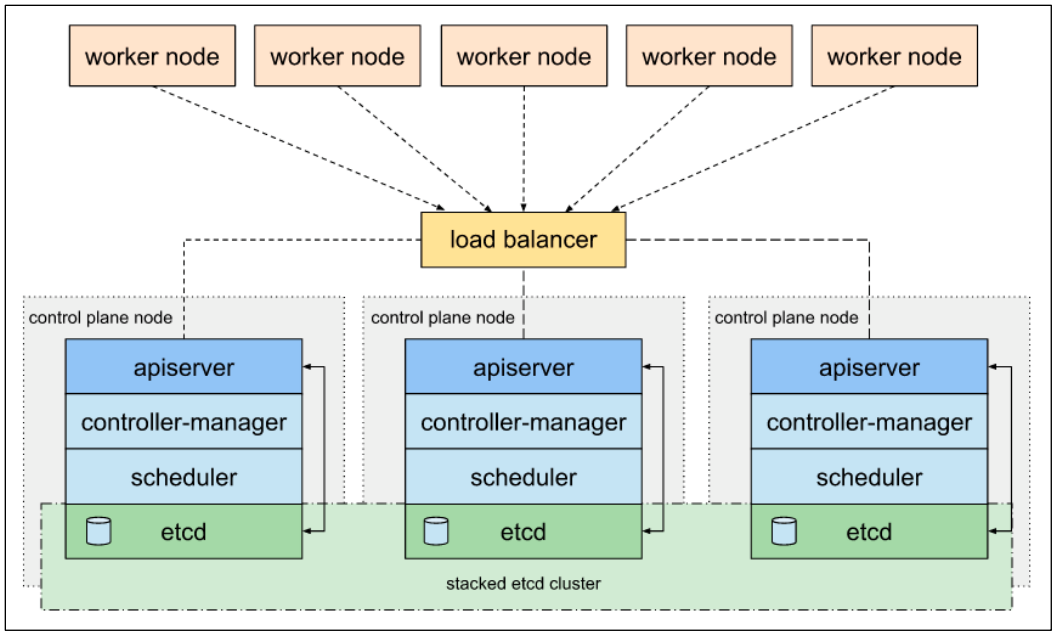

Delivery team
Version control
Build & unit tests
Automated acceptance tests
User acceptance tests
Release
Check in
Trigger
Feedback
Check in
Trigger
Feedback
Trigger
Feedback
Check in
Trigger
Feedback
Trigger
Feedback
Approval
Feedback
Approval

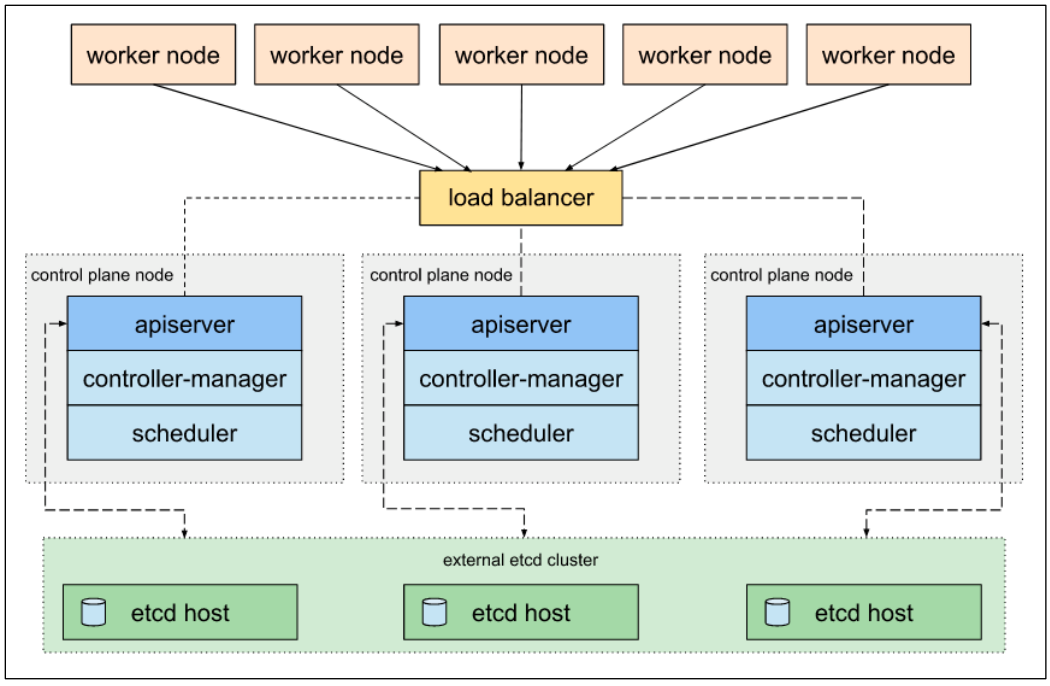
Chapter 2: Creating Kubernetes Clusters



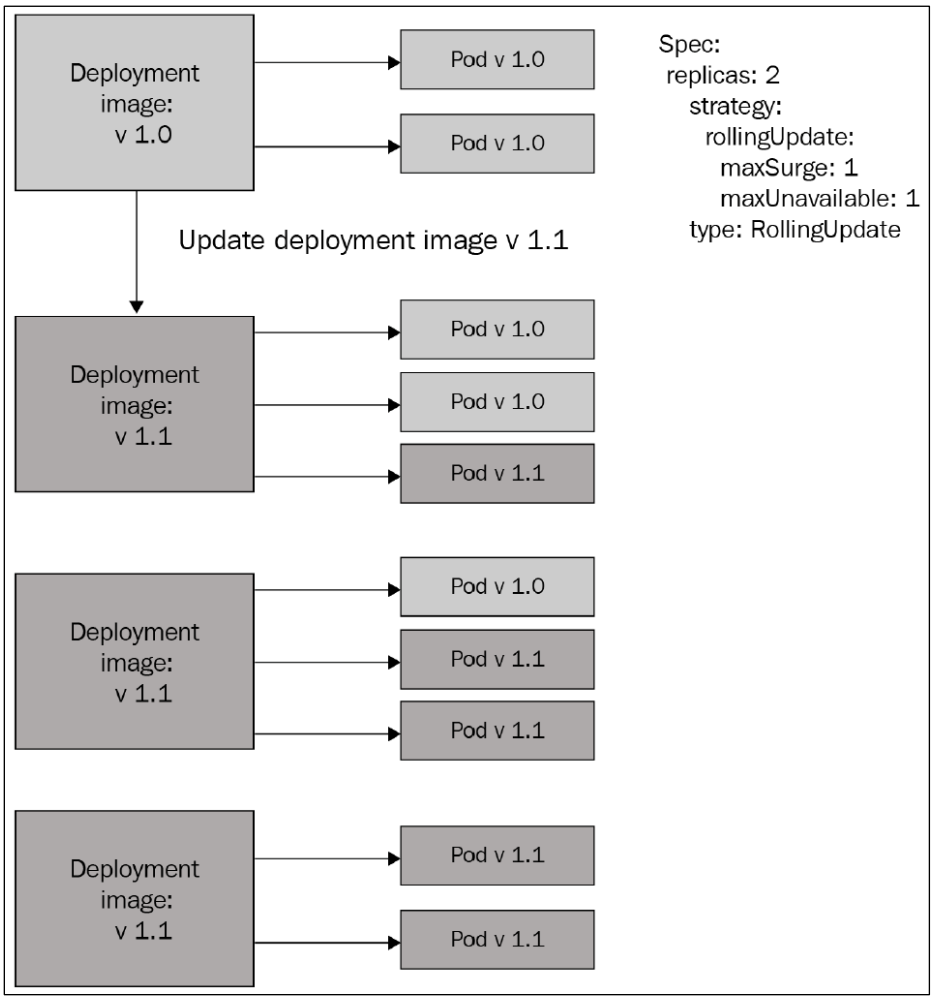


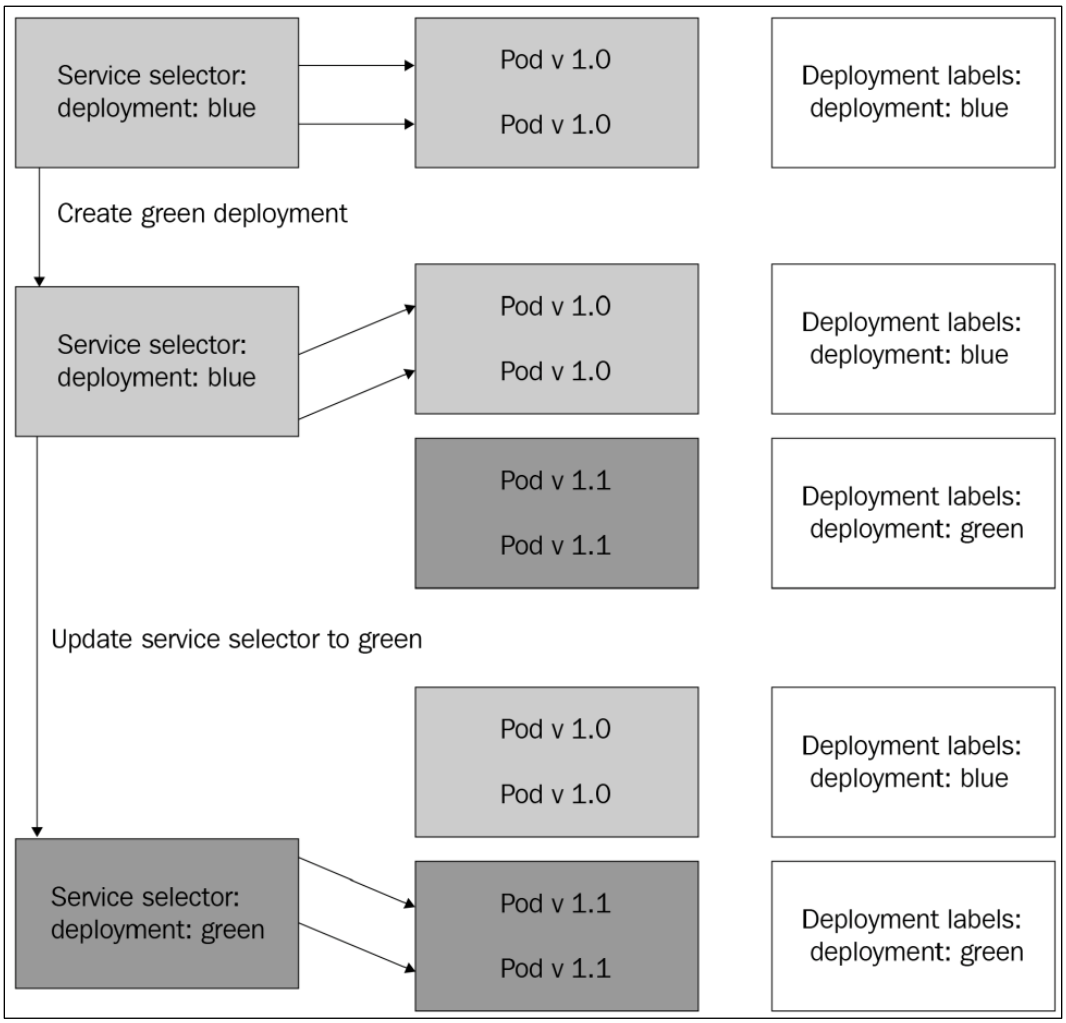
Chapter 3: High Availability and Reliability

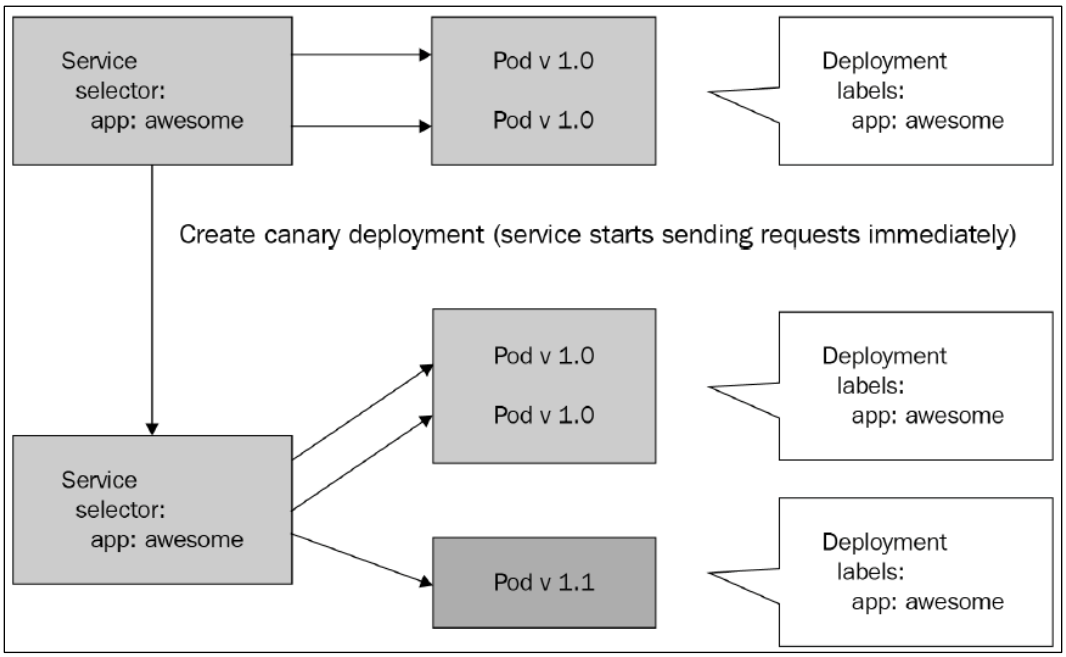




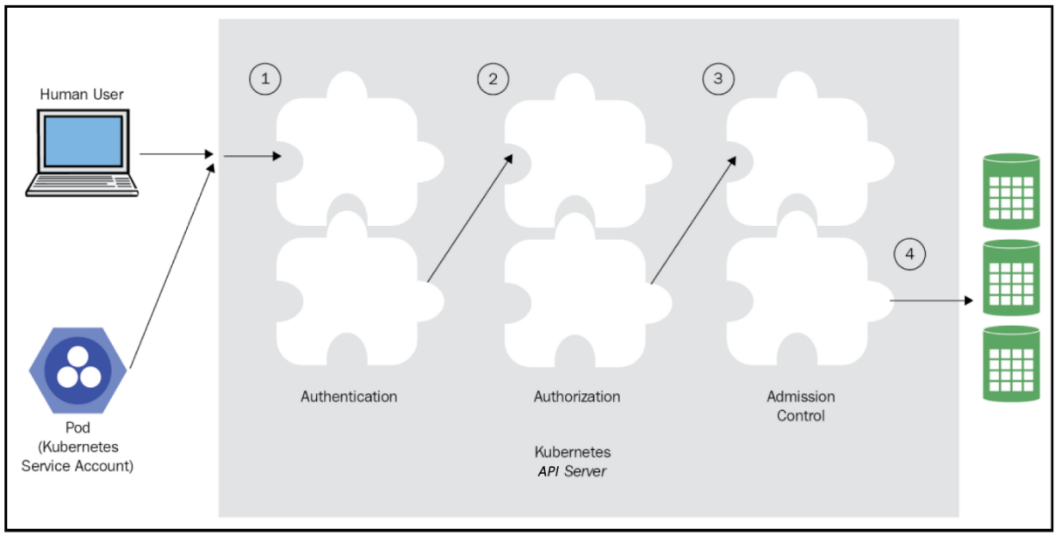

kubernetes
OPERATOR
etcd


Deployment
image:
v 1.0
Pod v 1.0
Pod v 1.0
Spec:
replicas: 2
strategy:
rollingUpdate:
maxSurge: 1
maxUnavailable: 1
type: RollingUpdate
Update deployment image v 1.1
Deployment
image:
v 1.1
Pod v 1.0
Pod v 1.0
Pod v 1.1
Deployment
image:
v 1.1
Pod v 1.0
Pod v 1.1
Pod v 1.1
Deployment
image:
v 1.1
Pod v 1.1
Pod v 1.1

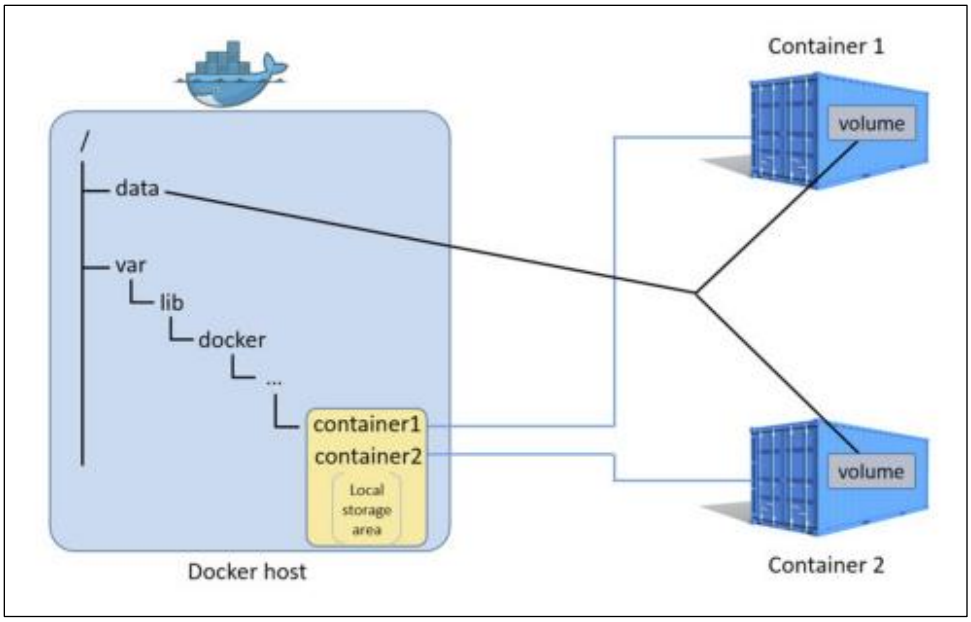

Service selector:
deployment: blue
Pod v 1.0
Pod v 1.0
Deployment labels:
deployment: blue
Create green deployment
Service selector:
deployment: blue
Pod v 1.0
Pod v 1.0
Deployment labels:
deployment: blue
Pod v 1.1
Pod v 1.1
Deployment labels:
deployment: green
Update service selector to green
Pod v 1.0
Pod v 1.0
Deployment labels:
deployment: blue
Service selector:
deployment: green
Pod v 1.1
Pod v 1.1
Deployment labels:
deployment: green

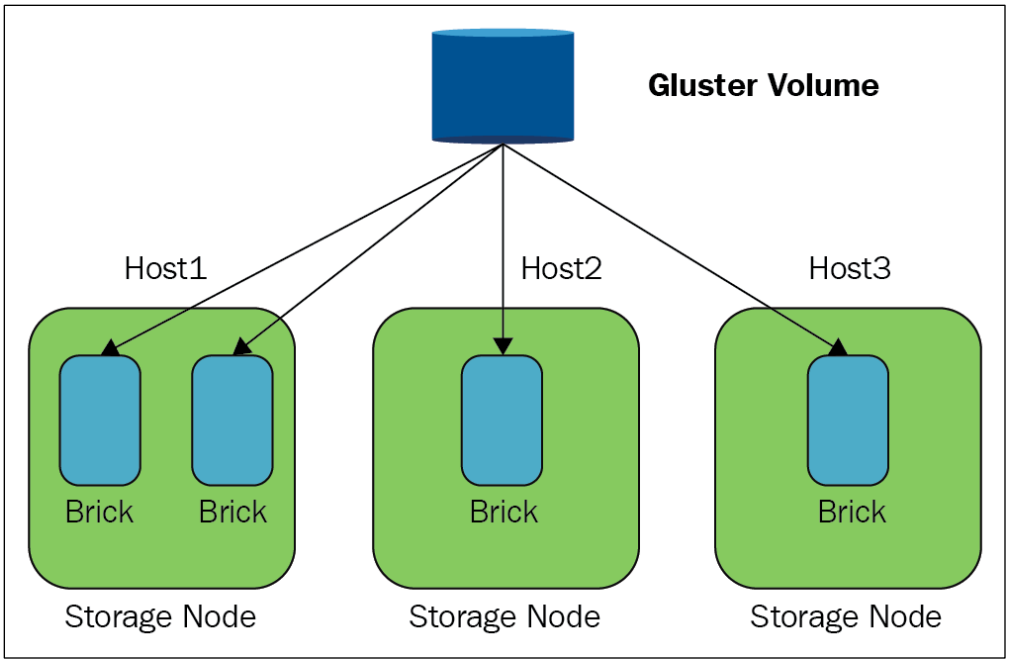

Service
selector:
app: awesome
Pod v 1.0
Pod v 1.0
Deployment
labels:
app: awesome
Create canary deployment (service starts sending requests immediately)
Service
selector:
app: awesome
Pod v 1.0
Pod v 1.0
Deployment
labels:
app: awesome
Pod v 1.1
Deployment
labels:
app: awesome

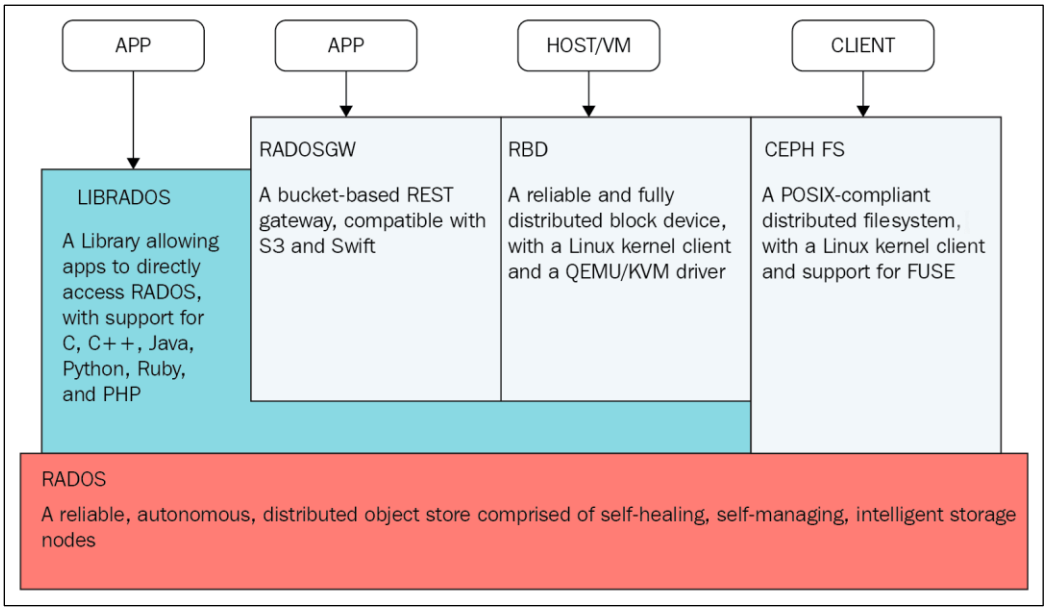
Chapter 4: Securing Kubernetes

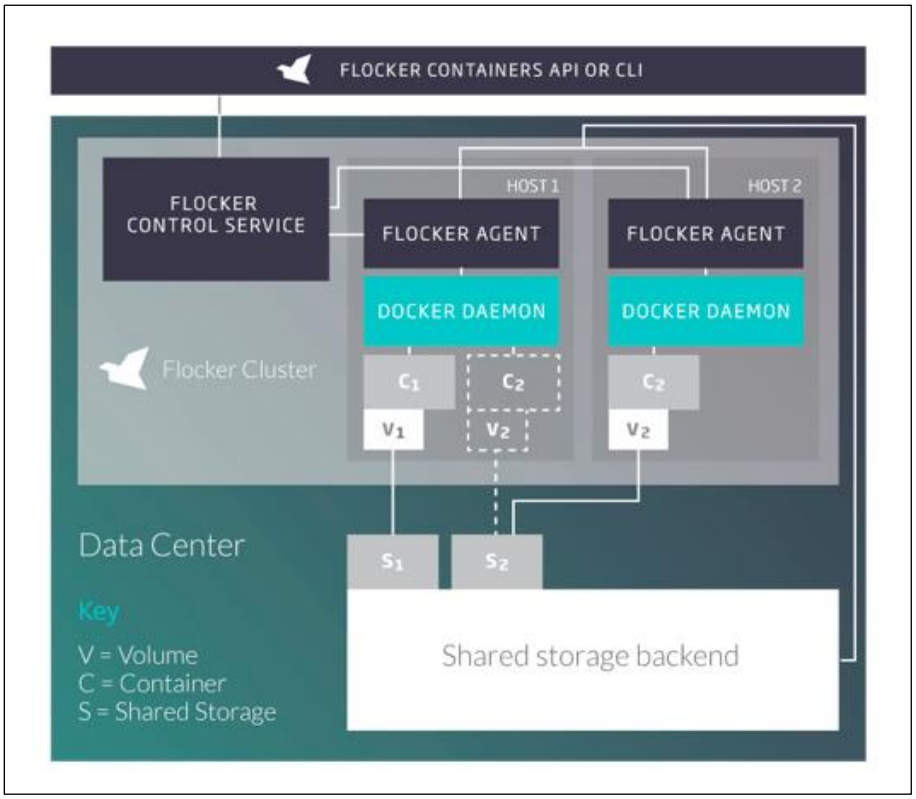


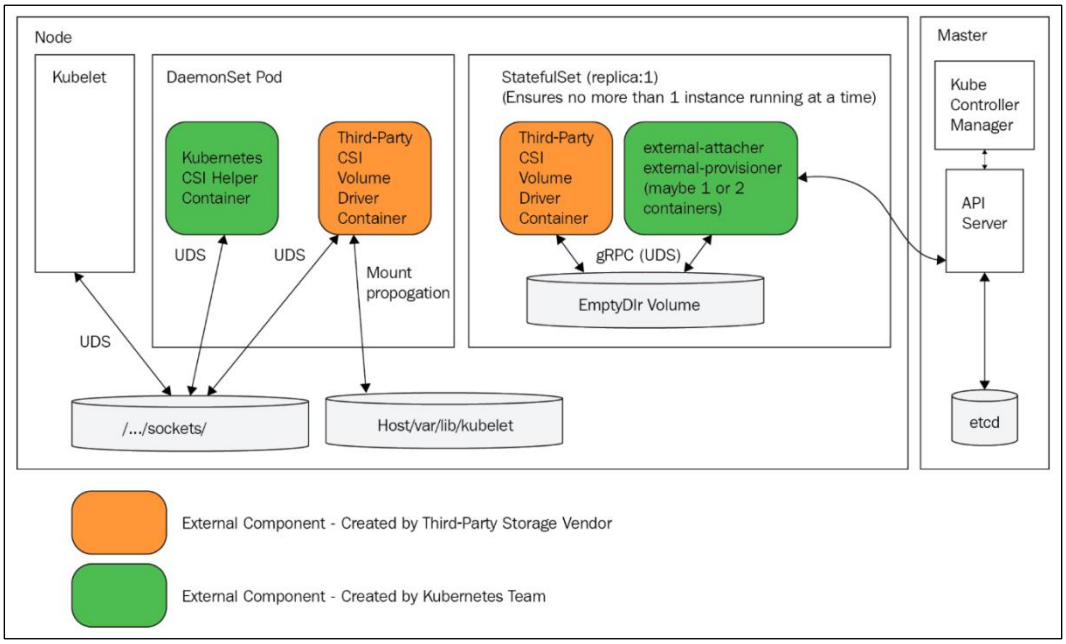
Chapter 6: Managing Storage



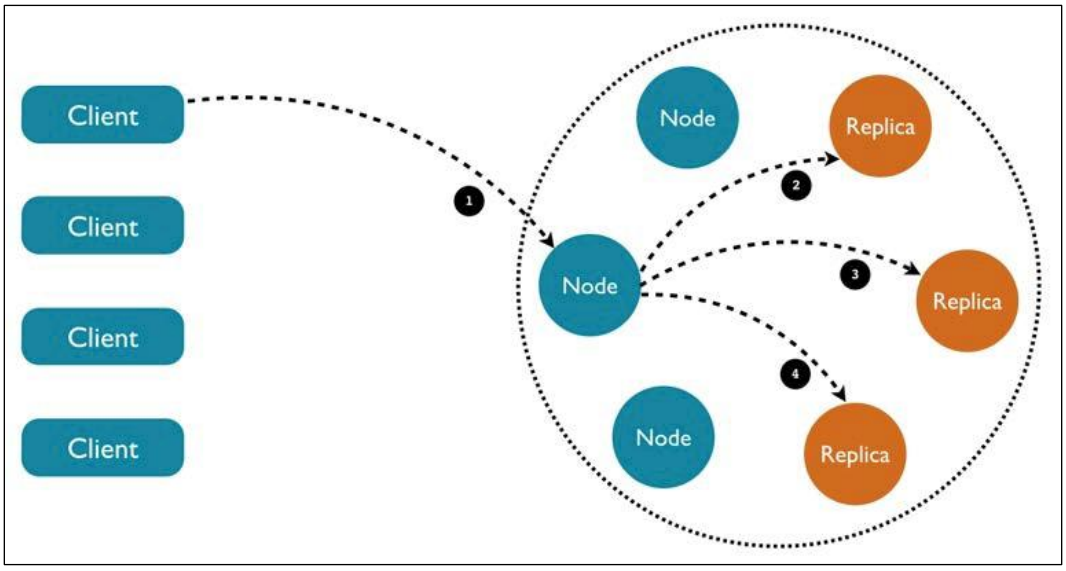



APP
APP
HOST/VM
CLIENT
LIBRADOS
A Library allowing apps to directly access RADOS, with support for C, C++, Java, Python, Ruby, and PHP
RADOSGW
A bucket-based REST gateway, compatible with S3 and Swift
RBD
A reliable and fully distributed block device, with a Linux kernel client and a QEMU/KVM driver
CEPH FS
A POSIX-compliant distributed filesystem, with a Linux kernel client and support for FUSE
RADOS
A reliable, autonomous, distributed object store comprised of self-healing, self-managing, intelligent storage nodes

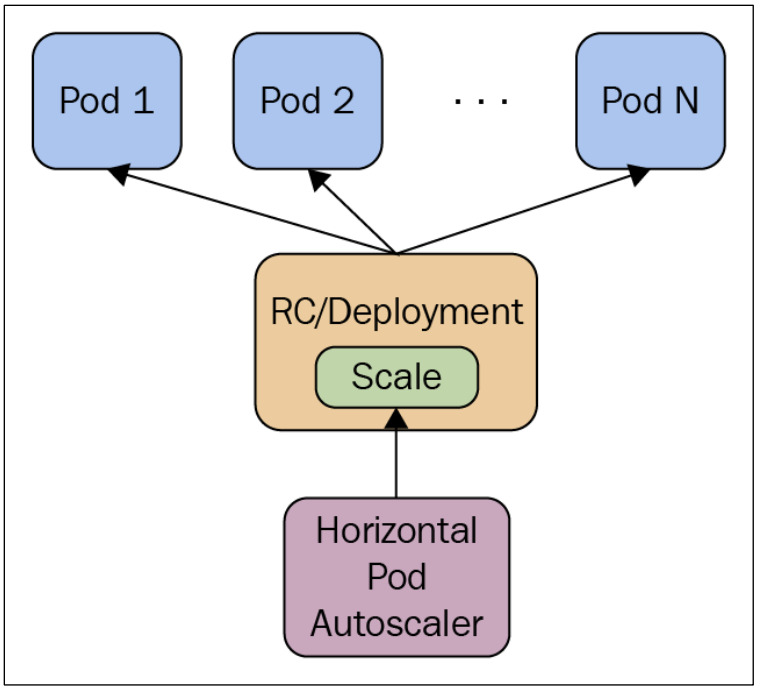

FLOCKER CONTAINERS API OR CLI
FLOCKER CONTROL SERVICE
HOST 1
HOST 2
FLOCKER AGENT
FLOCKER AGENT
DOCKER DAEMON
DOCKER DAEMON
Flocker Cluster
C1
C2
C2
V1
V2
V2
Data Center
S1
S2
Key
V = Volume
C = Container
S = Shared Storage
Shared storage backend

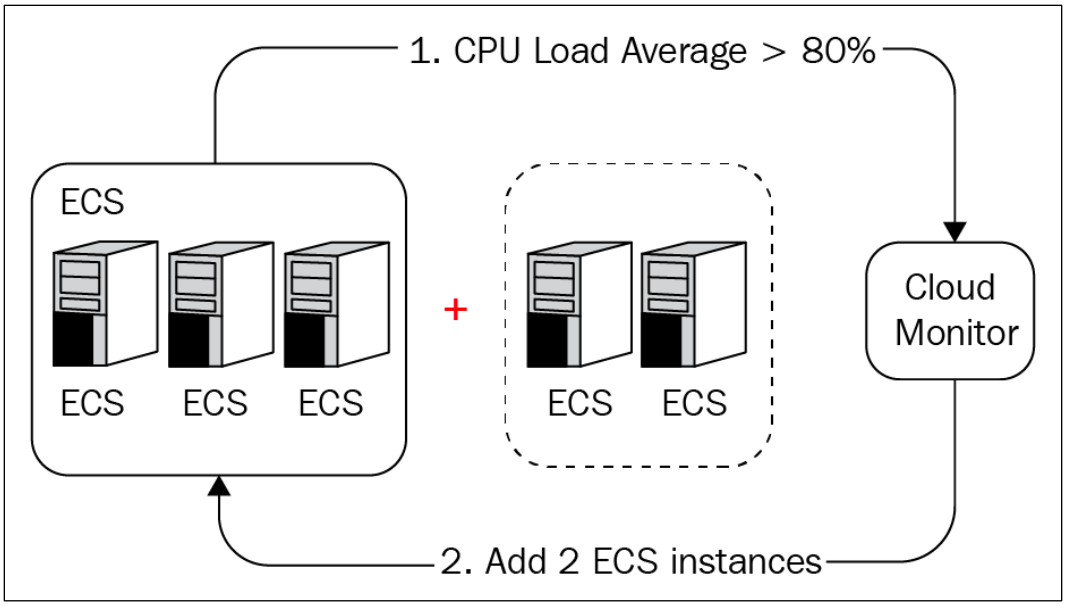

Node
Kubelet
DaemonSet Pod
Kubernetes CSI Helper Container
Third-Party CSI Volume Driver Container
UDS
UDS
Mount propogation
UDS
/.../sockets/
Host/var/lib/kubelet
StatefulSet (replica:1)
(Ensures no more than 1 instance running at a time)
Third-Party CSI Volume Driver Container
external-attacher external-provisioner (maybe 1 or 2 containers)
gRPC (UDS)
EmptyDir Volume
Master
Kube Controller Manager
API Server
etcd
External Component - Created by Third-Party Storage Vendor
External Component - Created by Kubernetes Team

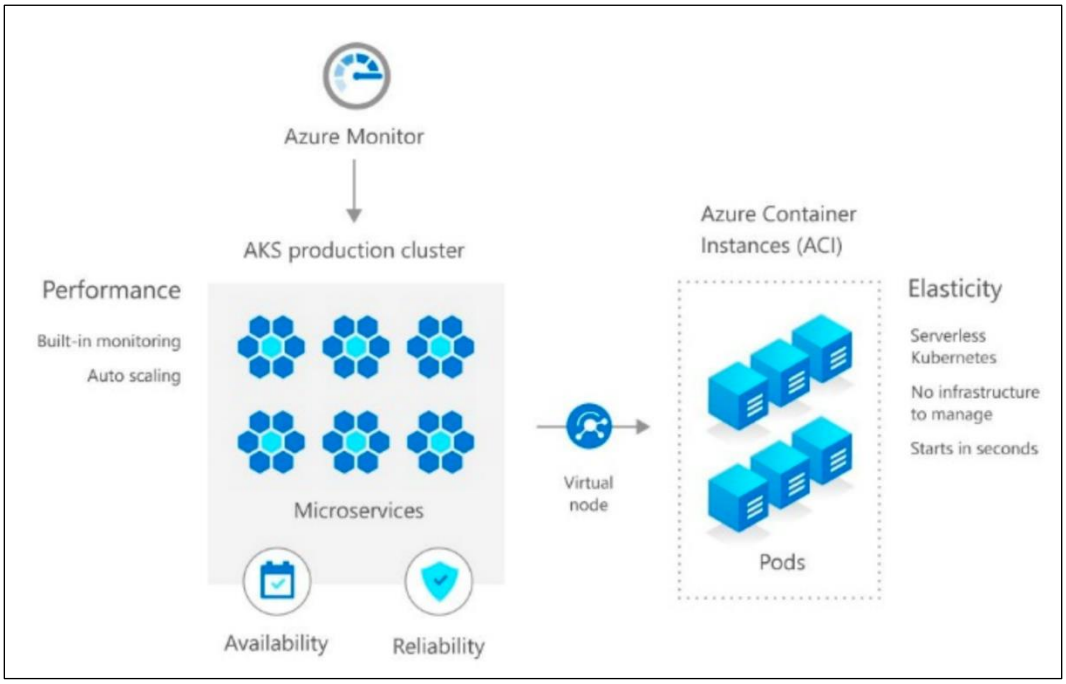
Chapter 7: Running Stateful Applications with Kubernetes

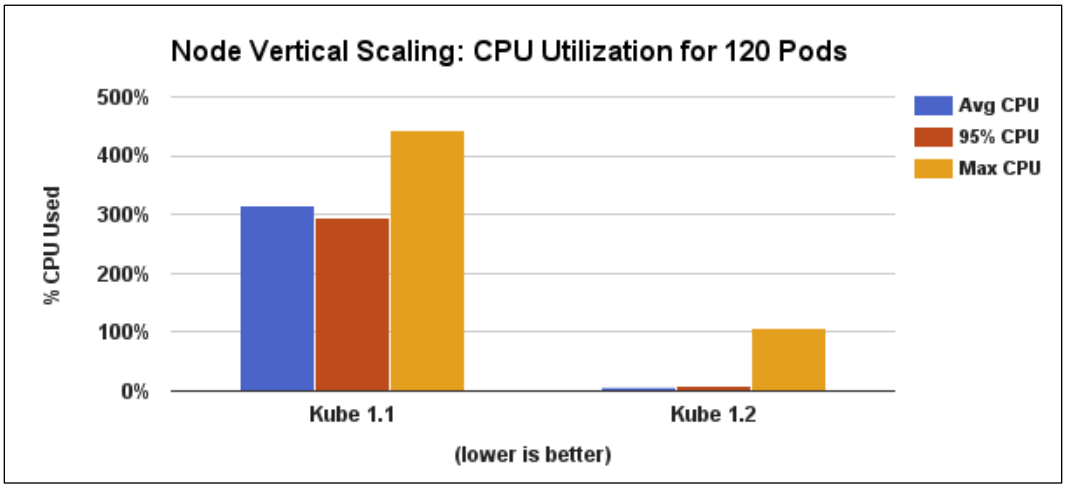


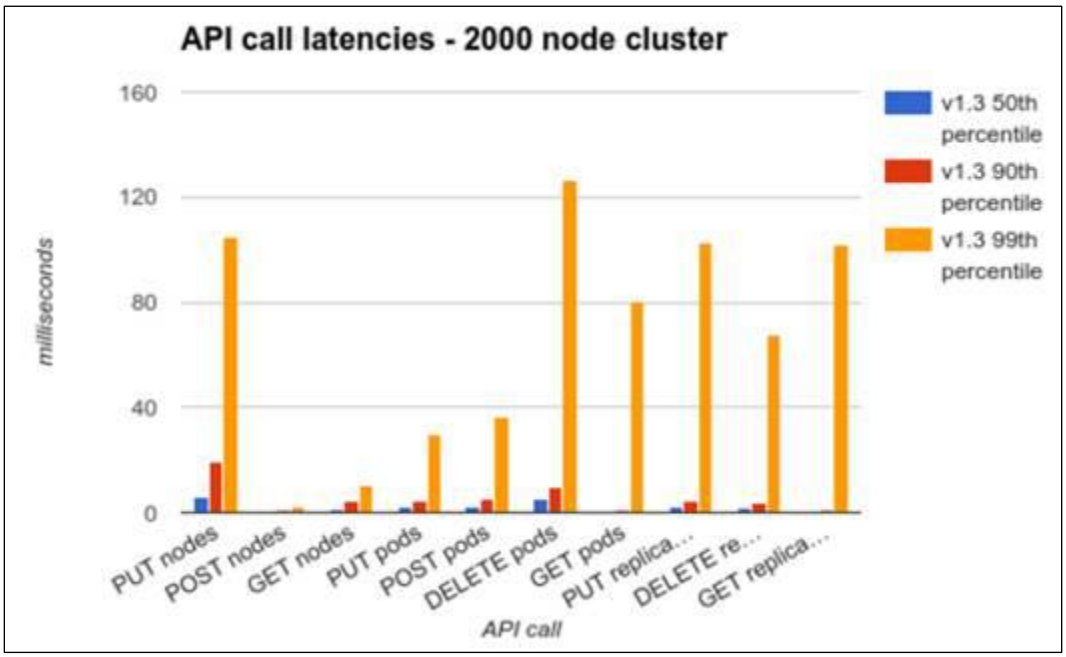
Chapter 8: Deploying and Updating Applications

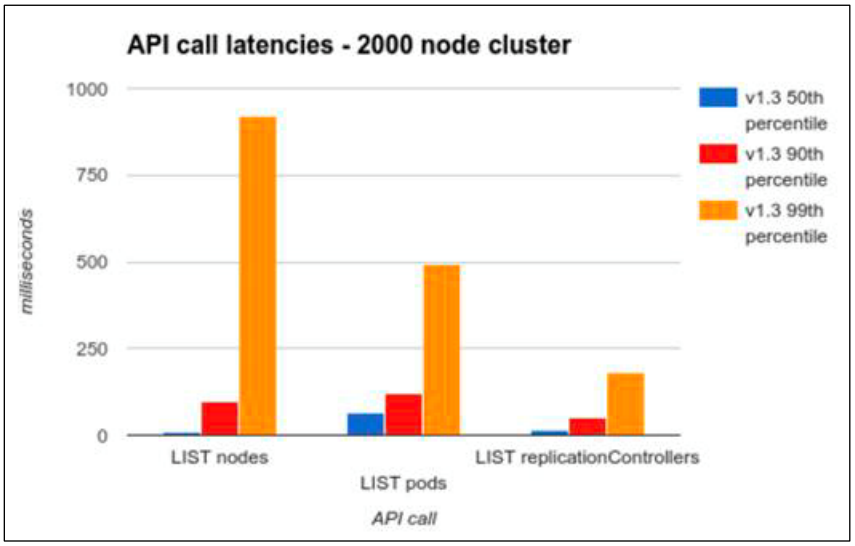


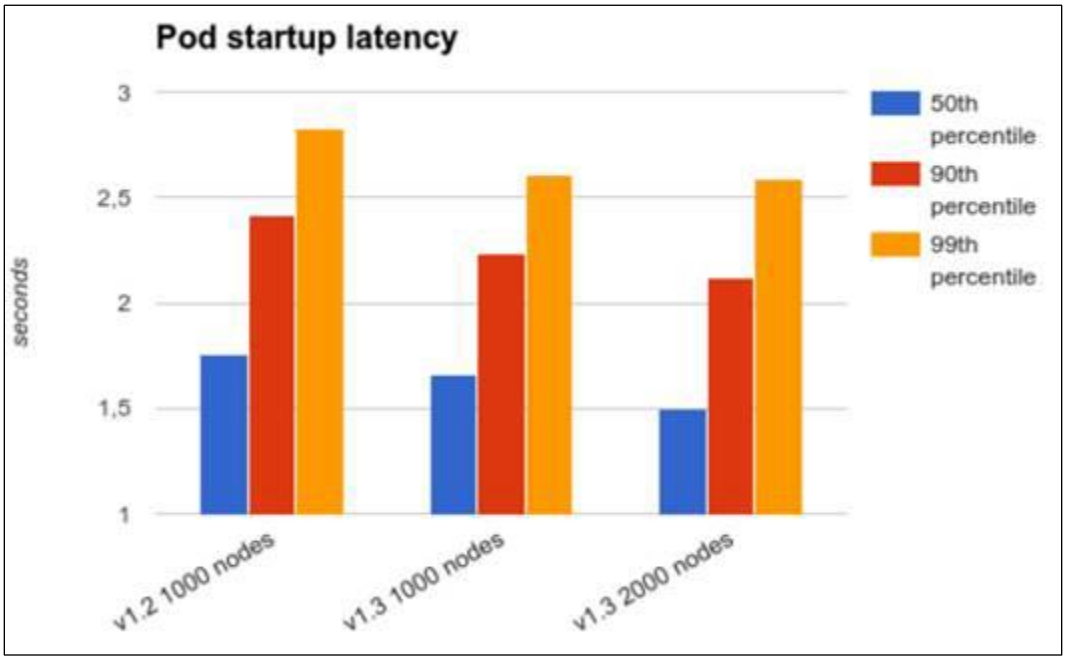


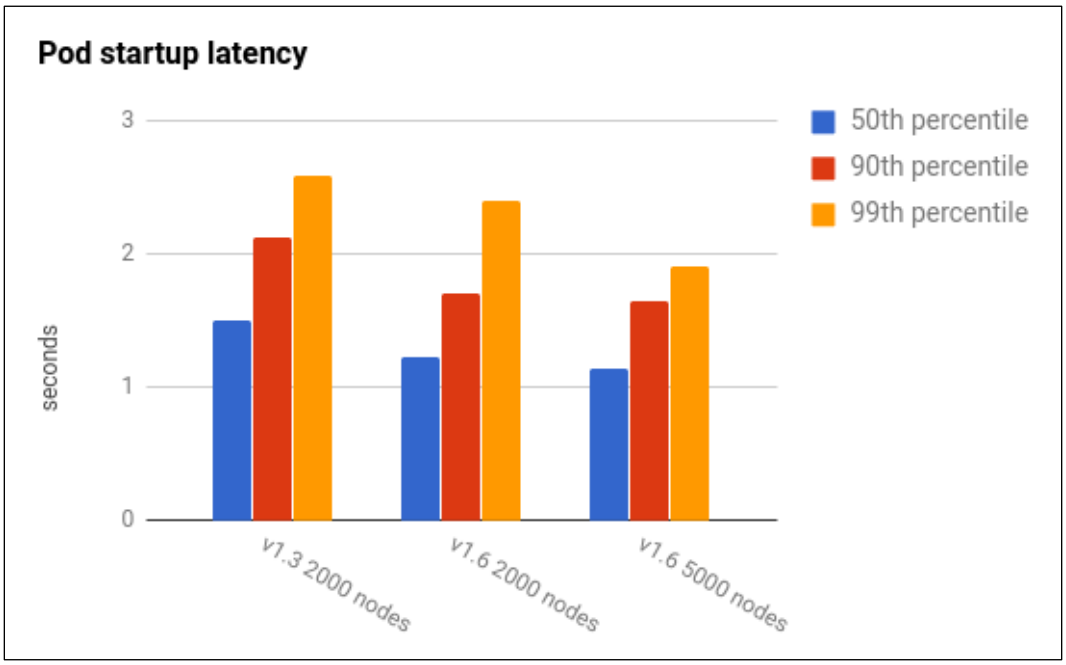

Azure Monitor
AKS production cluster
Performance
Built-in monitoring
Auto scaling
Microservices
Availability
Reliability
Virtual node
Azure Container Instances (ACI)
Pods
Elasticity
Serverless Kubernetes
No infrastructure to manage
Starts in seconds


Node Vertical Scaling: CPU Utilization for 120 Pods
% CPU Used
500%
400%
300%
200%
100%
0%
Kube 1.1
Kube 1.2
(lower is better)
Avg CPU
95% CPU
Max CPU

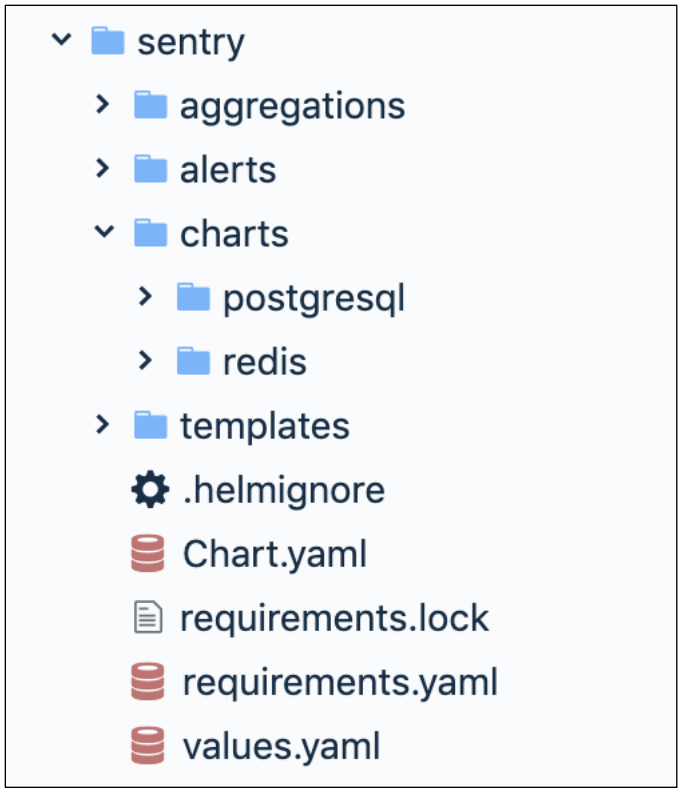

API call latencies - 2000 node cluster
160
120
80
40
0
milliseconds
v1.3 50th percentile
v1.3 90th percentile
v1.3 99th percentile
PUT nodes
POST nodes
GET nodes
PUT pods
POST pods
DELETE pods
GET pods
PUT replica...
DELETE re...
GET replica...
API call


API call latencies - 2000 node cluster
1000
750
500
250
0
milliseconds
v1.3 50th percentile
v1.3 90th percentile
v1.3 99th percentile
LIST nodes
LIST pods
LIST replicationControllers
API call

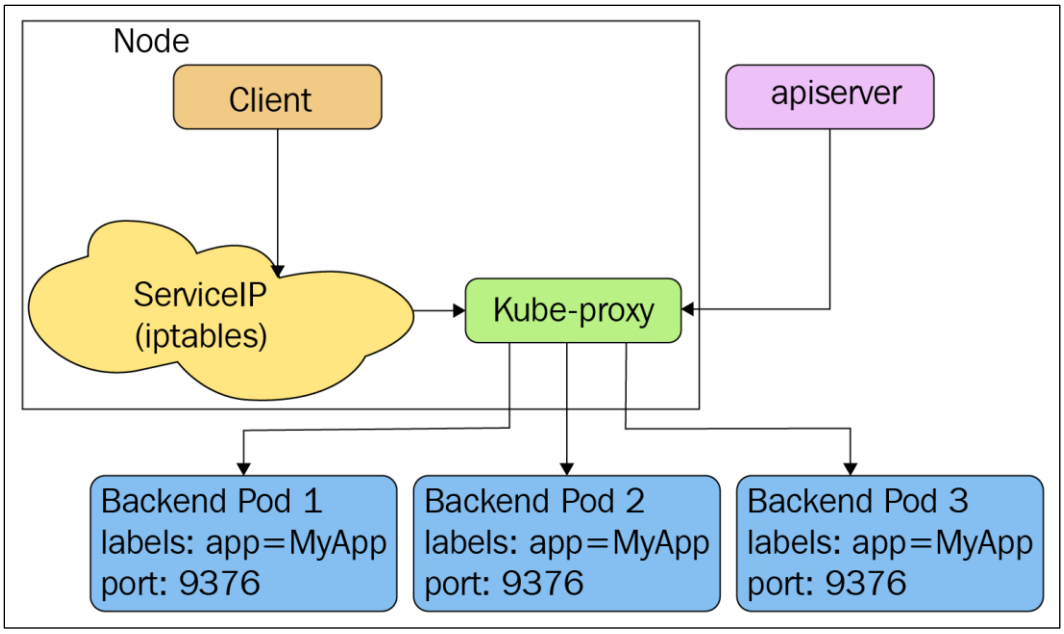

Pod startup latency
3
2,5
2
1,5
1
seconds
50th percentile
90th percentile
99th percentile
v1.2 1000 nodes
v1.3 1000 nodes
v1.3 2000 nodes

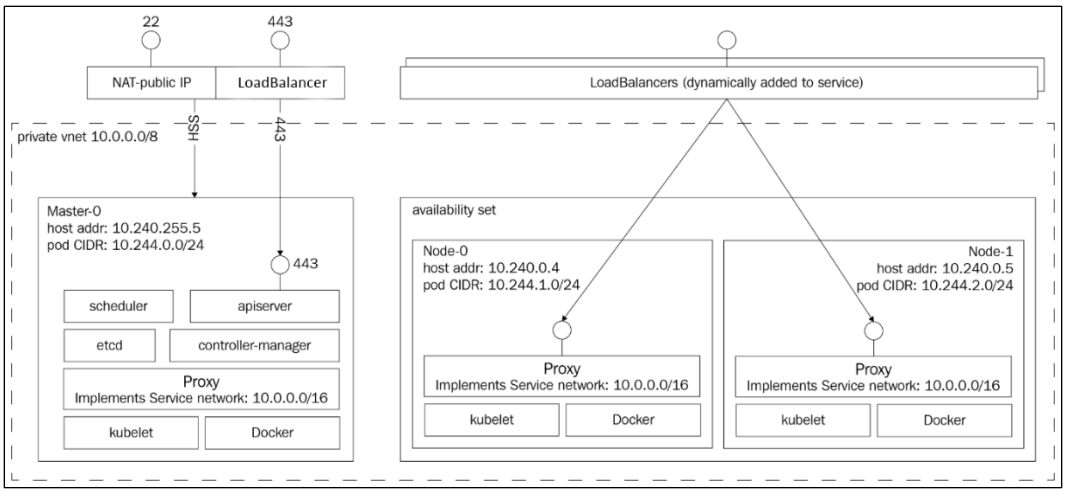

Pod startup latency
3
2
1
0
seconds
50th percentile
90th percentile
99th percentile
v1.3 2000 nodes
v1.6 2000 nodes
v1.6 5000 nodes

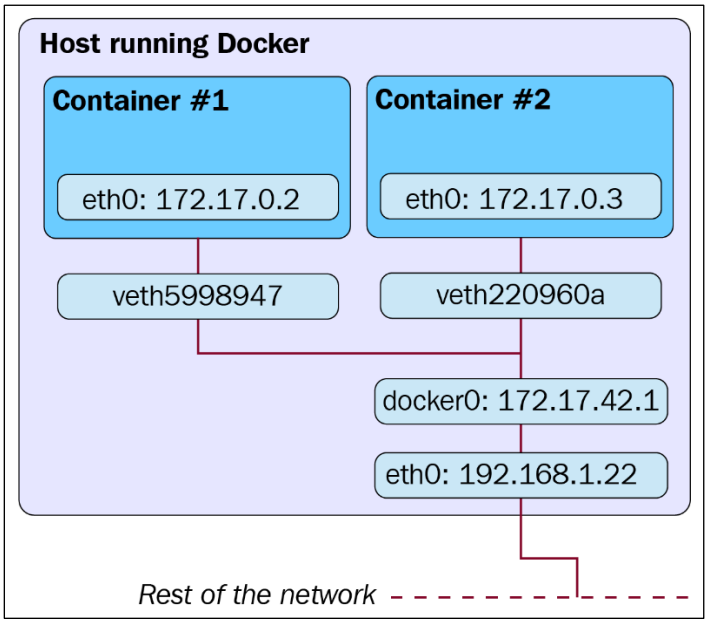
Chapter 9: Packaging Applications

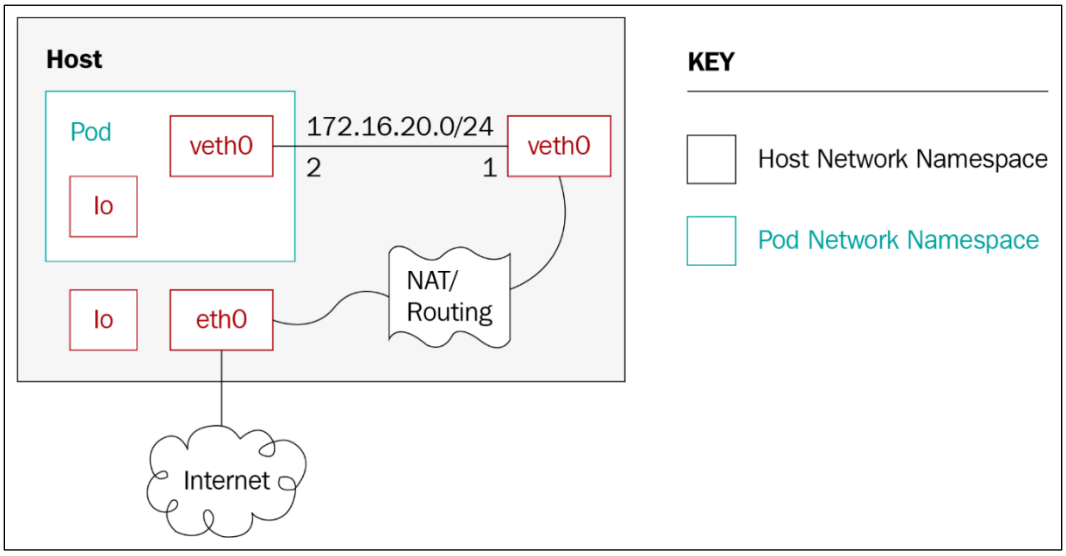


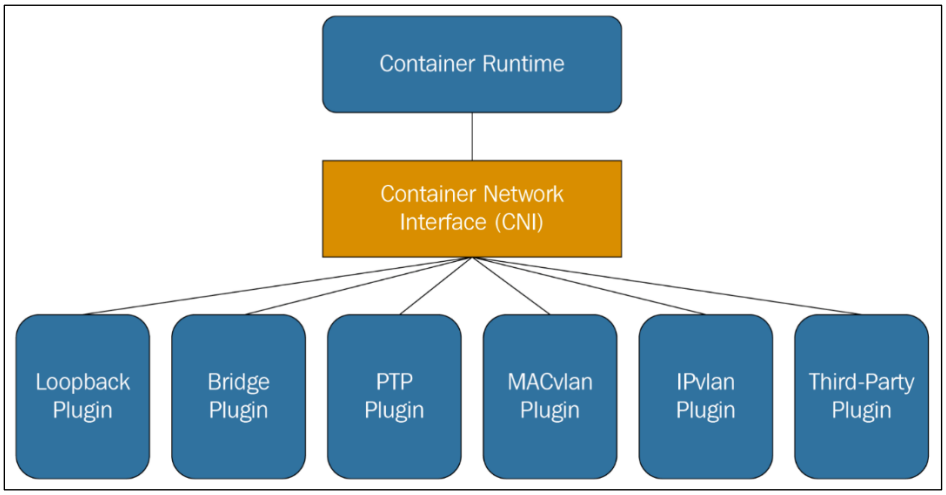
Chapter 10: Exploring Advanced Networking

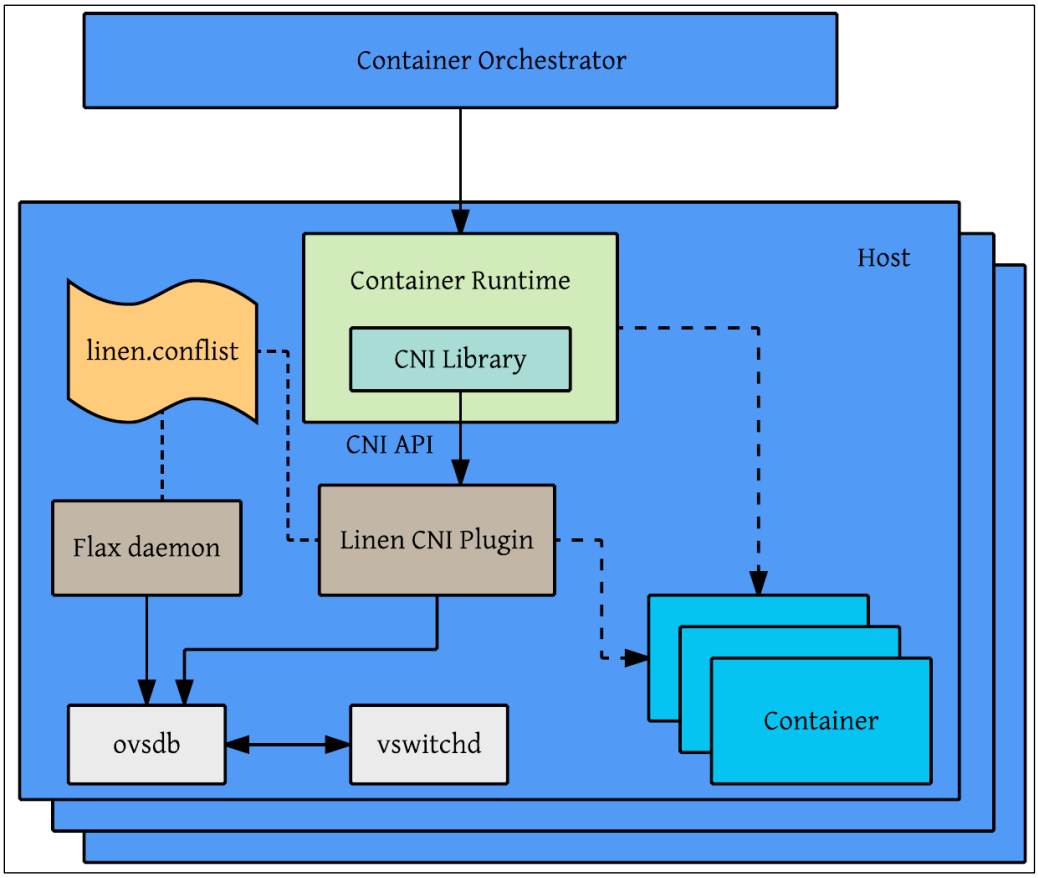


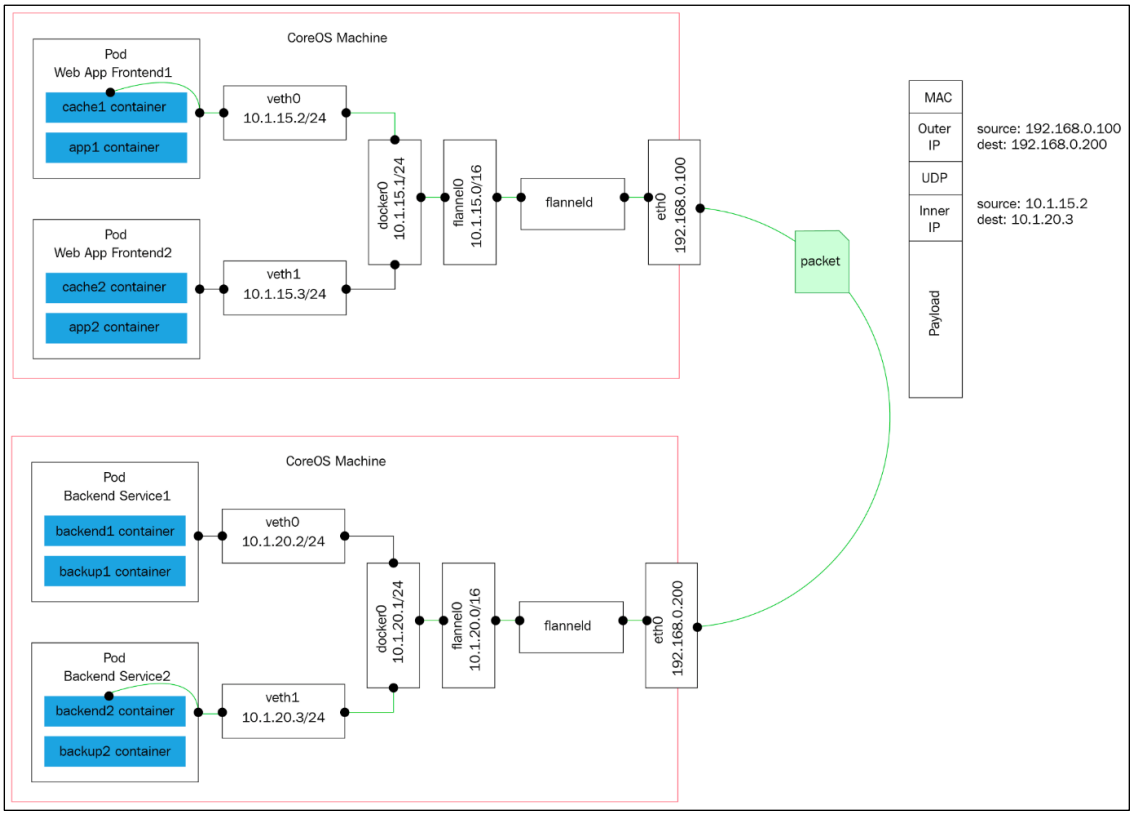


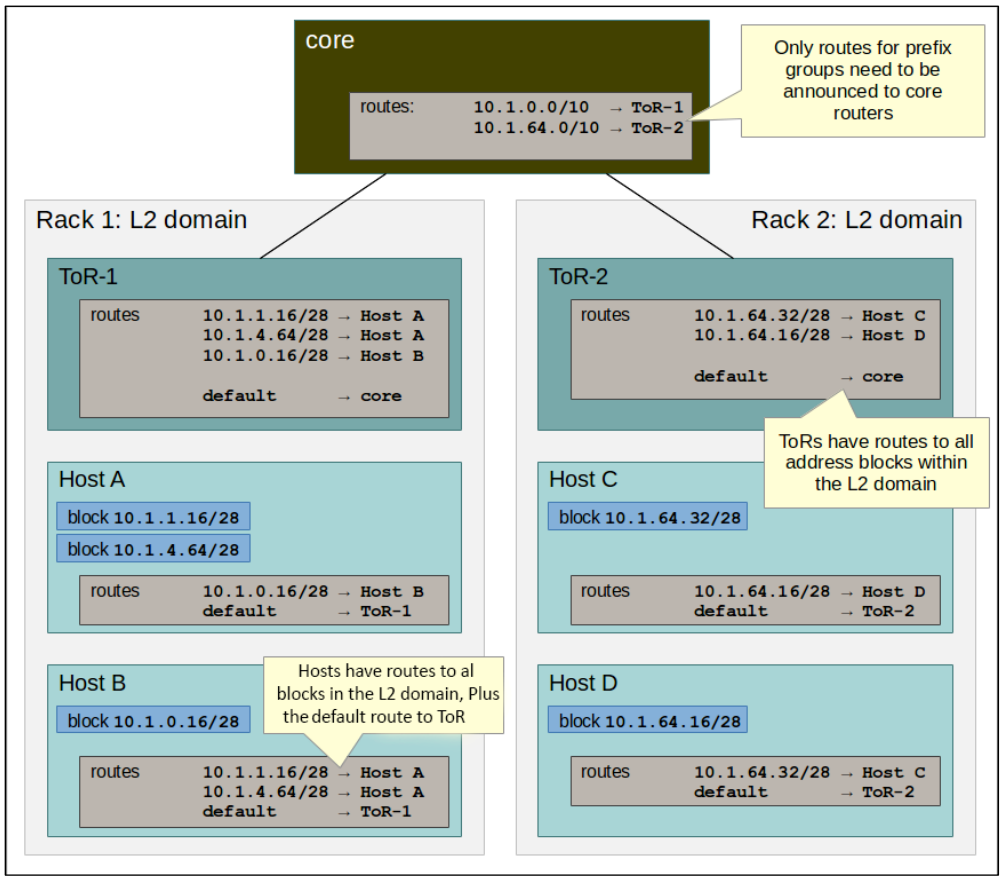

Host running Docker
Container #1
eth0: 172.17.0.2
Container #2
eth0: 172.17.0.3
veth5998947
veth220960a
docker0: 172.17.42.1
eth0: 192.168.1.22
Rest of the network

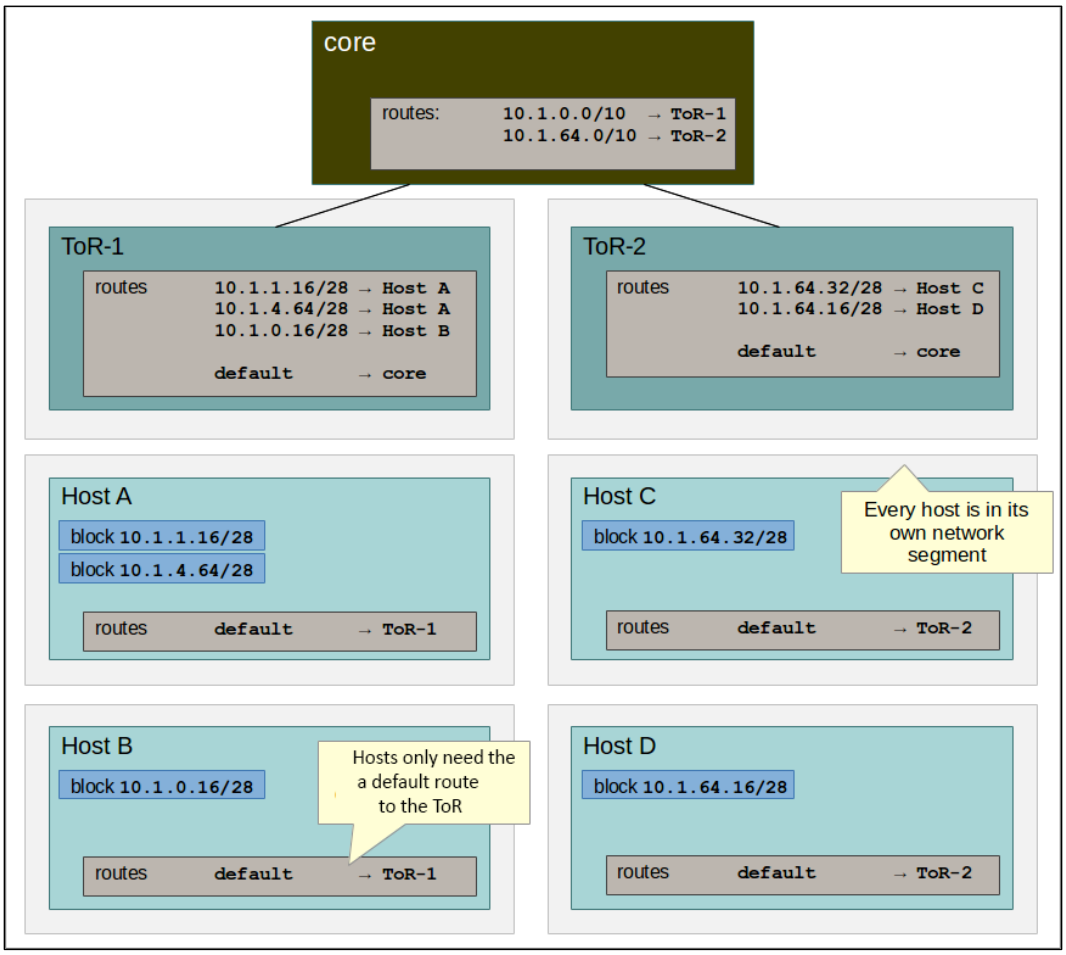

Host
Pod
veth0
lo
172.16.20.0/24
2
1
veth0
lo
eth0
NAT/
Routing
Internet
KEY
Host Network Namespace
Pod Network Namespace

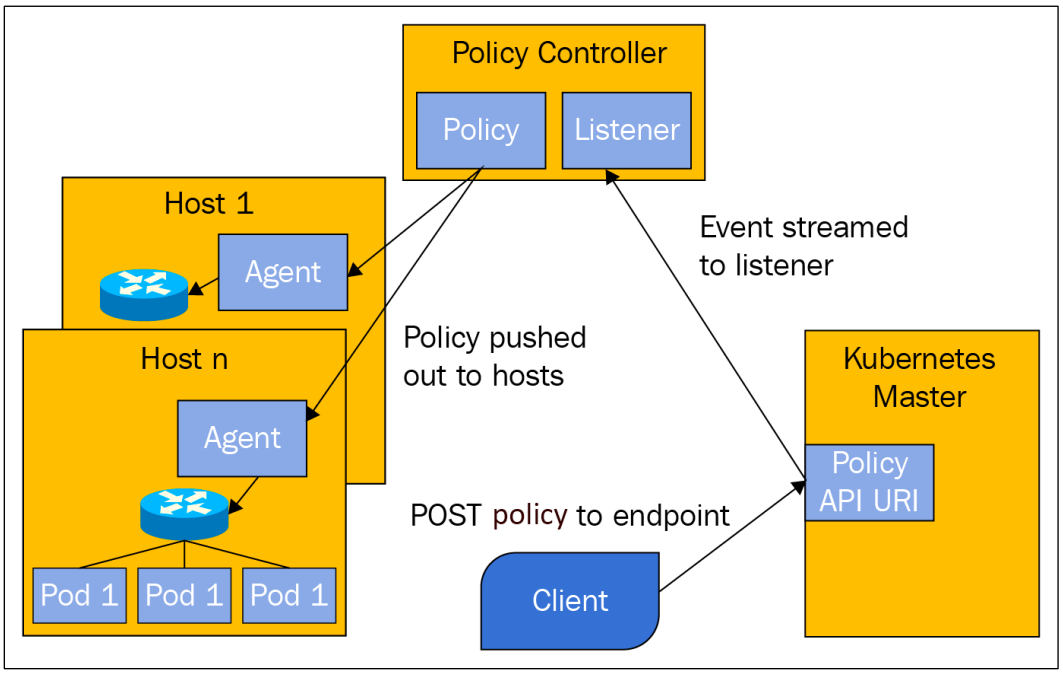

Container Runtime
Container Network Interface (CNI)
Loopback Plugin
Bridge Plugin
PTP Plugin
MACvlan Plugin
IPvlan Plugin
Third-Party Plugin

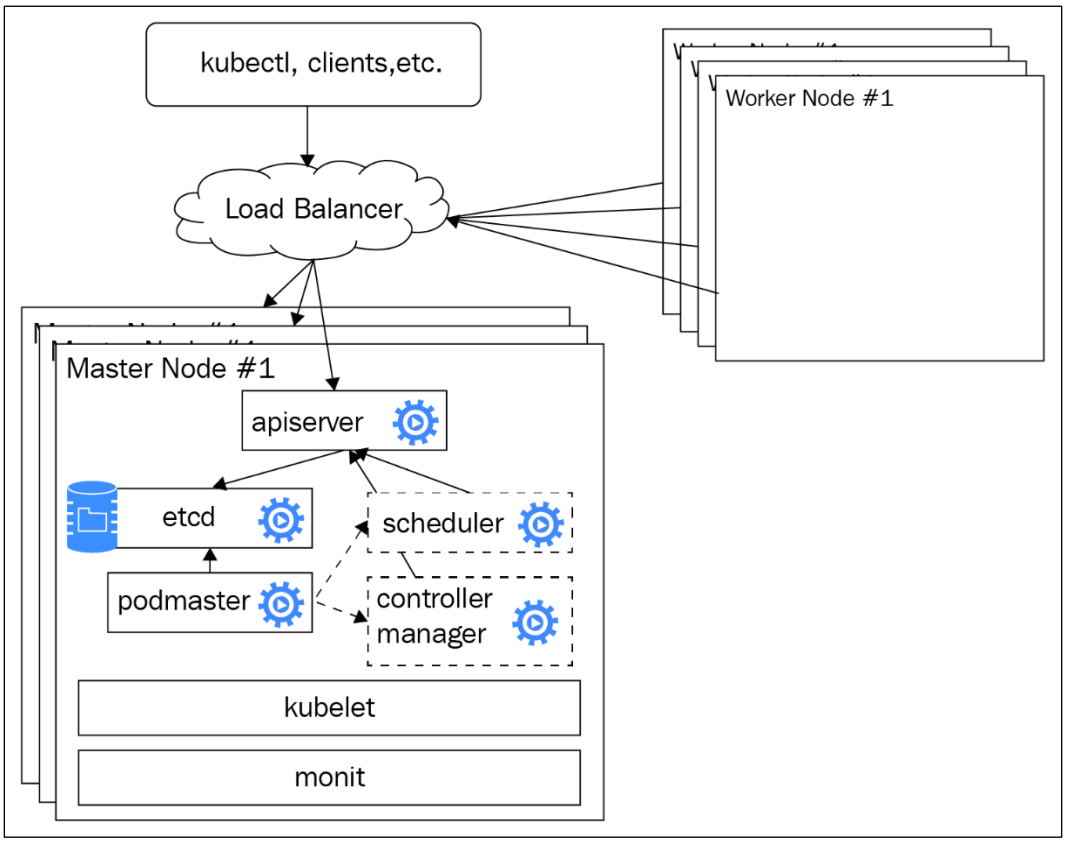

Container Orchestrator
Host
Container Runtime
CNI Library
linen.conflist
CNI API
Flax daemon
Linen CNI Plugin
Container
ovsdb
vswitchd

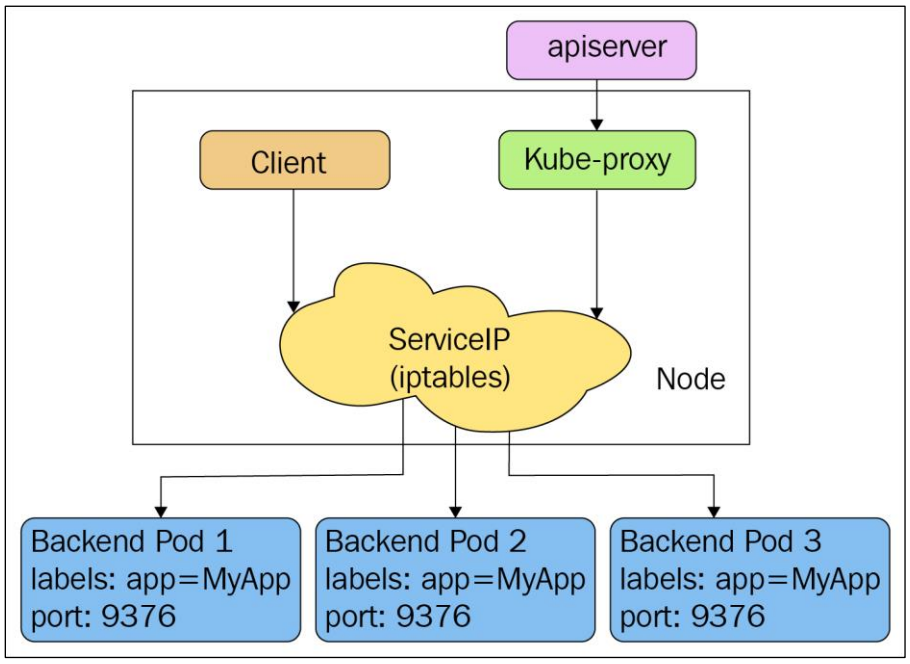

CoreOS Machine
Pod
Web App Frontend1
cache1 container
app1 container
Pod
Web App Frontend2
cache2 container
app2 container
veth0
10.1.15.2/24
veth1
10.1.15.3/24
docker0
10.1.15.1/24
flannel0
10.1.15.0/16
flanneld
eth0
192.168.0.100
packet
MAC
Outer IP
UDP
Inner IP
Payload
source: 192.168.0.100
dest: 192.168.0.200
source: 10.1.15.2
dest: 10.1.20.3
CoreOS Machine
Pod
Backend Service1
backend1 container
backup1 container
Pod
Backend Service2
backend2 container
backup2 container
veth0
10.1.20.2/24
veth1
10.1.20.3/24
docker0
10.1.20.1/24
flannel0
10.1.20.0/16
flanneld
eth0
192.168.0.200

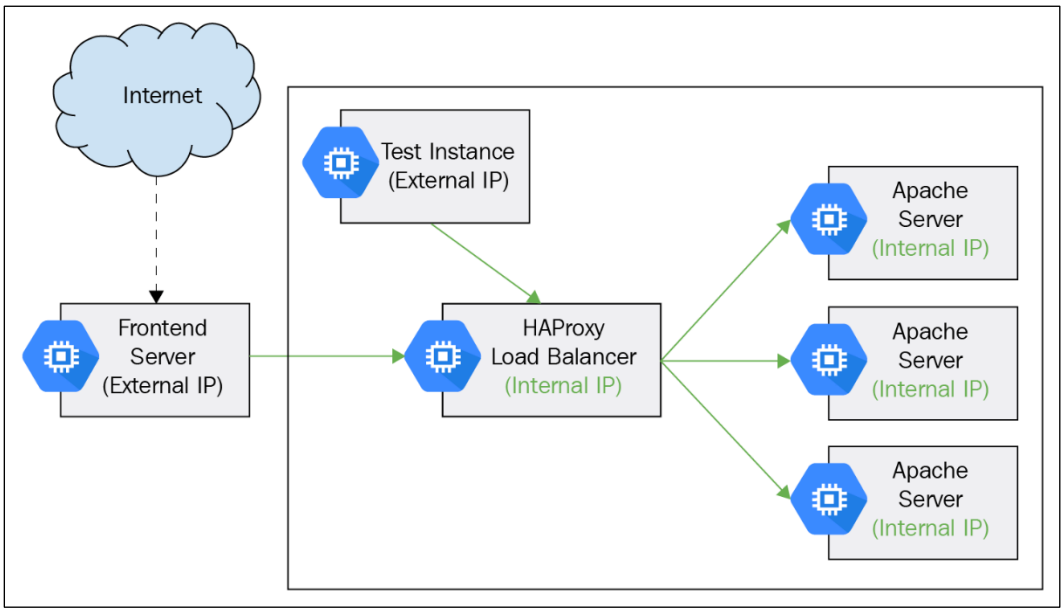

core
routes: 10.1.0.0/10 → ToR-1
10.1.64.0/10 → ToR-2
Only routes for prefix groups need to be announced to core routers
Rack 1: L2 domain
Rack 2: L2 domain
ToR-1
routes 10.1.1.16/28 → Host A
10.1.4.64/28 → Host A
10.1.0.16/28 → Host B
default → core
ToR-2
routes 10.1.64.32/28 → Host C
10.1.64.16/28 → Host D
default → core
ToRs have routes to all address blocks within the L2 domain
Host A
block 10.1.1.16/28
block 10.1.4.64/28
routes 10.1.0.16/28 → Host B
default → ToR-1
Host C
block 10.1.64.32/28
routes 10.1.64.16/28 → Host D
default → ToR-2
Host B
block 10.1.0.16/28
Hosts have routes to al blocks in the L2 domain, Plus the default route to ToR
routes 10.1.1.16/28 → Host A
10.1.4.64/28 → Host A
default → ToR-1
Host D
block 10.1.64.16/28
routes 10.1.64.32/28 → Host C
default → ToR-2


core
routes: 10.1.0.0/10 → ToR-1
10.1.64.0/10 → ToR-2
ToR-1
routes 10.1.1.16/28 → Host A
10.1.4.64/28 → Host A
10.1.0.16/28 → Host B
default → core
ToR-2
routes 10.1.64.32/28 → Host C
10.1.64.16/28 → Host D
default → core
Host A
block 10.1.1.16/28
block 10.1.4.64/28
routes default → ToR-1
Host C
block 10.1.64.32/28
routes default → ToR-2
Every host is in its own network segment
Host B
block 10.1.0.16/28
routes default → ToR-1
Hosts only need the a default route to the ToR
Host D
block 10.1.64.16/28
routes default → ToR-2

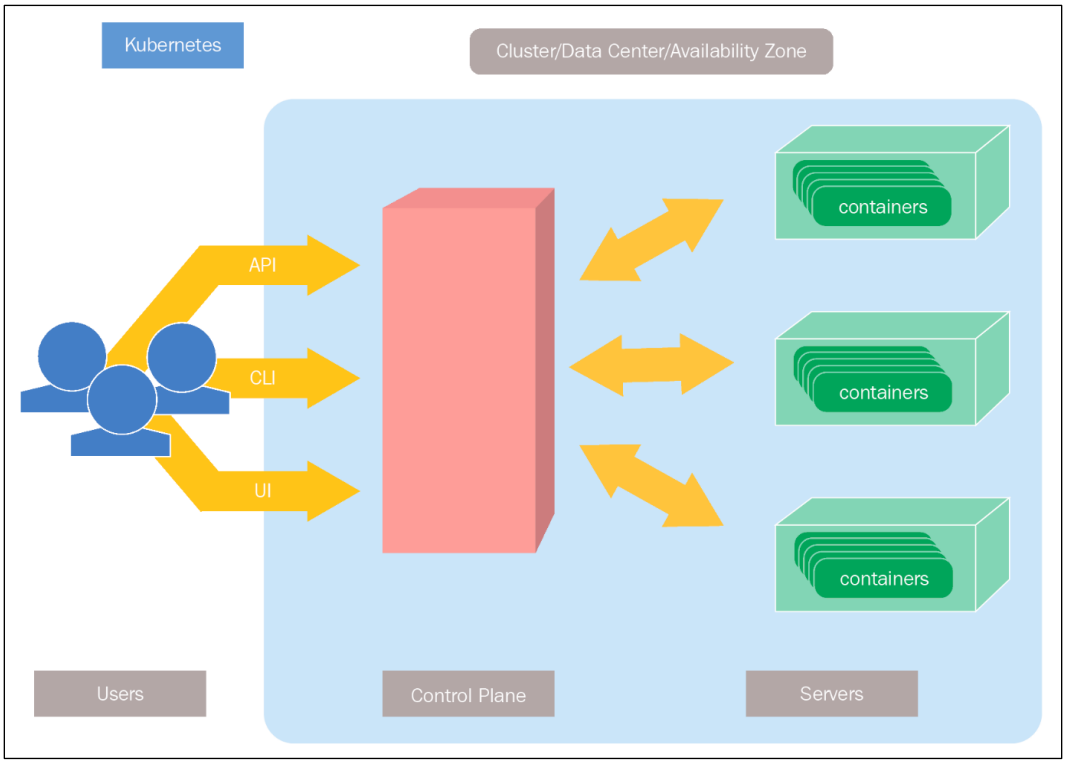

Policy Controller
Policy
Listener
Host 1
Agent
Host n
Agent
Pod 1
Pod 1
Pod 1
Policy pushed
out to hosts
Event streamed
to listener
Kubernetes
Master
Policy
API URI
POST policy to endpoint
Client

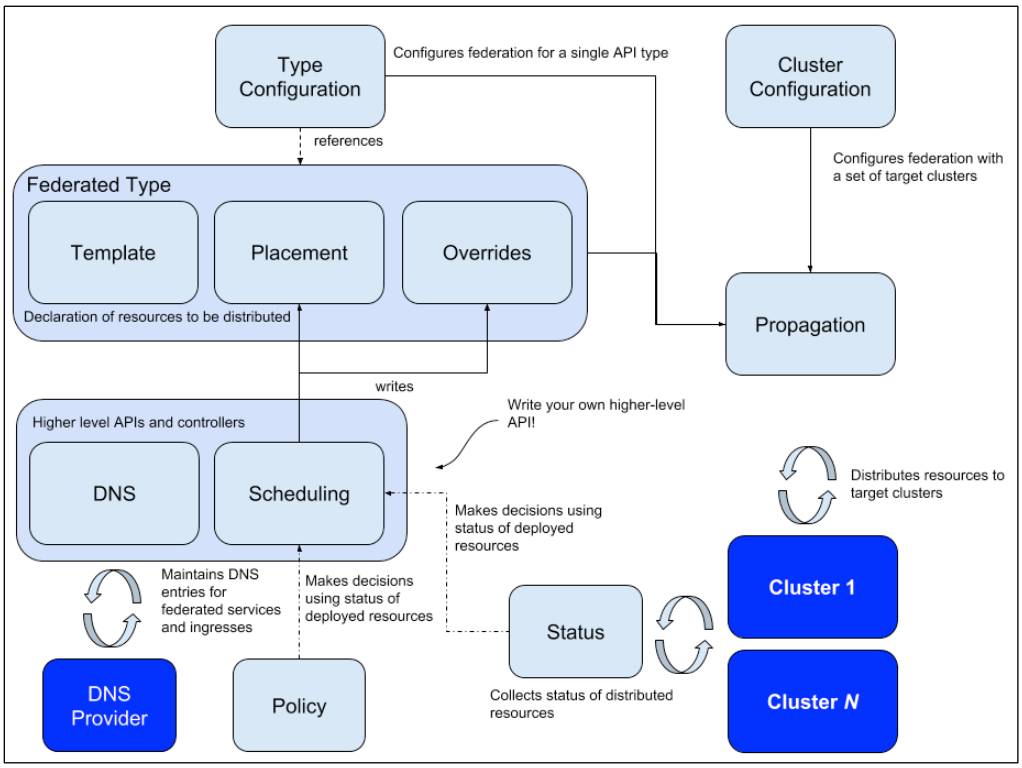

kubectl, clients,etc.
Load Balancer
Worker Node #1
Master Node #1
apiserver
etcd
scheduler
podmaster
controller
manager
kubelet
monit

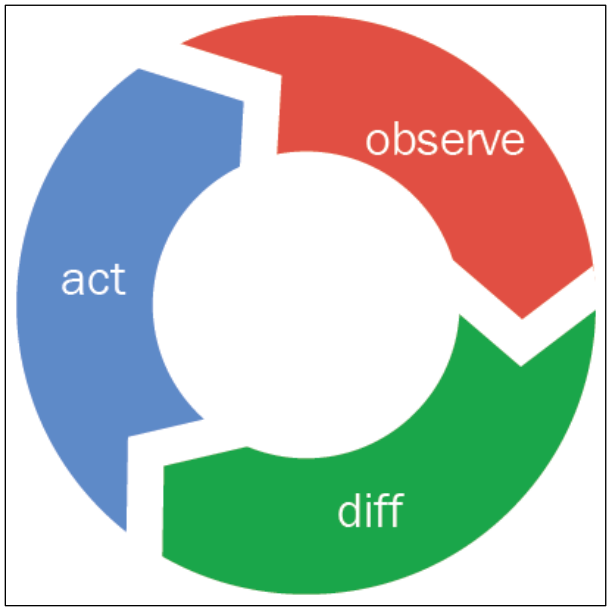

apiserver
Client
Kube-proxy
ServiceIP
(iptables)
Node
Backend Pod 1
labels: app=MyApp
port: 9376
Backend Pod 2
labels: app=MyApp
port: 9376
Backend Pod 3
labels: app=MyApp
port: 9376

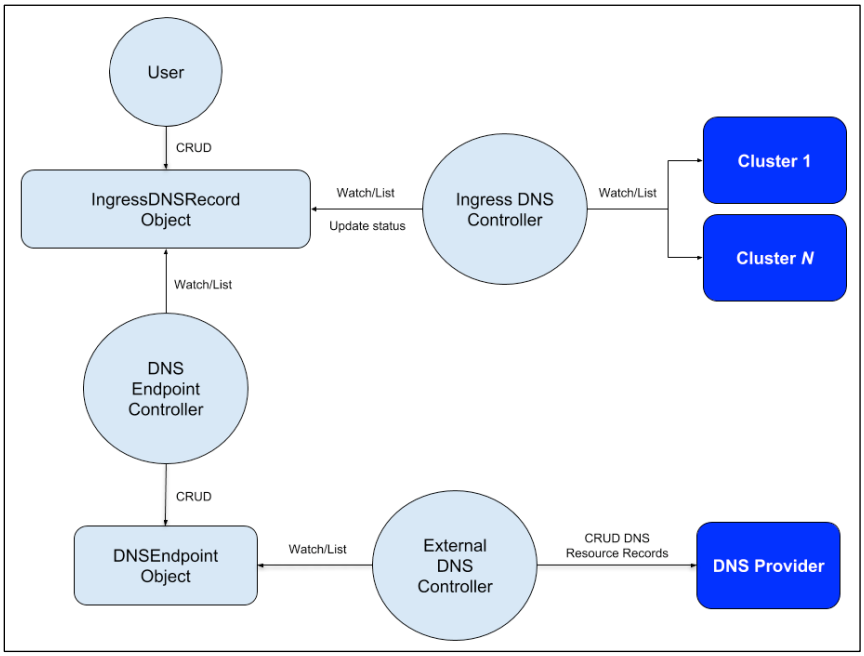

Internet
Test Instance
(External IP)
Frontend
Server
(External IP)
HAProxy
Load Balancer
(Internal IP)
Apache
Server
(Internal IP)
Apache
Server
(Internal IP)
Apache
Server
(Internal IP)

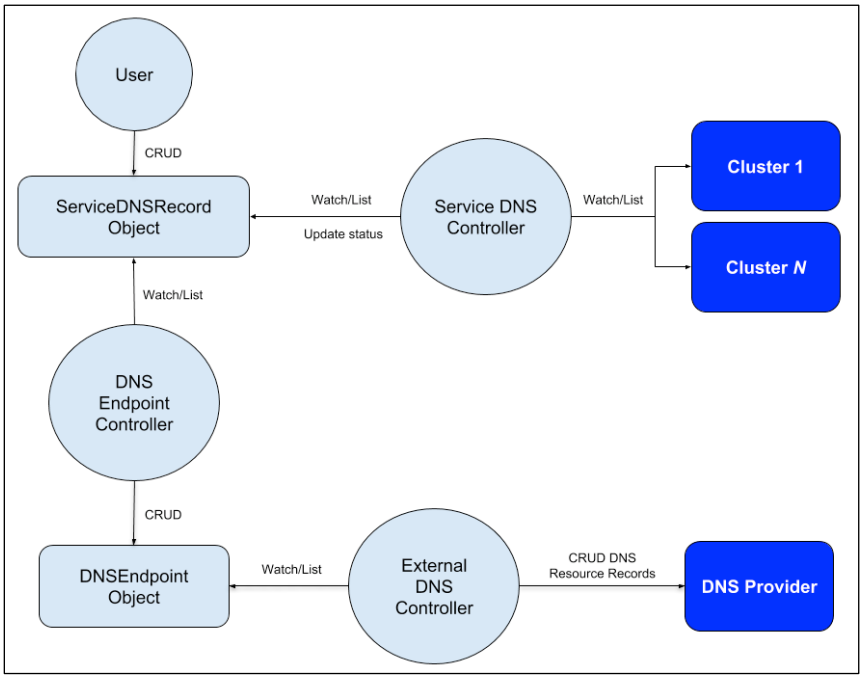
Chapter 11: Running Kubernetes on Multiple Clouds and Cluster Federation

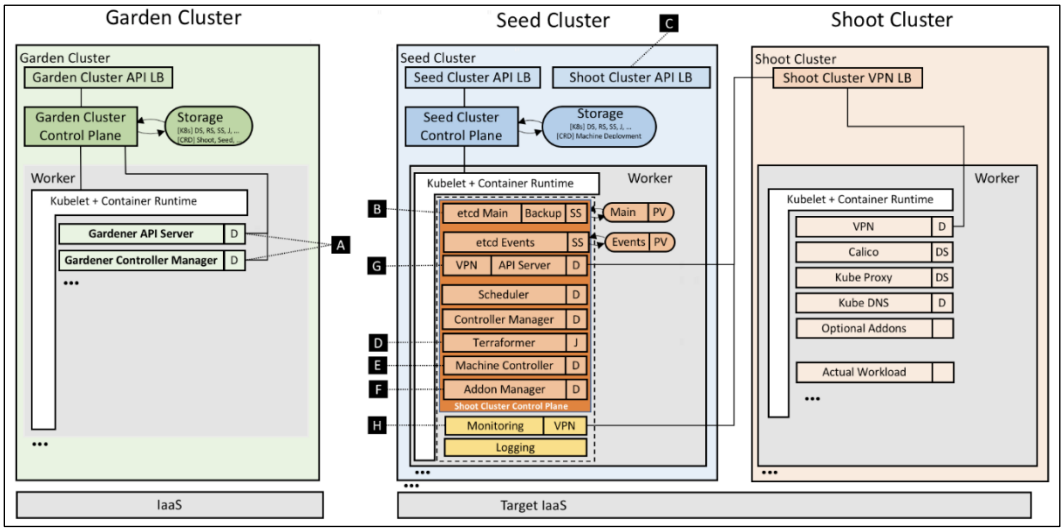


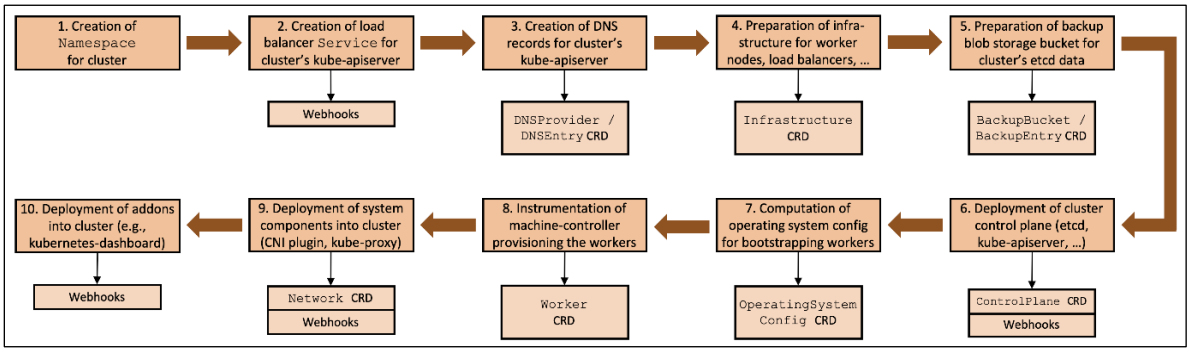

Type Configuration
Configures federation for a single API type
Cluster Configuration
references
Configures federation with a set of target clusters
Federated Type
Template
Placement
Overrides
Declaration of resources to be distributed
Propagation
writes
Higher level APIs and controllers
Write your own higher-level API!
DNS
Scheduling
Distributes resources to target clusters
Makes decisions using status of deployed resources
Cluster 1
Maintains DNS entries for federated services and ingresses
Makes decisions using status of deployed resources
Status
DNS Provider
Policy
Collects status of distributed resources
Cluster N

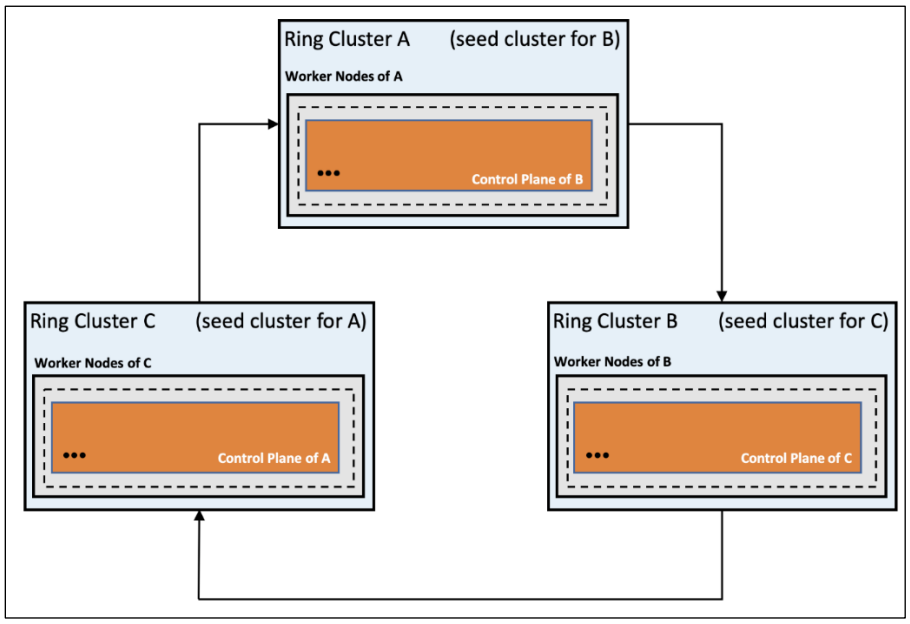

observe
act
diff


User
CRUD
IngressDNSRecord Object
Watch/List
Update status
Ingress DNS Controller
Watch/List
Cluster 1
Cluster N
Watch/List
DNS Endpoint Controller
CRUD
DNSEndpoint Object
Watch/List
External DNS Controller
CRUD DNS Resource Records
DNS Provider

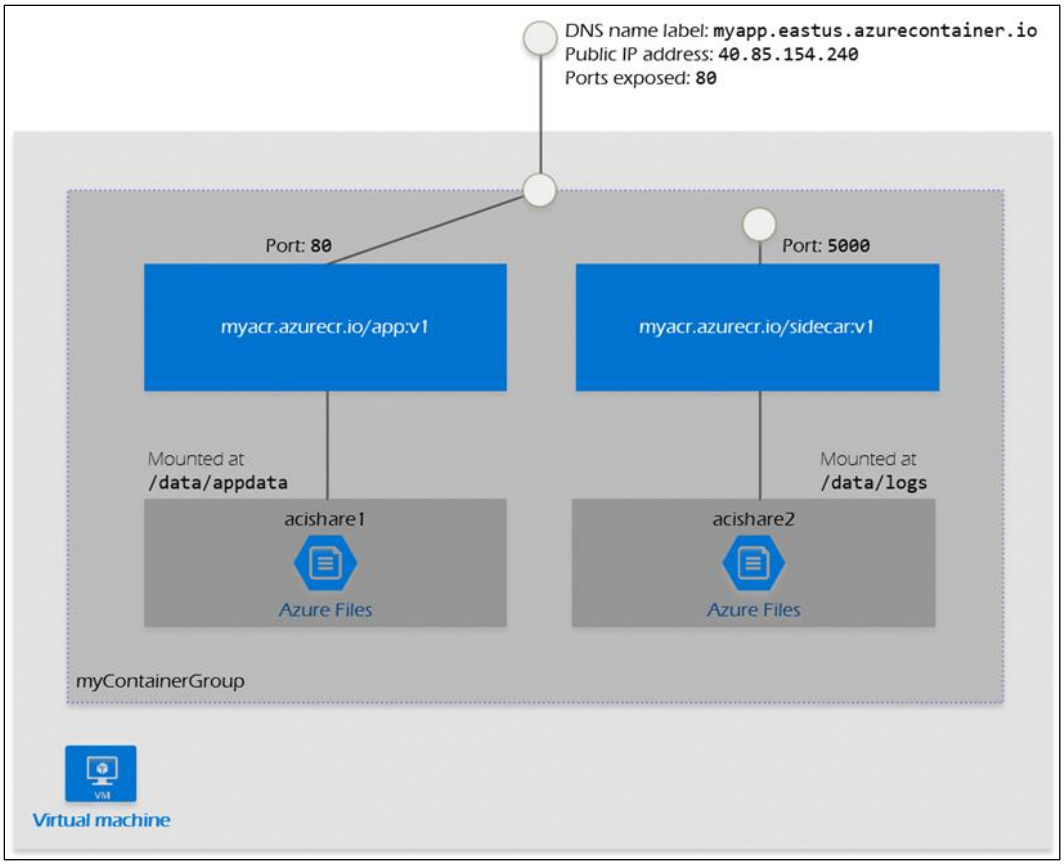

User
CRUD
ServiceDNSRecord Object
Watch/List
Update status
Service DNS Controller
Watch/List
Cluster 1
Cluster N
Watch/List
DNS Endpoint Controller
CRUD
DNSEndpoint Object
Watch/List
External DNS Controller
CRUD DNS Resource Records
DNS Provider


Garden Cluster
Seed Cluster
Shoot Cluster
Garden Cluster
Garden Cluster API LB
Garden Cluster Control Plane
Storage
Worker
Kubelet + Container Runtime
Gardener API Server
Gardener Controller Manager
Seed Cluster
Seed Cluster API LB
Shoot Cluster API LB
Seed Cluster Control Plane
Storage
Kubelet + Container Runtime
Worker
etcd Main
Backup
etcd Events
VPN
API Server
Scheduler
Controller Manager
Terraformer
Machine Controller
Addon Manager
Shoot Cluster Control Plane
Monitoring
Logging
Main
Events
PV
Shoot Cluster
Shoot Cluster VPN LB
Kubelet + Container Runtime
Worker
VPN
Calico
Kube Proxy
Kube DNS
Optional Addons
Actual Workload
IaaS
Target IaaS
A
B
C
D
E
F
G
H

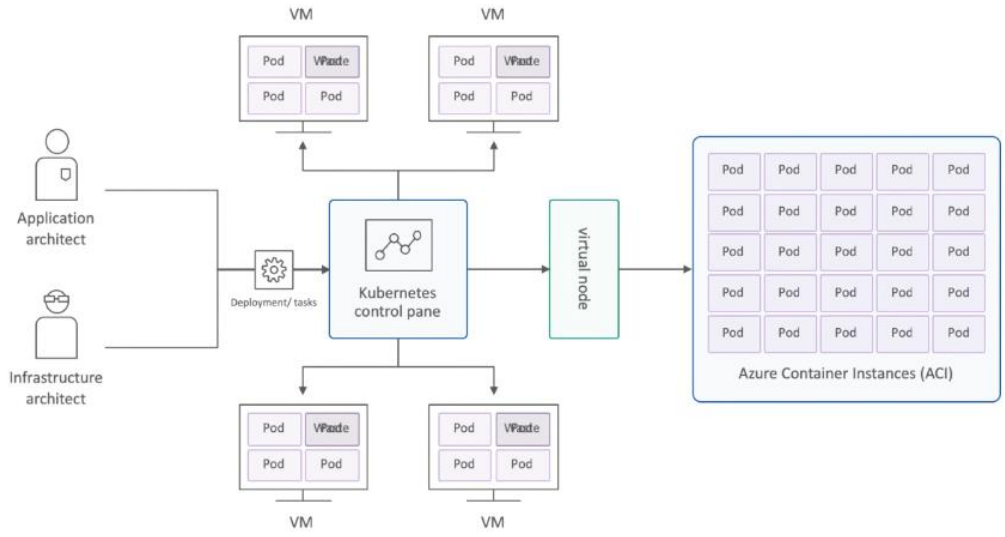

1. Creation of Namespace for cluster
2. Creation of load balancer Service for cluster's kube-apiserver
Webhooks
3. Creation of DNS records for cluster's kube-apiserver
DNSProvider / DNSEntry CRD
4. Preparation of infra-structure for worker nodes, load balancers, ...
Infrastructure CRD
5. Preparation of backup blob storage bucket for cluster's etcd data
BackupBucket / BackupEntry CRD
6. Deployment of cluster control plane (etcd, kube-apiserver, ...)
ControlPlane CRD
Webhooks
7. Computation of operating system config for bootstrapping workers
OperatingSystem Config CRD
8. Instrumentation of machine-controller provisioning the workers
Worker CRD
9. Deployment of system components into cluster (CNI plugin, kube-proxy)
Network CRD
Webhooks
10. Deployment of addons into cluster (e.g., kubernetes-dashboard)
Webhooks

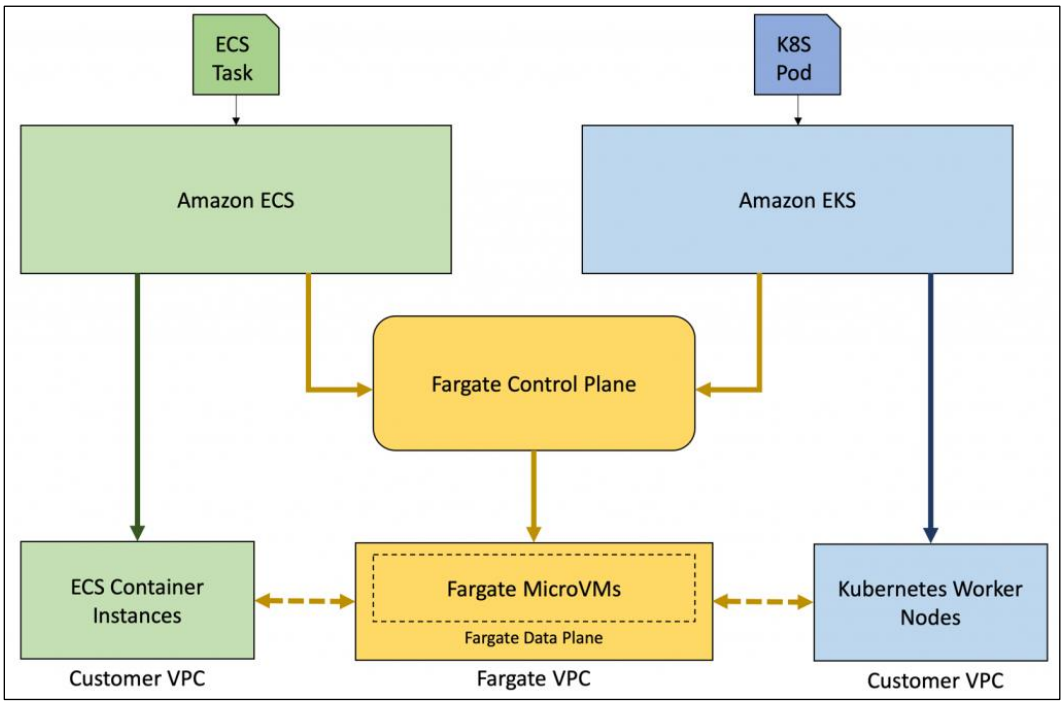

Ring Cluster A (seed cluster for B)
Worker Nodes of A
Control Plane of B
Ring Cluster C (seed cluster for A)
Worker Nodes of C
Control Plane of A
Ring Cluster B (seed cluster for C)
Worker Nodes of B
Control Plane of C

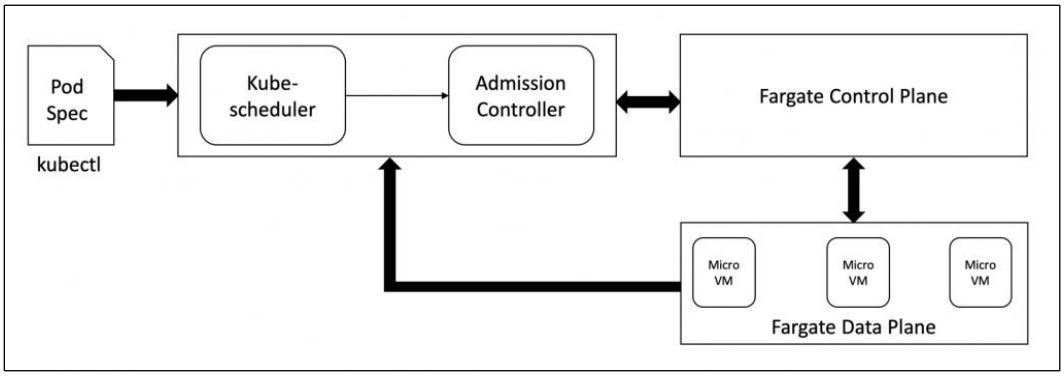
Chapter 12: Serverless Computing on Kubernetes

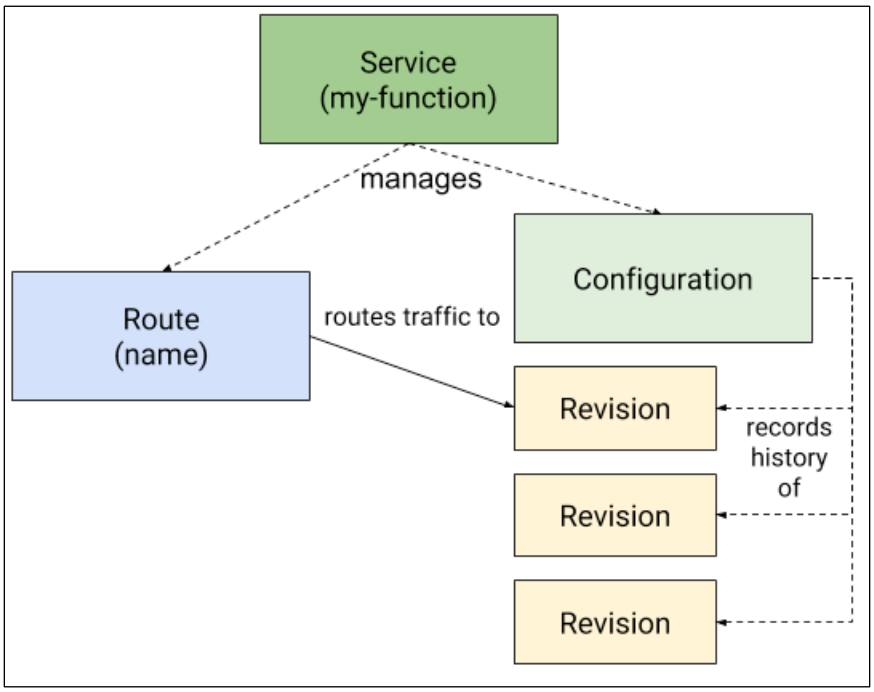


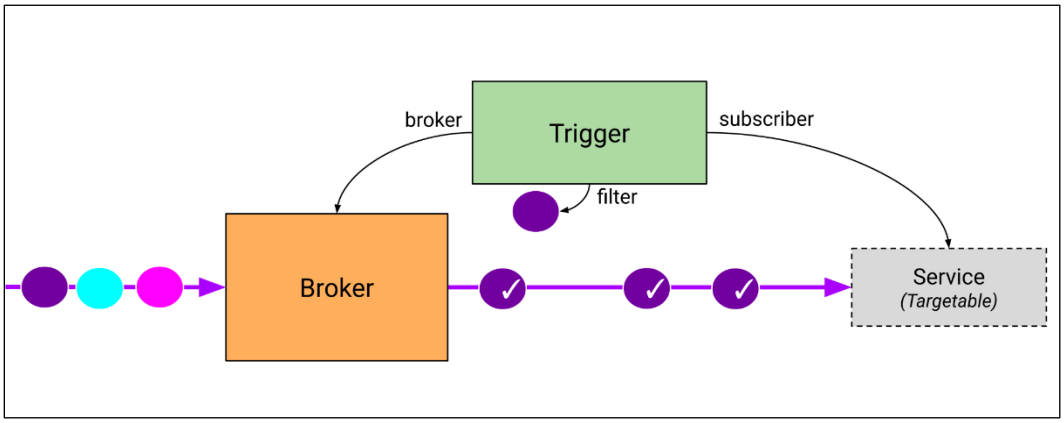
Virtual node architecture in AKS

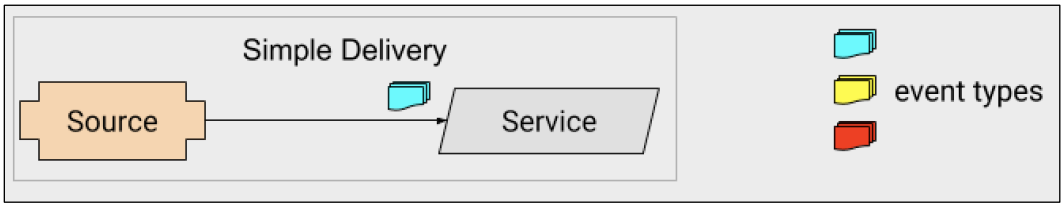


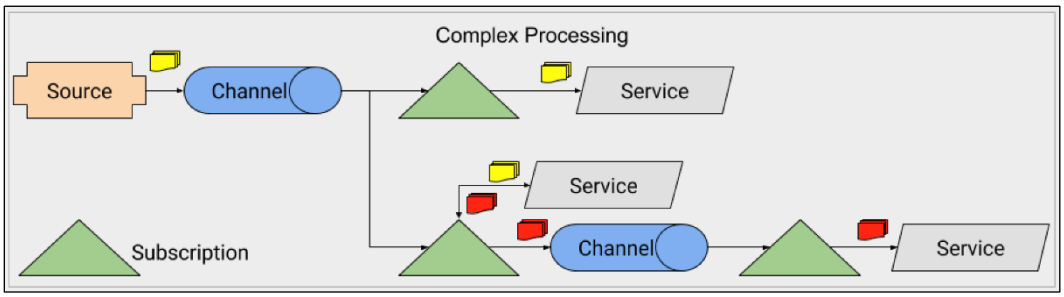

ECS
Task
K8S
Pod
Amazon ECS
Amazon EKS
Fargate Control Plane
ECS Container
Instances
Fargate MicroVMs
Fargate Data Plane
Kubernetes Worker
Nodes
Customer VPC
Fargate VPC
Customer VPC

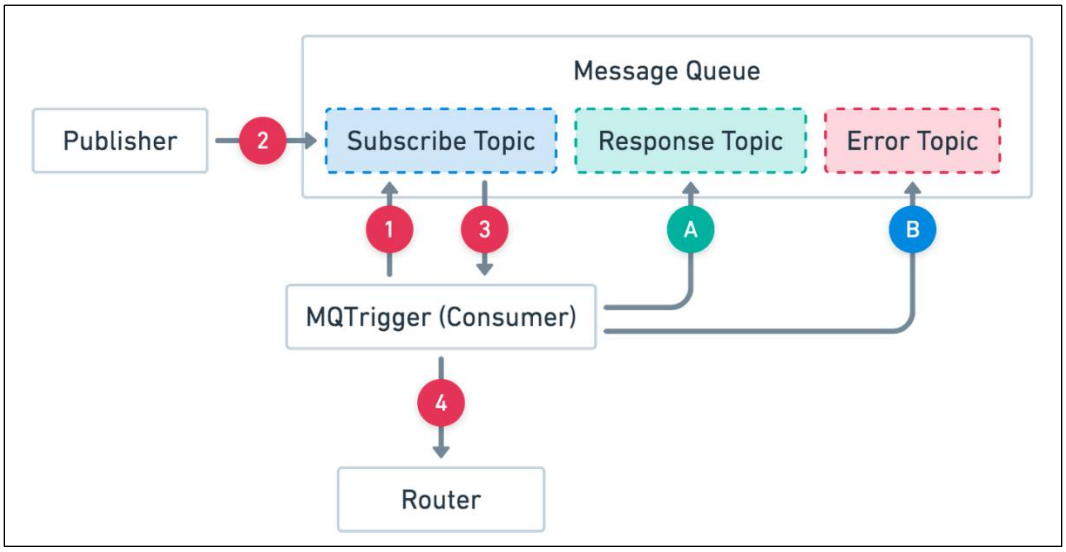

Pod
Spec
kubectl
Kube-
scheduler
Admission
Controller
Fargate Control Plane
Micro
VM
Micro
VM
Micro
VM
Fargate Data Plane

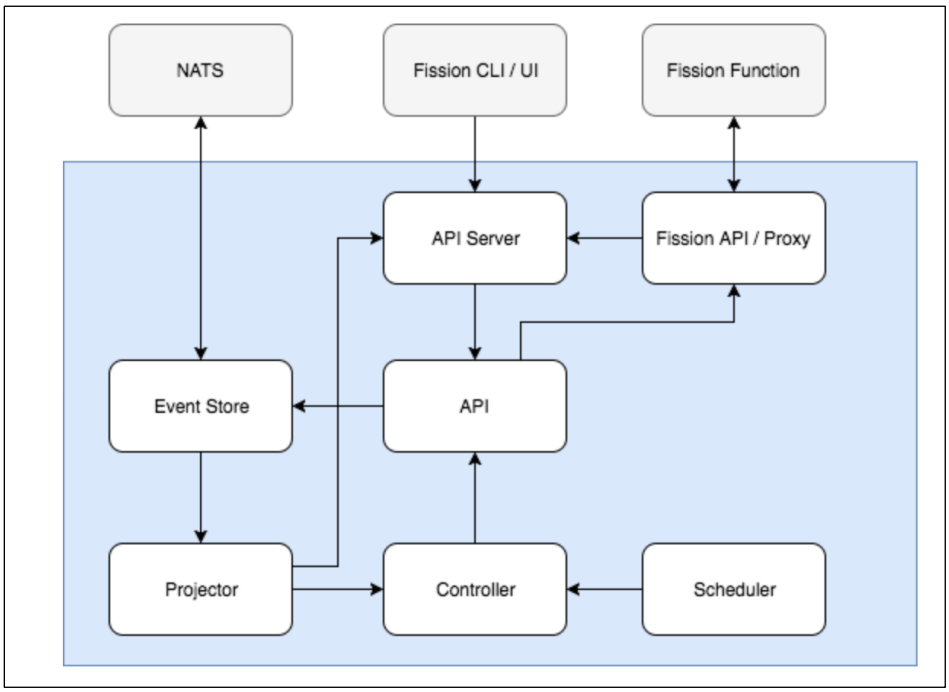

Service
(my-function)
manages
Route
(name)
routes traffic to
Configuration
Revision
Revision
Revision
records
history
of

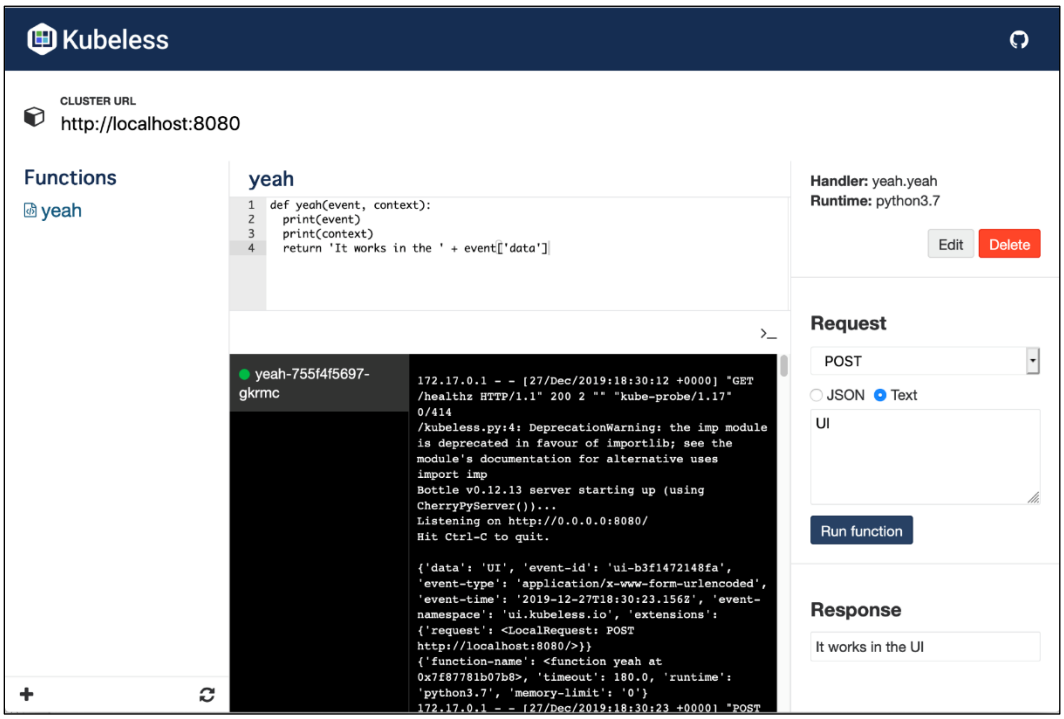

broker
Trigger
subscriber
filter
Broker
Service
(Targetable)


Simple Delivery
Source
Service
event types

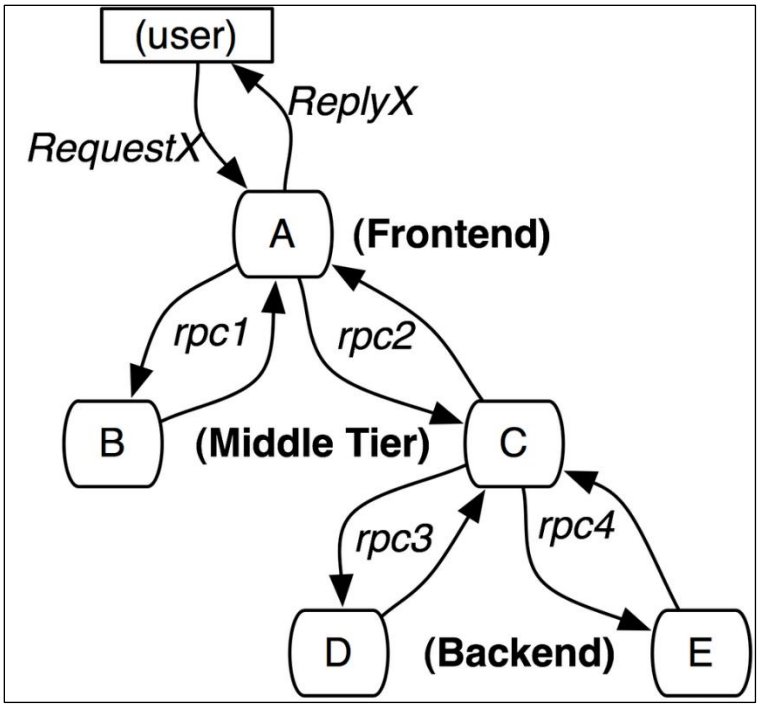

Complex Processing
Source
Channel
Service
Service
Subscription
Channel
Service

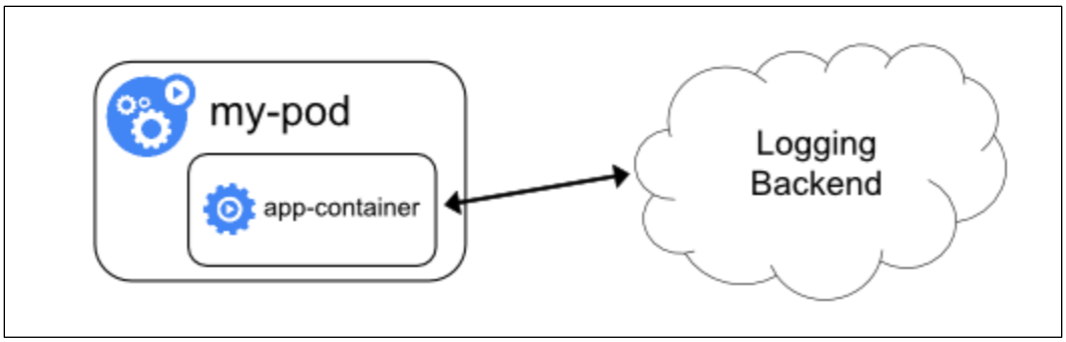

Message Queue
Publisher
2
Subscribe Topic
Response Topic
Error Topic
1
3
A
B
MQTrigger (Consumer)
4
Router

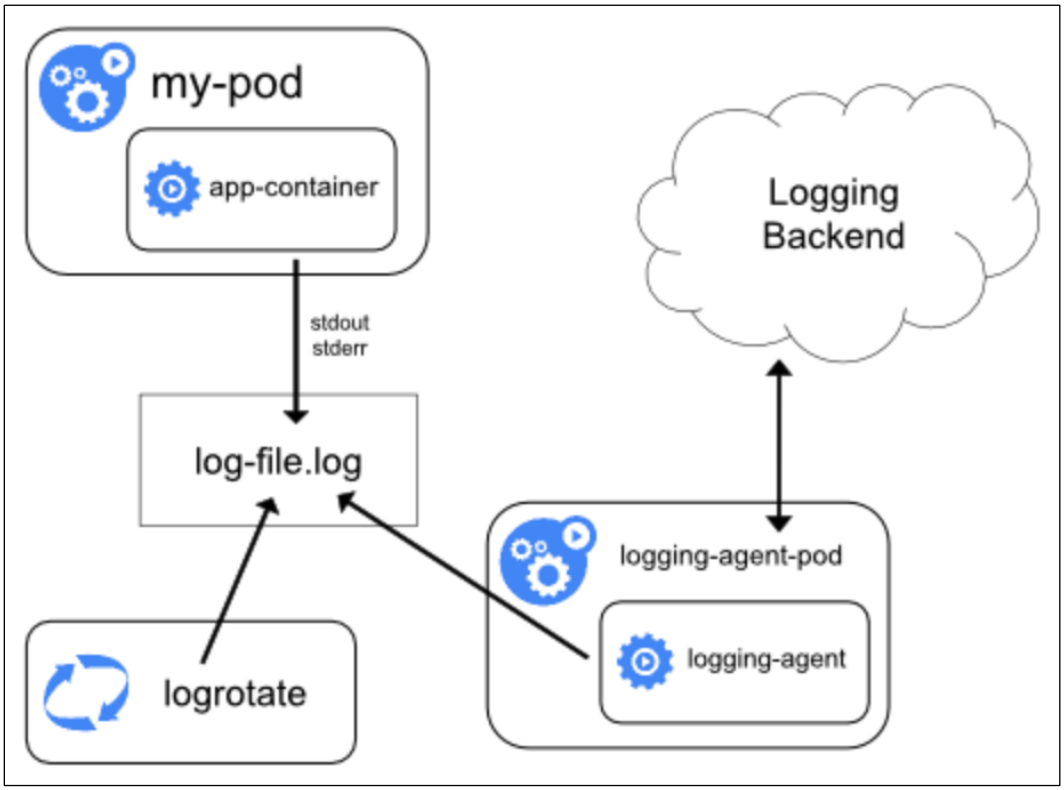

NATS
Fission CLI / UI
Fission Function
API Server
Fission API / Proxy
Event Store
API
Projector
Controller
Scheduler

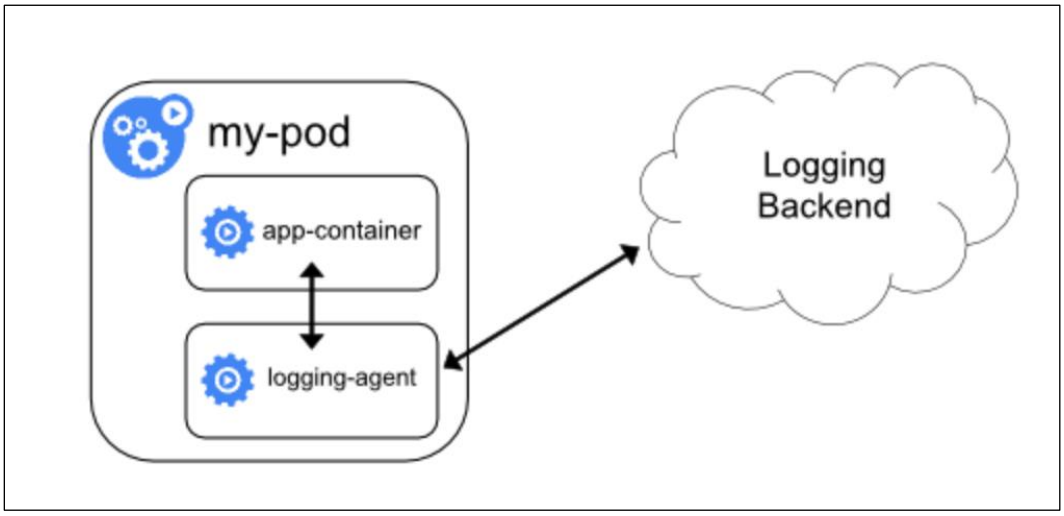

Kubeless
CLUSTER URL
http://localhost:8080
Functions
yeah
yeah
1 def yeah(event, context):
2 print(event)
3 print(context)
4 return 'It works in the ' + event['data']
>_
yeah-755f4f5697-gkrmc
172.17.0.1 - - [27/Dec/2019:18:30:12 +0000] "GET /healthz HTTP/1.1" 200 2 "" "kube-probe/1.17" 0/414
/kubeless.py:4: DeprecationWarning: the imp module is deprecated in favour of importlib; see the module's documentation for alternative uses
import imp
Bottle v0.12.13 server starting up (using CherryPyServer())...
Listening on http://0.0.0.0:8080/
Hit Ctrl-C to quit.
{'data': 'UI', 'event-id': 'ui-b3f1472148fa', 'event-type': 'application/x-www-form-urlencoded', 'event-time': '2019-12-27T18:30:23.156Z', 'event-namespace': 'ui.kubeless.io', 'extensions': {'request': <LocalRequest: POST http://localhost:8080/>}}
{'function-name': <function yeah at 0x7f87781b07b8>, 'timeout': 180.0, 'runtime': 'python3.7', 'memory-limit': '0'}
172.17.0.1 - - [27/Dec/2019:18:30:23 +0000] "POST
Handler: yeah.yeah
Runtime: python3.7
Edit
Delete
Request
POST
JSON
Text
UI
Run function
Response
It works in the UI

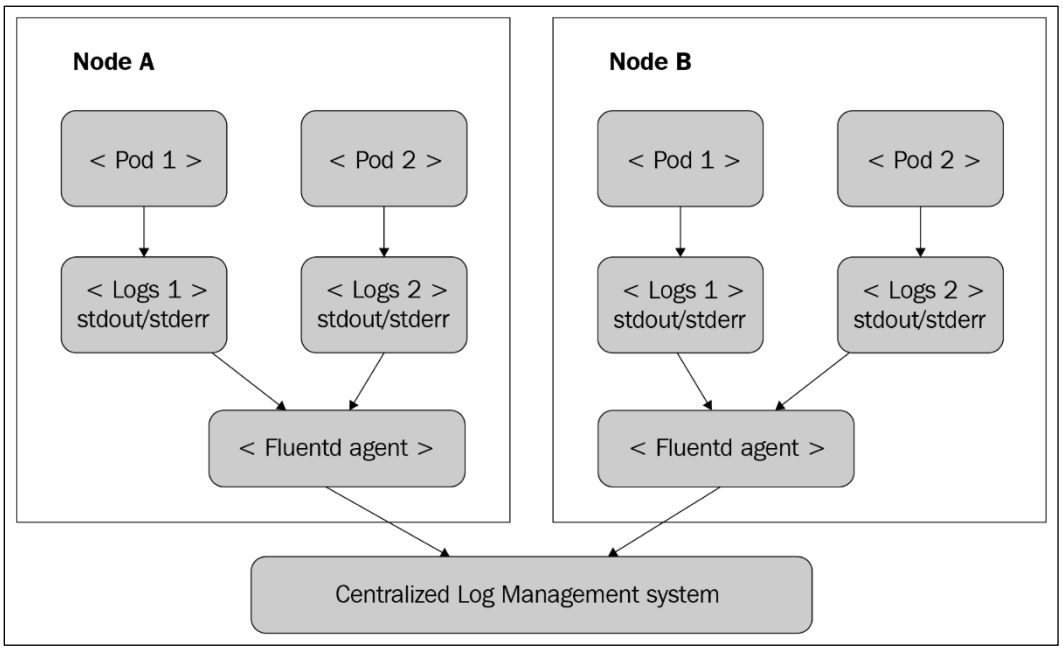
Chapter 13: Monitoring Kubernetes Clusters

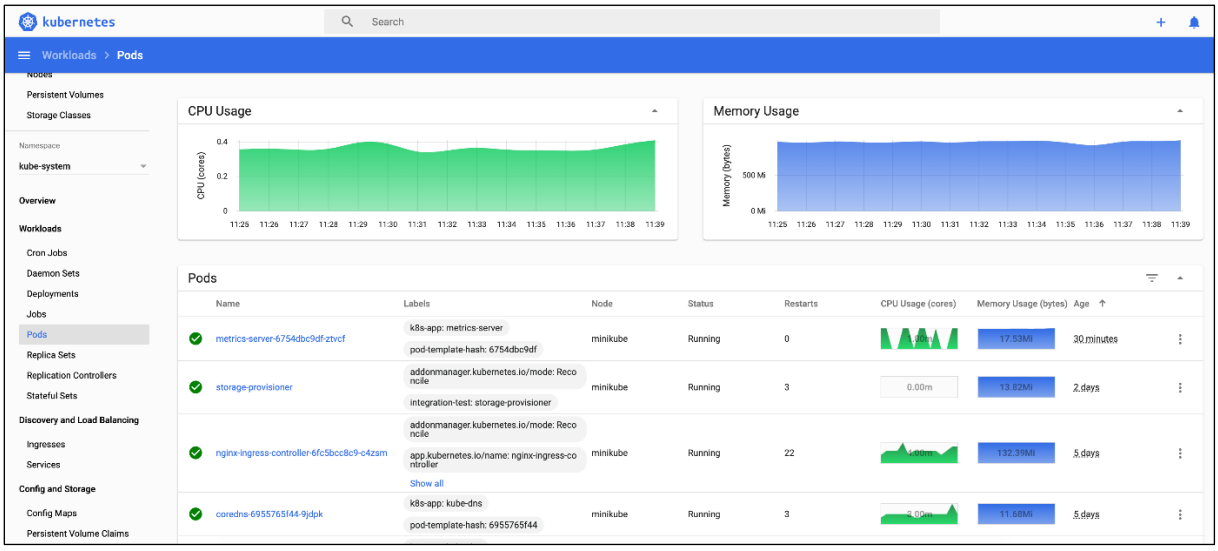


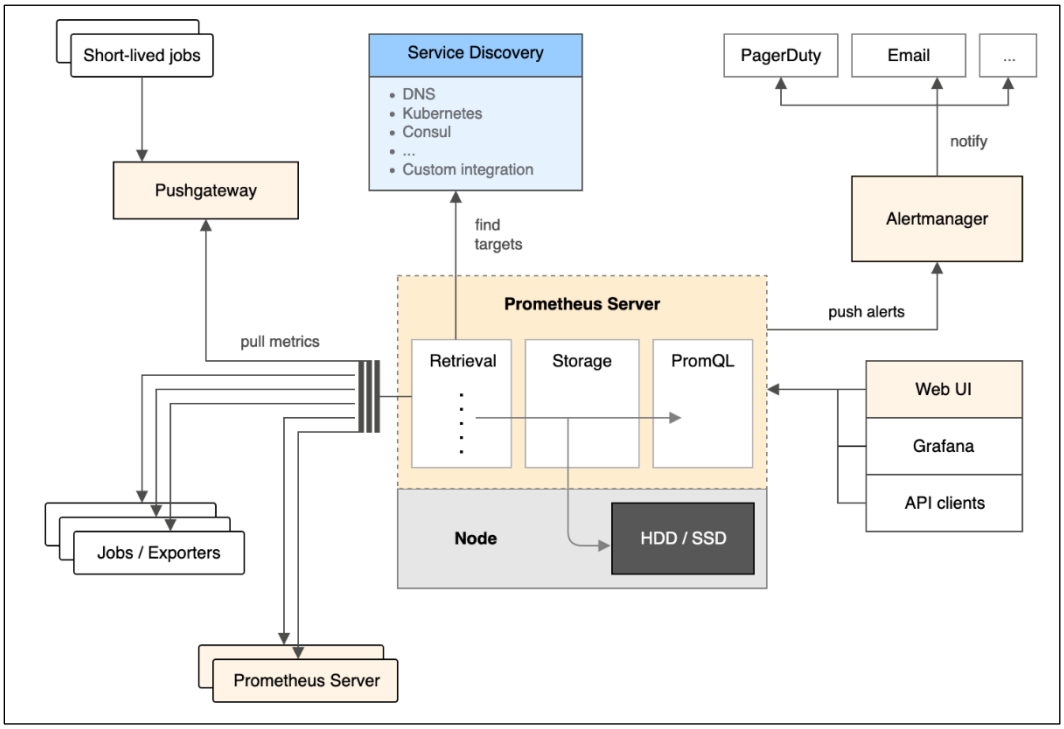



my-pod
app-container
Logging
Backend
stdout
stderr
log-file.log
logging-agent-pod
logging-agent
logrotate

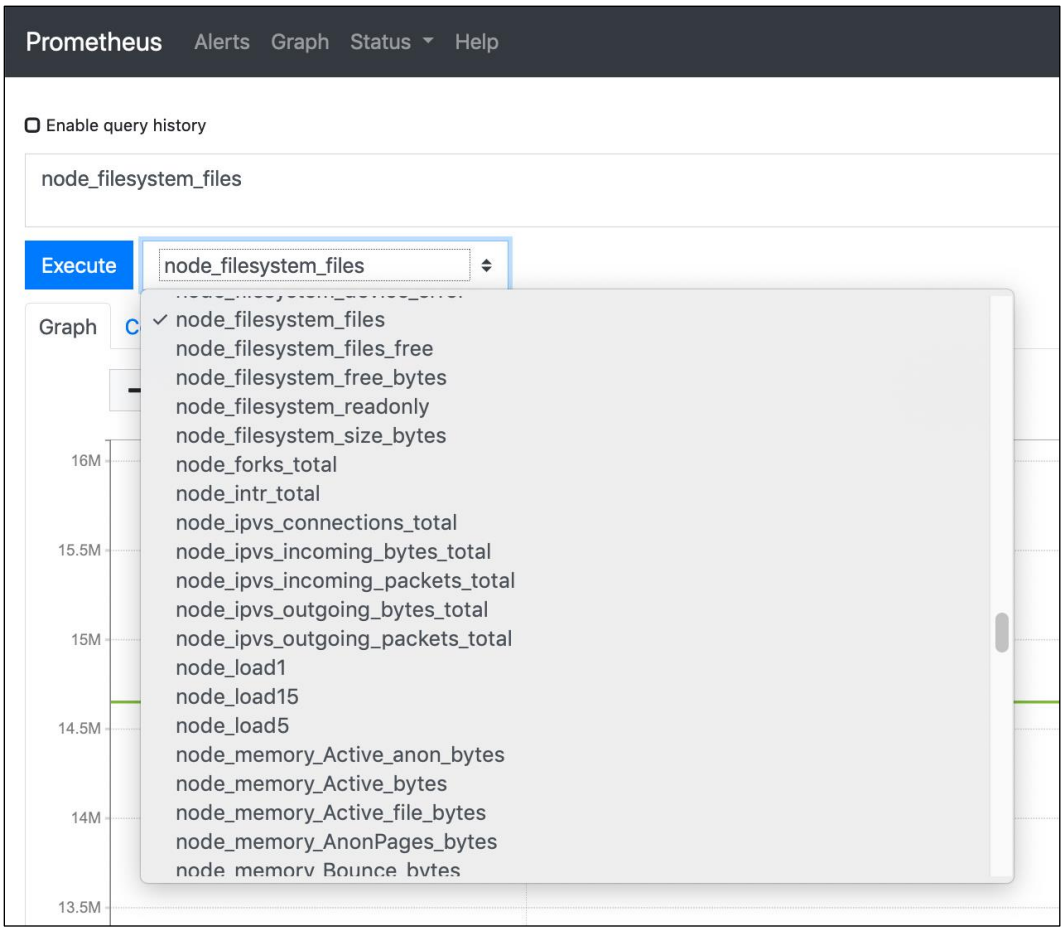

my-pod
app-container
logging-agent
Logging
Backend


Node A
< Pod 1 >
< Pod 2 >
< Logs 1 >
stdout/stderr
< Logs 2 >
stdout/stderr
< Fluentd agent >
Node B
< Pod 1 >
< Pod 2 >
< Logs 1 >
stdout/stderr
< Logs 2 >
stdout/stderr
< Fluentd agent >
Centralized Log Management system

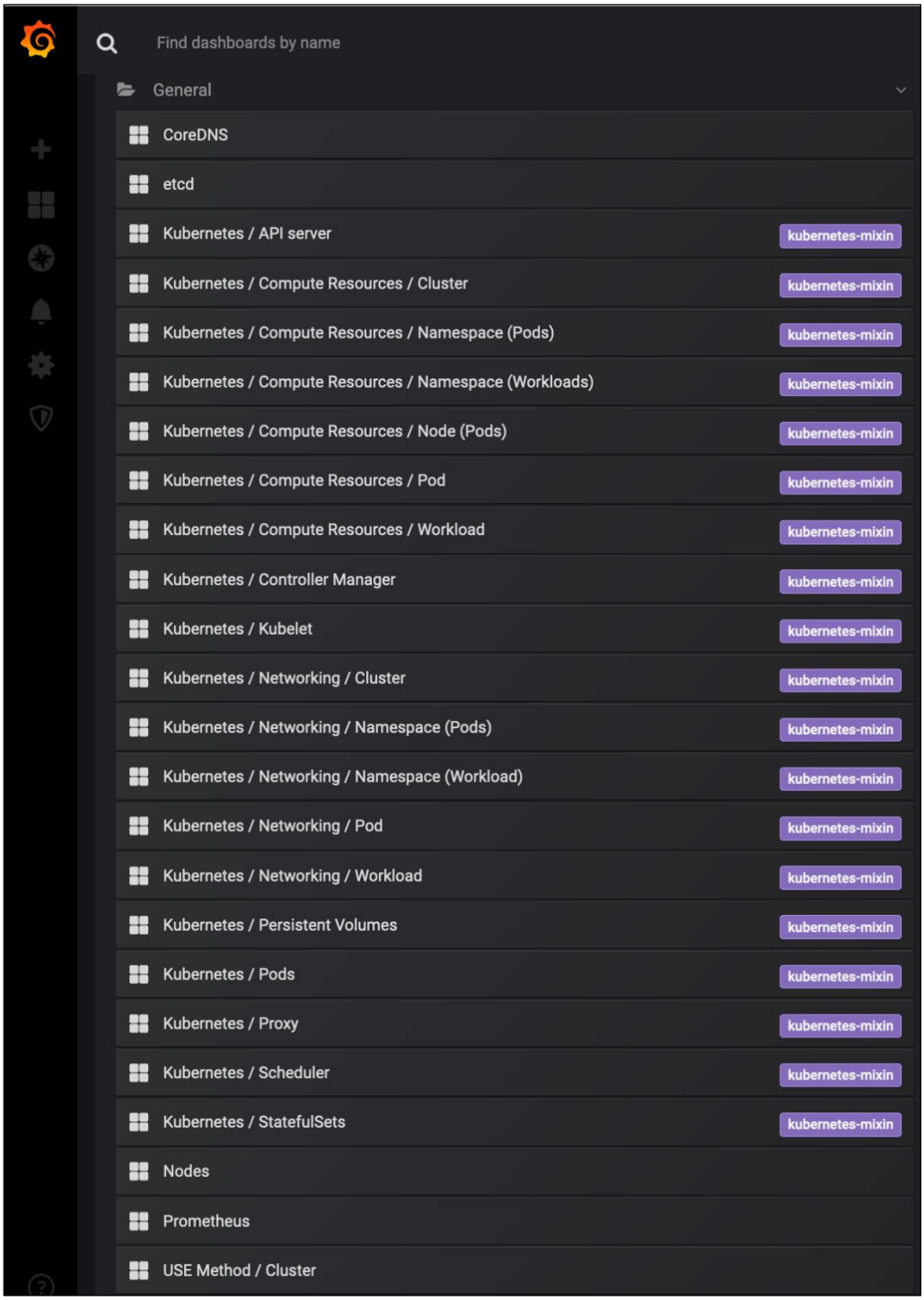

kubernetes
Search
Workloads > Pods
Persistent Volumes
Storage Classes
Namespace
kube-system
Overview
Workloads
Cron Jobs
Daemon Sets
Deployments
Jobs
Pods
Replica Sets
Replication Controllers
Stateful Sets
Discovery and Load Balancing
Ingresses
Services
Config and Storage
Config Maps
Persistent Volume Claims
CPU Usage
CPU (cores)
Memory Usage
Memory (bytes)
Pods
Name
Labels
Node
Status
Restarts
CPU Usage (cores)
Memory Usage (bytes)
Age
metrics-server-6754dbc9df-ztvcf
k8s-app: metrics-server
pod-template-hash: 6754dbc9df
minikube
Running
0
17.53Mi
30 minutes
storage-provisioner
addonmanager.kubernetes.io/mode: Reconcile
integration-test: storage-provisioner
minikube
Running
3
0.00m
13.62Mi
2 days
nginx-ingress-controller-6fc5bcc8c9-c4zsm
addonmanager.kubernetes.io/mode: Reconcile
app.kubernetes.io/name: nginx-ingress-controller
Show all
minikube
Running
22
132.39Mi
5 days
coredns-6955765f44-9jdpk
k8s-app: kube-dns
pod-template-hash: 6955765f44
minikube
Running
3
11.68Mi
5 days

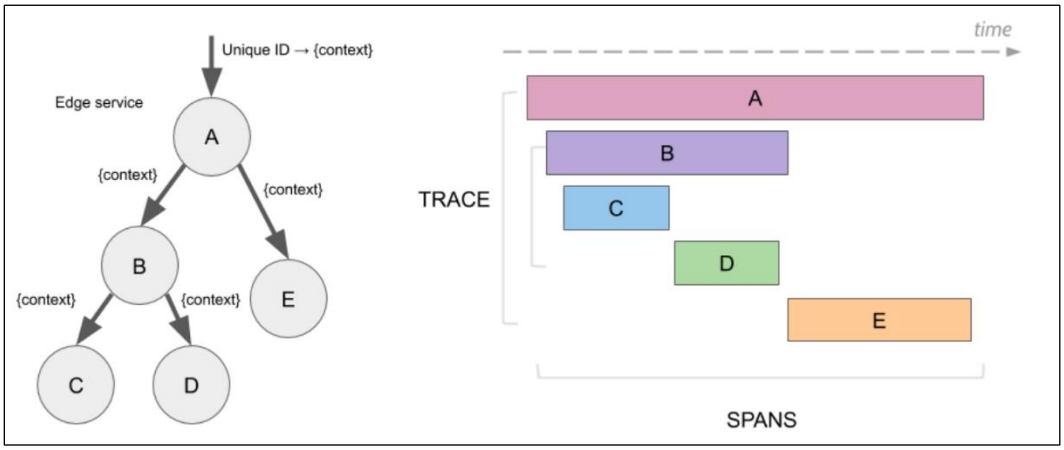

Short-lived jobs
Service Discovery
• DNS
• Kubernetes
• Consul
• ...
• Custom integration
PagerDuty
Email
...
notify
Pushgateway
find
targets
Alertmanager
Prometheus Server
push alerts
pull metrics
Retrieval
Storage
PromQL
Web UI
Grafana
API clients
Node
HDD / SSD
Jobs / Exporters
Prometheus Server

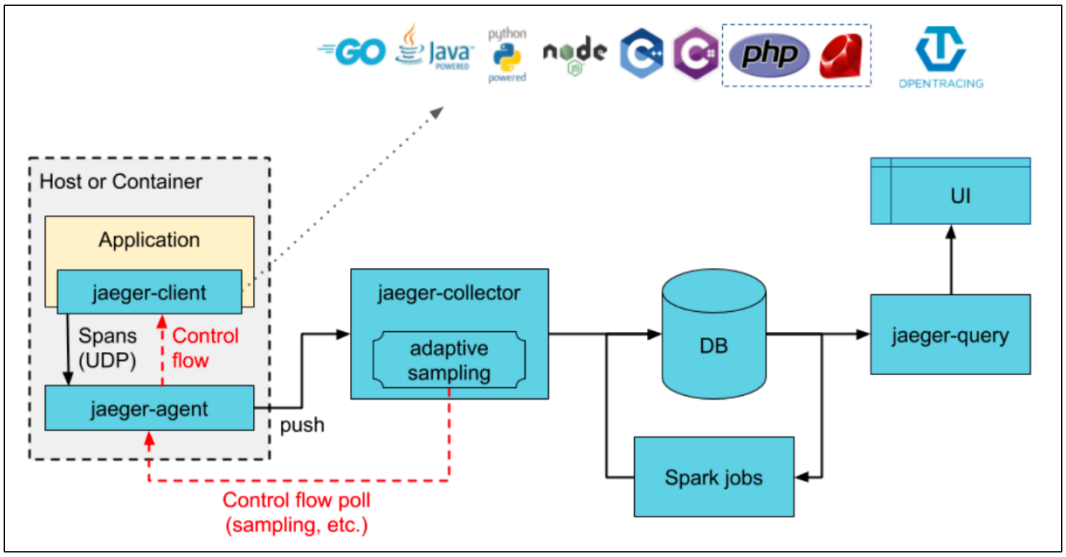

Prometheus
Alerts
Graph
Status
Help
Execute
go_memstats_heap_alloc_bytes
Graph
Console
1h
Until
Res. (s)
stacked
500M
400M
300M
200M
100M
0
20:30
20:45
21:00
21:15

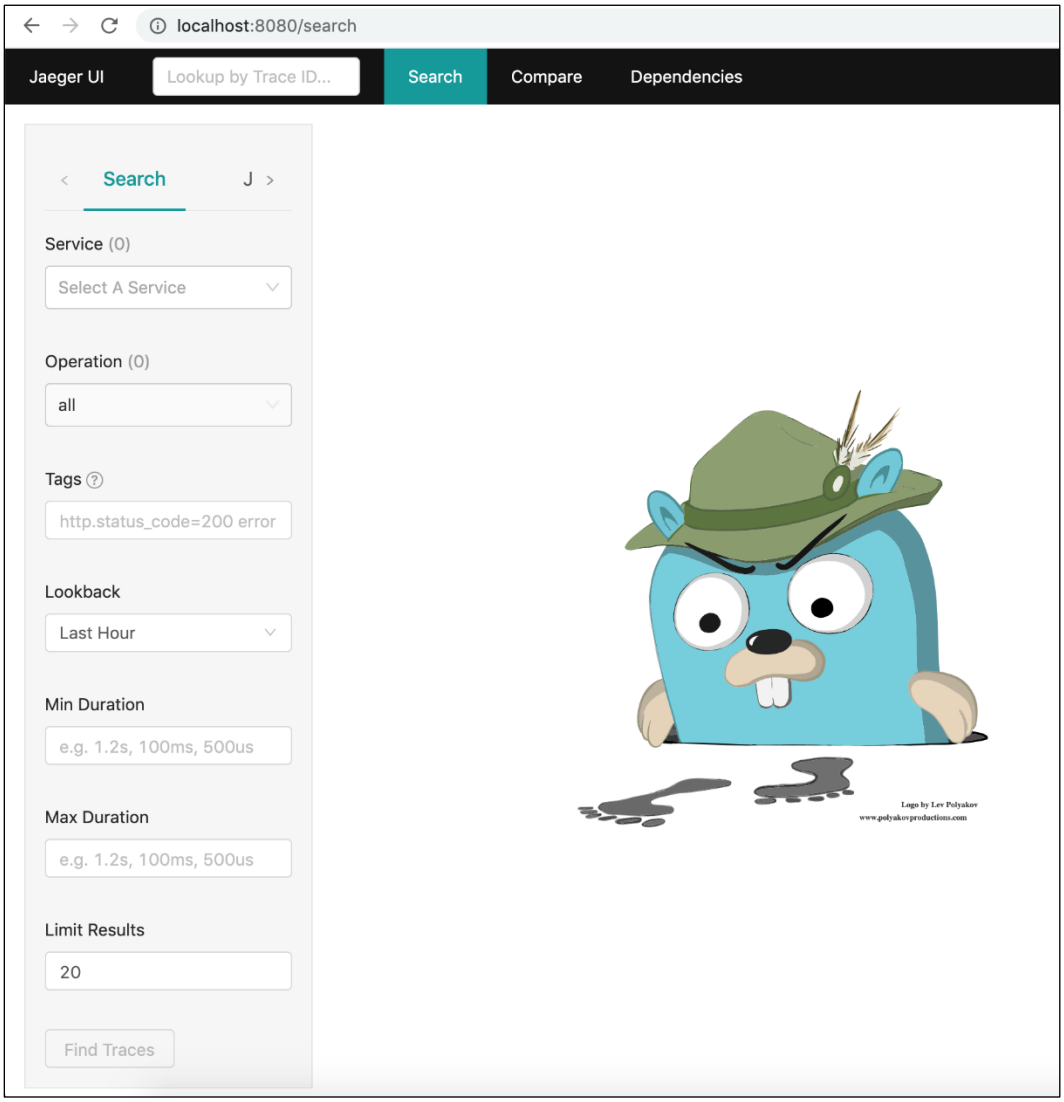

Prometheus
Alerts
Graph
Status
Help
Enable query history
node_filesystem_files
Execute
node_filesystem_files
Graph
node_filesystem_files
node_filesystem_files_free
node_filesystem_free_bytes
node_filesystem_readonly
node_filesystem_size_bytes
node_forks_total
node_intr_total
node_ipvs_connections_total
node_ipvs_incoming_bytes_total
node_ipvs_incoming_packets_total
node_ipvs_outgoing_bytes_total
node_ipvs_outgoing_packets_total
node_load1
node_load15
node_load5
node_memory_Active_anon_bytes
node_memory_Active_bytes
node_memory_Active_file_bytes
node_memory_AnonPages_bytes
16M
15.5M
15M
14.5M
14M
13.5M


Kubernetes / Networking / Namespace (Pods) -
Current Rate of Bytes Received
11.5 kBs
kube-system
Current Rate of Bytes Transmitted
3.502 kBs
kube-system
Current Status
Pod
Bandwidth Received
Bandwidth Transmitted
Rate of Received Packets
Rate of Transmitted Packets
Rate of Received Packets Dropped
Rate of Transmitted Packets Dropped
kube-scheduler-minikube 630.89 Bs 71.43 Bs 1.04 pps 0.75 pps 0 pps 0 pps
metrics-server-6754dbc9df-zbvcf 3.27 kBs 1.54 kBs 4.22 pps 4.53 pps 0 pps 0 pps
etcd-minikube 40.51 Bs 27.85 Bs 0.53 pps 0.43 pps 0 pps 0 pps
nginx-ingress-controller-6fc5bcc8c9-c4zsm 2.73 kBs 662.52 Bs 6.88 pps 6.12 pps 0 pps 0 pps
coredns-6955765f44-dk4g2 1.32 kBs 618.46 Bs 4.70 pps 4.11 pps 0 pps 0 pps
kube-addon-manager-minikube 631.79 Bs 72.53 Bs 1.05 pps 0.77 pps 0 pps 0 pps
coredns-6955765f44-fkjbk 1.31 kBs 625.47 Bs 4.71 pps 4.47 pps 0 pps 0 pps
kube-controller-manager-minikube 40.51 Bs 27.85 Bs 0.53 pps 0.43 pps 0 pps 0 pps
Bandwidth
Receive Bandwidth
Transmit Bandwidth

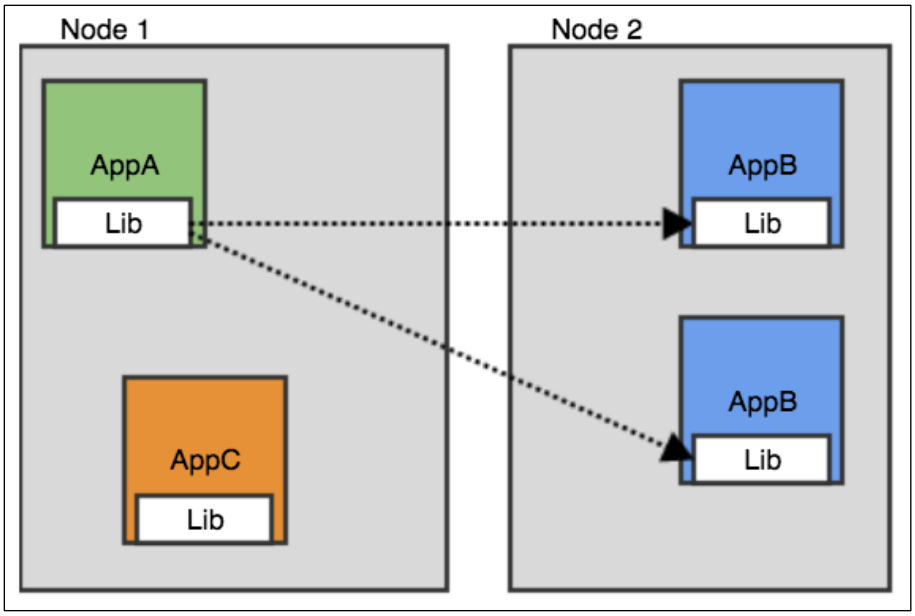

Find dashboards by name
General
CoreDNS
etcd
Kubernetes / API server
kubernetes-mixin
Kubernetes / Compute Resources / Cluster
kubernetes-mixin
Kubernetes / Compute Resources / Namespace (Pods)
kubernetes-mixin
Kubernetes / Compute Resources / Namespace (Workloads)
kubernetes-mixin
Kubernetes / Compute Resources / Node (Pods)
kubernetes-mixin
Kubernetes / Compute Resources / Pod
kubernetes-mixin
Kubernetes / Compute Resources / Workload
kubernetes-mixin
Kubernetes / Controller Manager
kubernetes-mixin
Kubernetes / Kubelet
kubernetes-mixin
Kubernetes / Networking / Cluster
kubernetes-mixin
Kubernetes / Networking / Namespace (Pods)
kubernetes-mixin
Kubernetes / Networking / Namespace (Workload)
kubernetes-mixin
Kubernetes / Networking / Pod
kubernetes-mixin
Kubernetes / Networking / Workload
kubernetes-mixin
Kubernetes / Persistent Volumes
kubernetes-mixin
Kubernetes / Pods
kubernetes-mixin
Kubernetes / Proxy
kubernetes-mixin
Kubernetes / Scheduler
kubernetes-mixin
Kubernetes / StatefulSets
kubernetes-mixin
Nodes
Prometheus
USE Method / Cluster

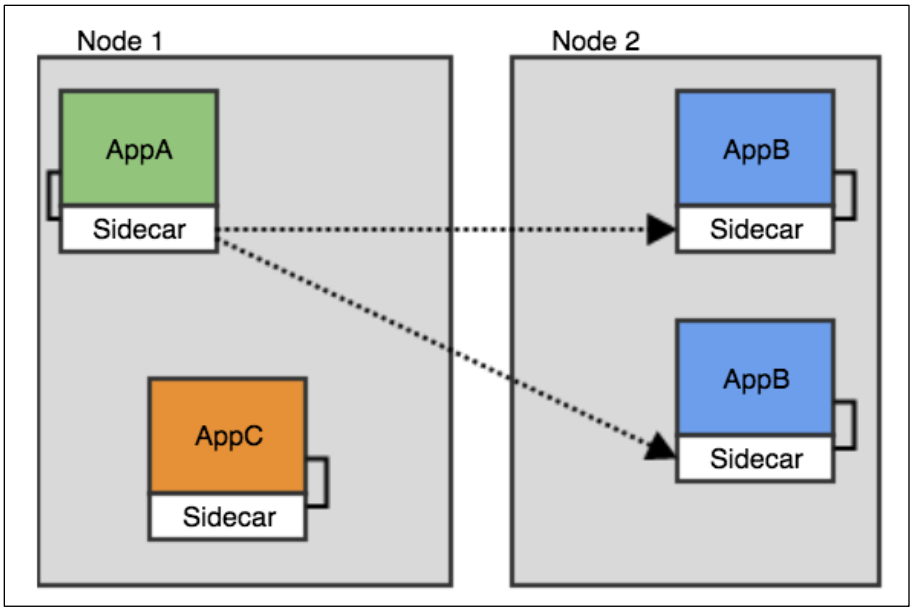

Unique ID → {context}
Edge service
A
{context}
{context}
B
E
{context}
{context}
C
D
time
TRACE
A
B
C
D
E
SPANS

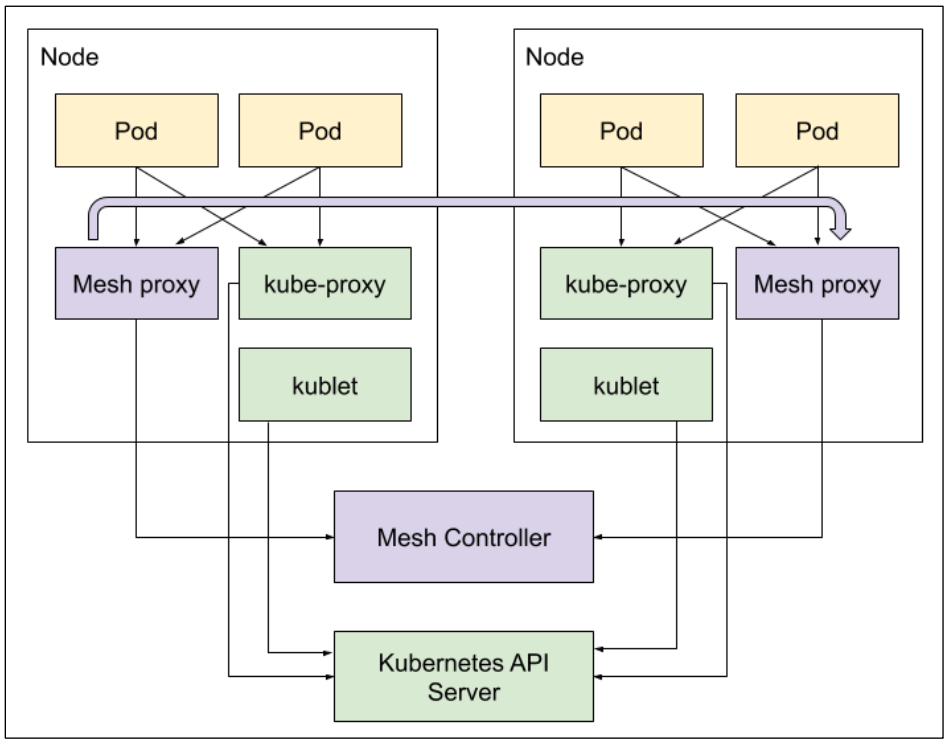

Host or Container
Application
jaeger-client
Spans
(UDP)
Control
flow
jaeger-agent
push
jaeger-collector
adaptive
sampling
DB
UI
jaeger-query
Spark jobs
Control flow poll
(sampling, etc.)
OPENTRACING

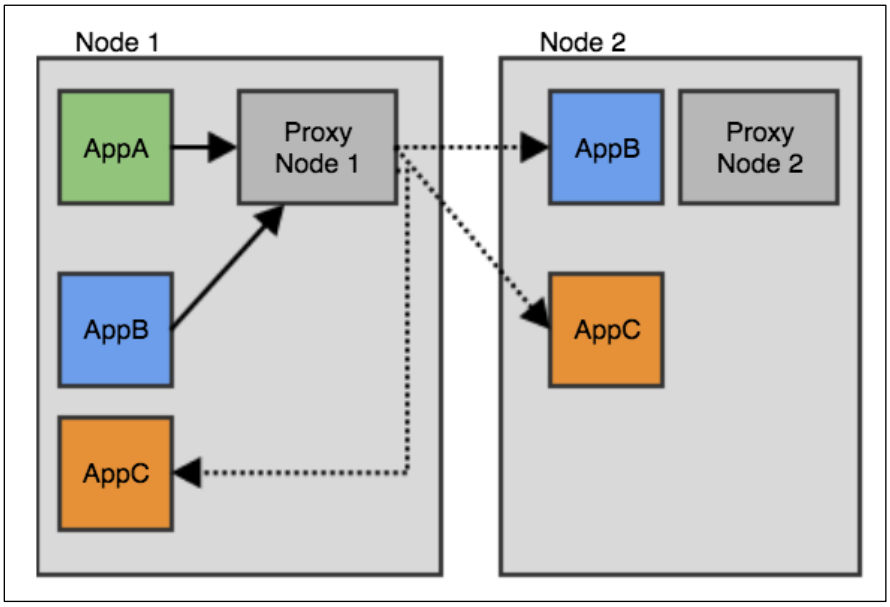

localhost:8080/search
Jaeger UI
Lookup by Trace ID...
Search
Compare
Dependencies
Search
J
Service (0)
Select A Service
Operation (0)
all
Tags
http.status_code=200 error
Lookback
Last Hour
Min Duration
e.g. 1.2s, 100ms, 500us
Max Duration
e.g. 1.2s, 100ms, 500us
Limit Results
20
Find Traces
Logo by Lev Polyakov
www.polyakovproductions.com

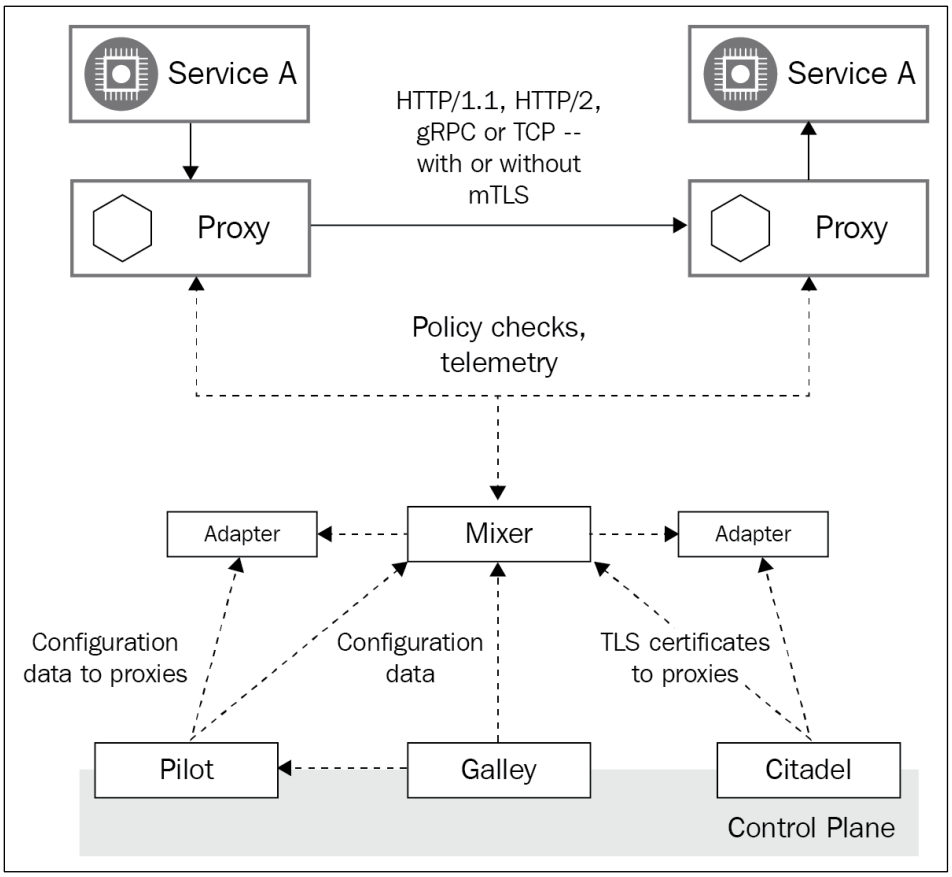
Chapter 14: Utilizing Service Meshes

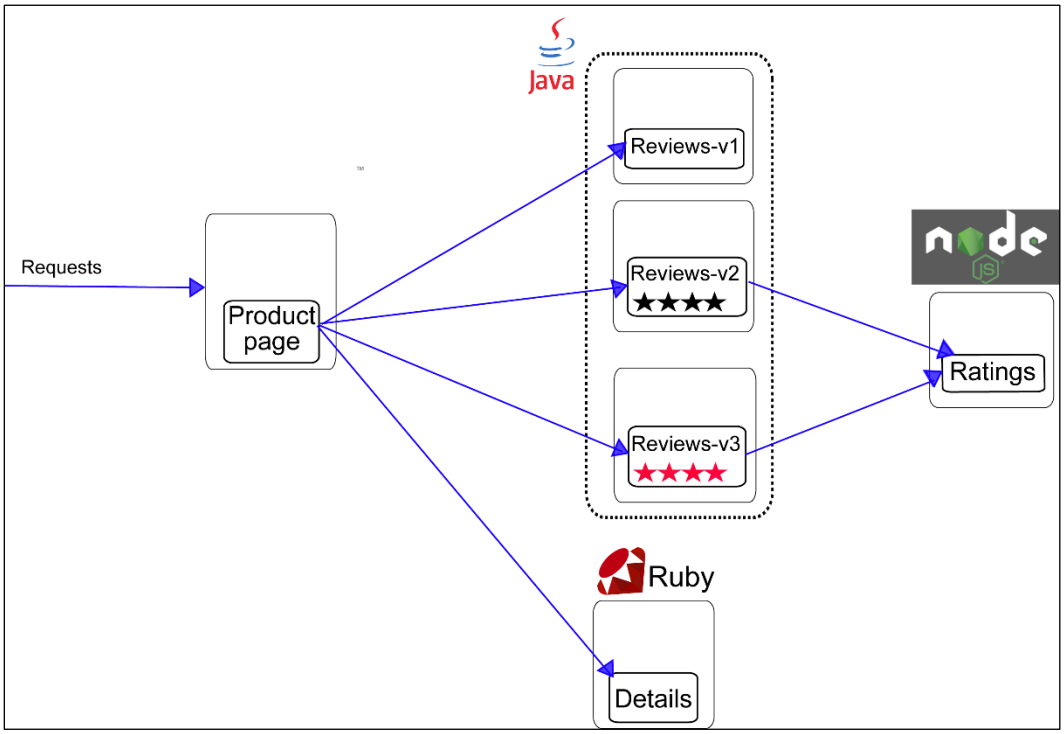


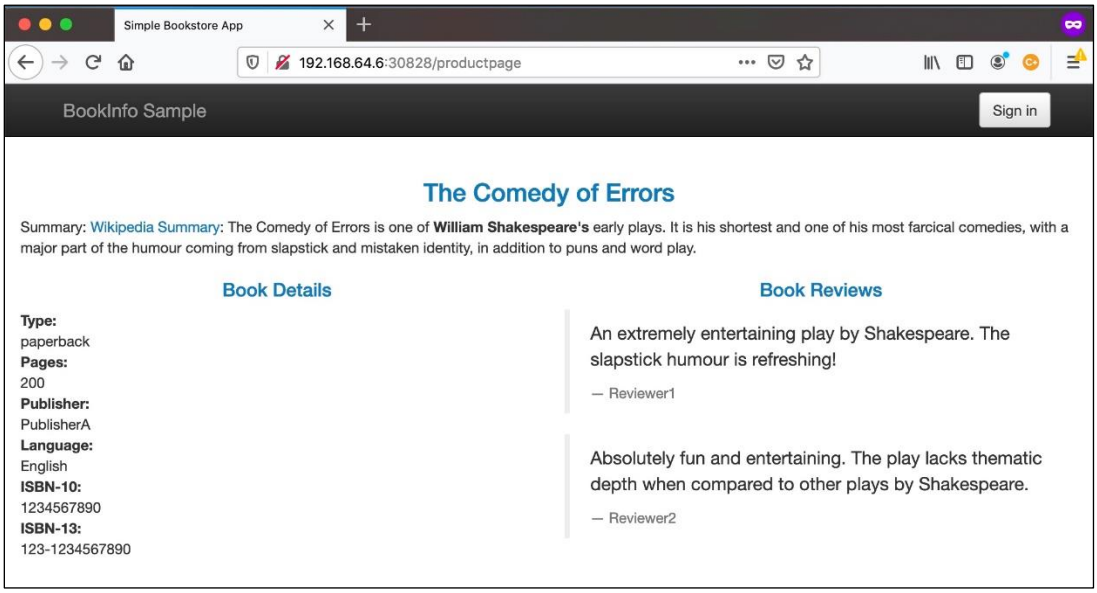


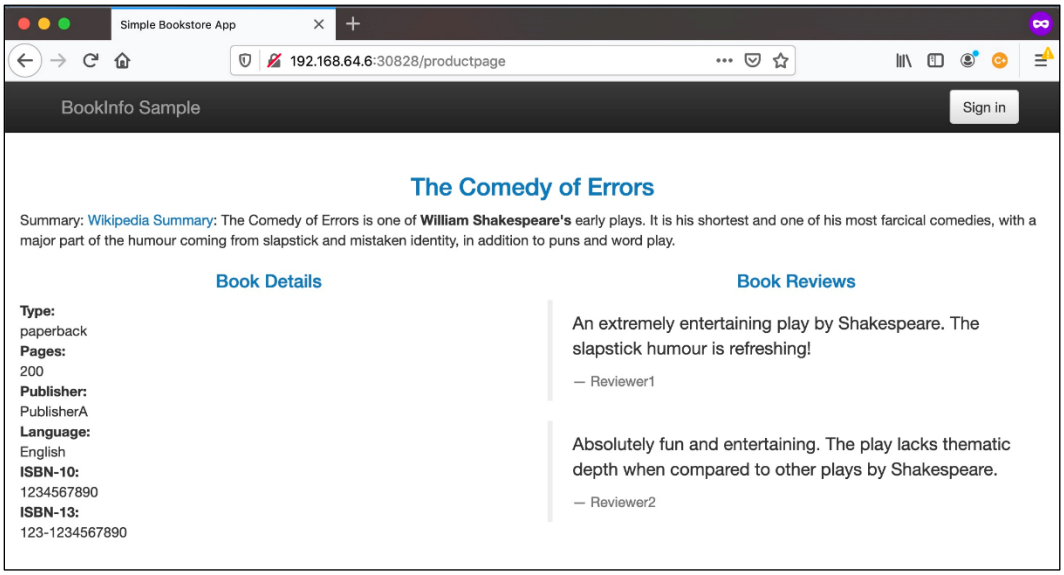

Node
Pod
Pod
Mesh proxy
kube-proxy
kublet
Node
Pod
Pod
kube-proxy
Mesh proxy
kublet
Mesh Controller
Kubernetes API Server

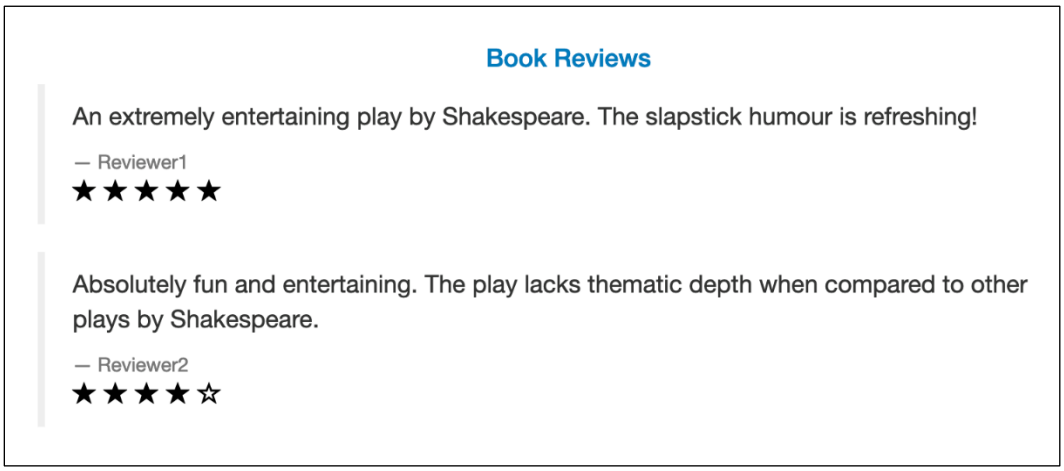

Node 1
AppA
Proxy Node 1
AppB
AppC
Node 2
AppB
Proxy Node 2
AppC

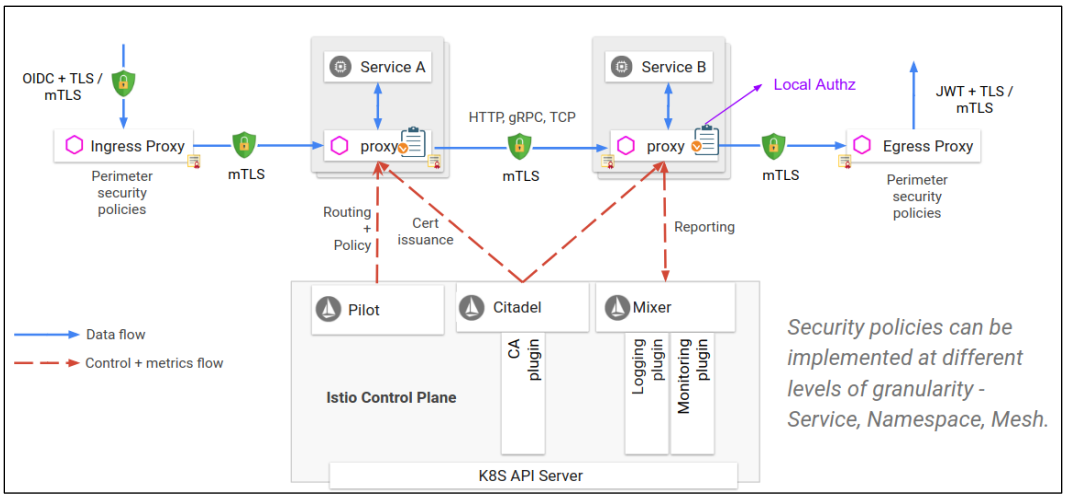

Service A
Service A
HTTP/1.1, HTTP/2,
gRPC or TCP --
with or without
mTLS
Proxy
Proxy
Policy checks,
telemetry
Adapter
Mixer
Adapter
Configuration
data to proxies
Configuration
data
TLS certificates
to proxies
Pilot
Galley
Citadel
Control Plane

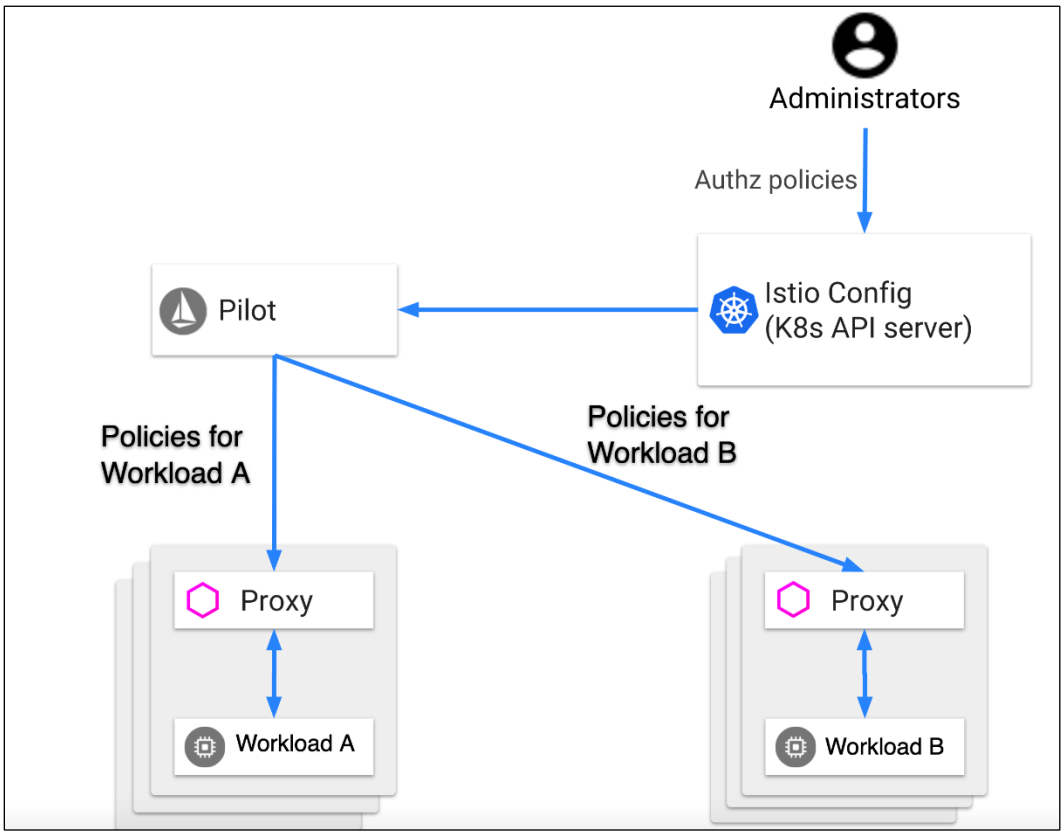

Java
Reviews-v1
Reviews-v2
★★★★
Reviews-v3
★★★★
Requests
Product page
Ratings
Ruby
Details

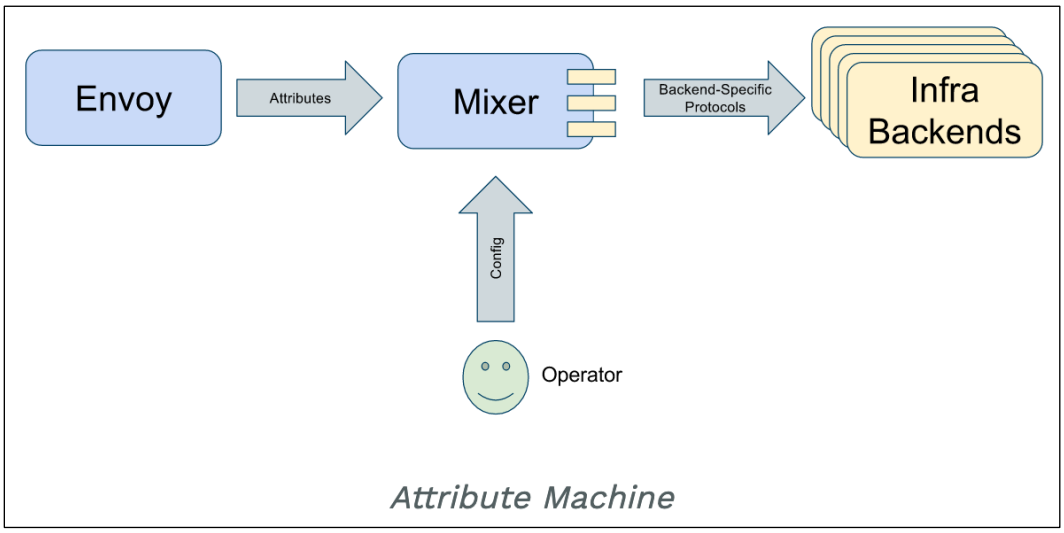

Simple Bookstore App
192.168.64.6:30828/productpage
BookInfo Sample
Sign in
The Comedy of Errors
Summary: Wikipedia Summary: The Comedy of Errors is one of William Shakespeare's early plays. It is his shortest and one of his most farcical comedies, with a major part of the humour coming from slapstick and mistaken identity, in addition to puns and word play.
Book Details
Type:
paperback
Pages:
200
Publisher:
PublisherA
Language:
English
ISBN-10:
1234567890
ISBN-13:
123-1234567890
Book Reviews
An extremely entertaining play by Shakespeare. The slapstick humour is refreshing!
— Reviewer1
Absolutely fun and entertaining. The play lacks thematic depth when compared to other plays by Shakespeare.
— Reviewer2


Simple Bookstore App
192.168.64.6:30828/productpage
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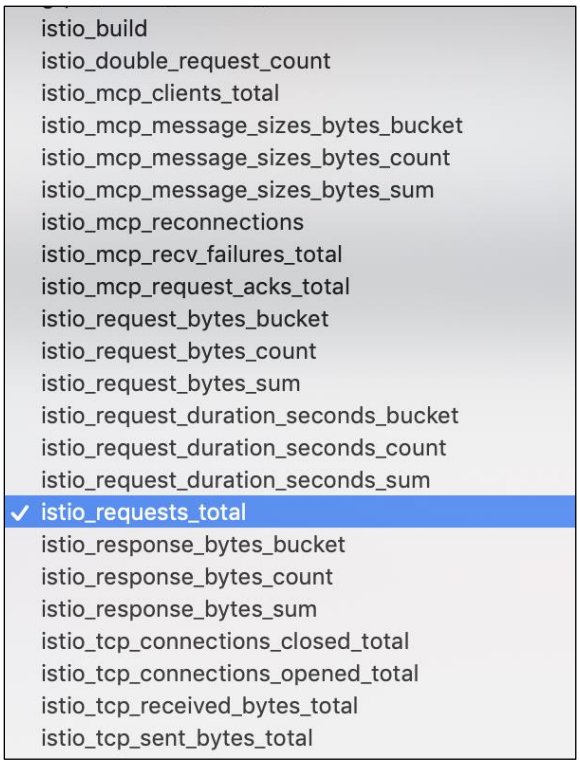

Book Reviews
An extremely entertaining play by Shakespeare. The slapstick humour is refreshing!
— Reviewer1
★★★★★
Absolutely fun and entertaining. The play lacks thematic depth when compared to other plays by Shakespeare.
— Reviewer2
★★★★☆

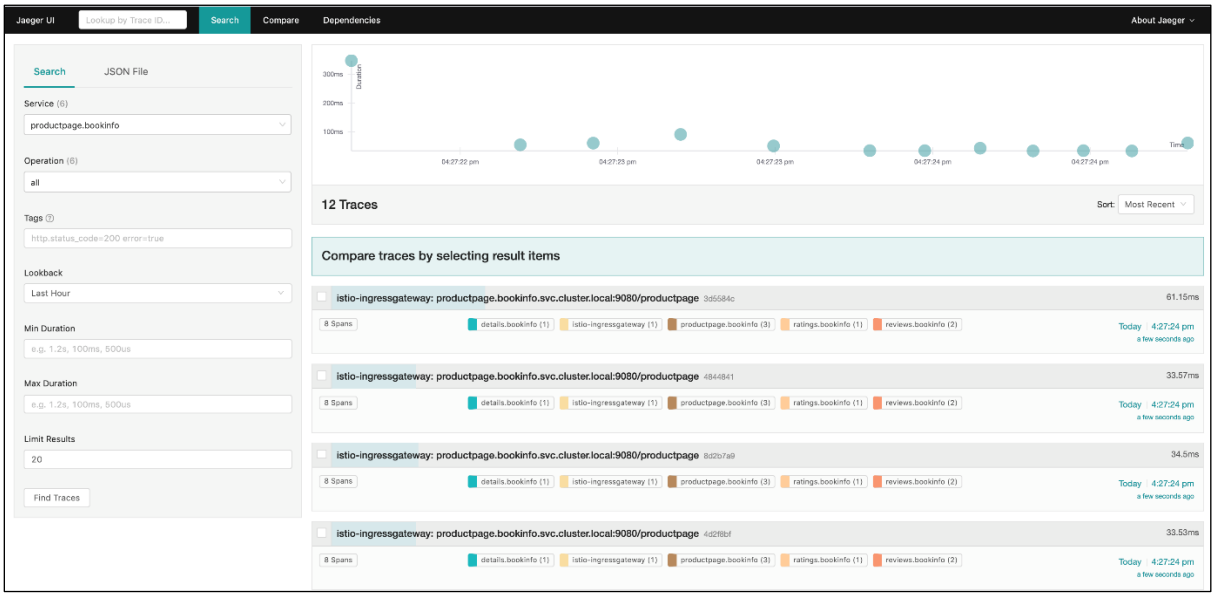

OIDC + TLS / mTLS
Ingress Proxy
Perimeter security policies
mTLS
Service A
proxy
HTTP, gRPC, TCP
mTLS
Service B
proxy
Local Authz
mTLS
Egress Proxy
JWT + TLS / mTLS
Perimeter security policies
Routing + Policy
Cert issuance
Reporting
Pilot
Citadel
Mixer
CA plugin
Logging plugin
Monitoring plugin
Istio Control Plane
K8S API Server
Data flow
Control + metrics flow
Security policies can be implemented at different levels of granularity - Service, Namespace, Mesh.

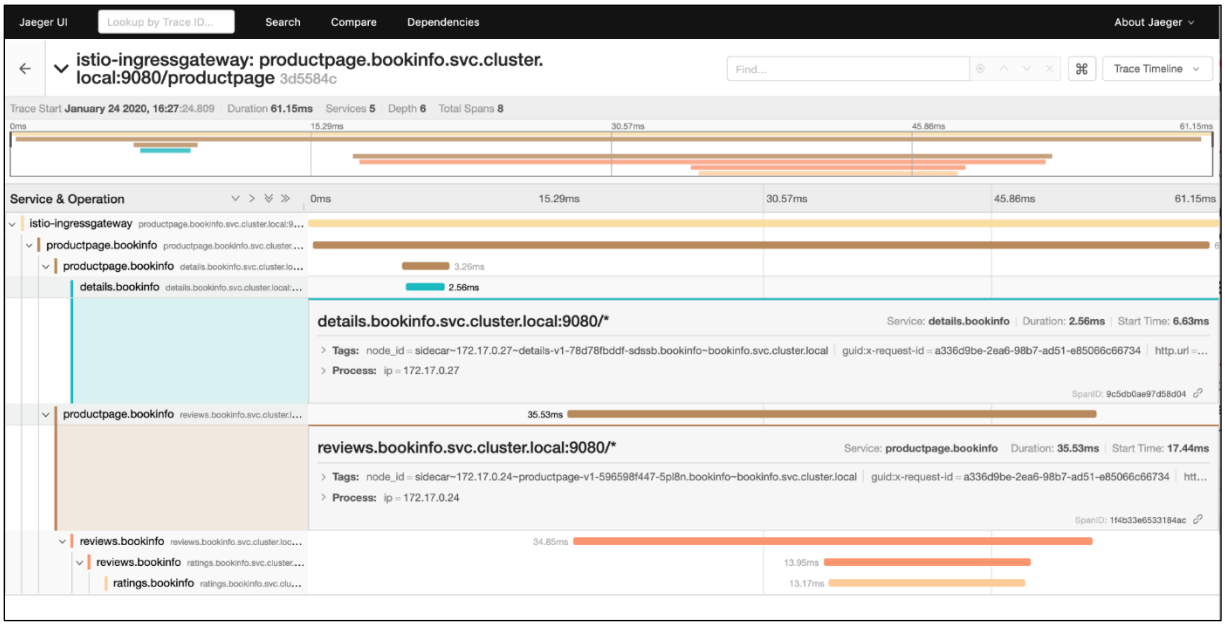

Administrators
Authz policies
Pilot
Istio Config (K8s API server)
Policies for Workload A
Policies for Workload B
Proxy
Workload A
Proxy
Workload B

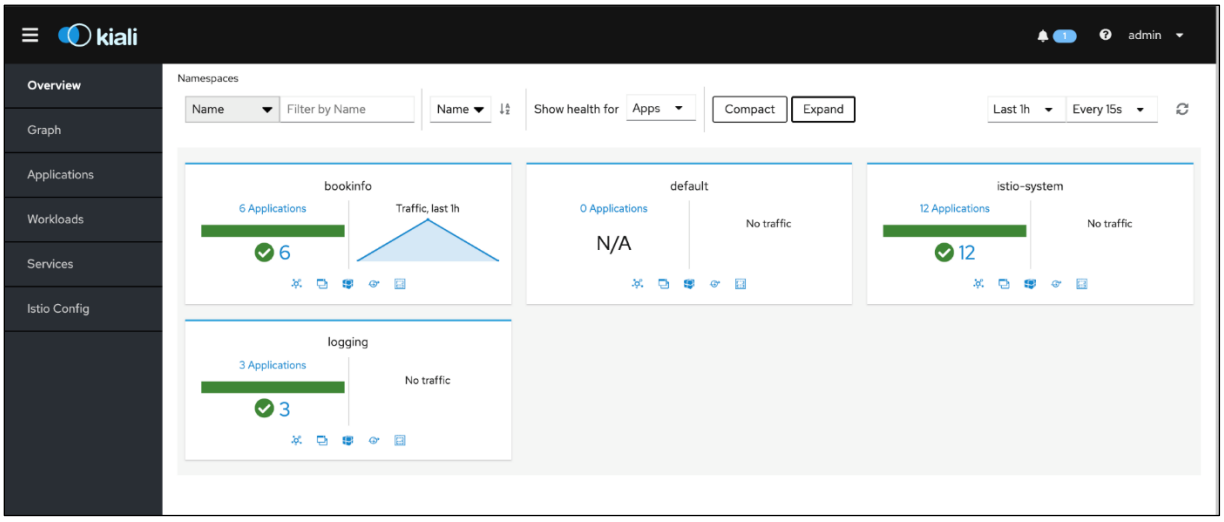

Envoy
Attributes
Mixer
Backend-Specific Protocols
Infra Backends
Config
Operator
Attribute Machine

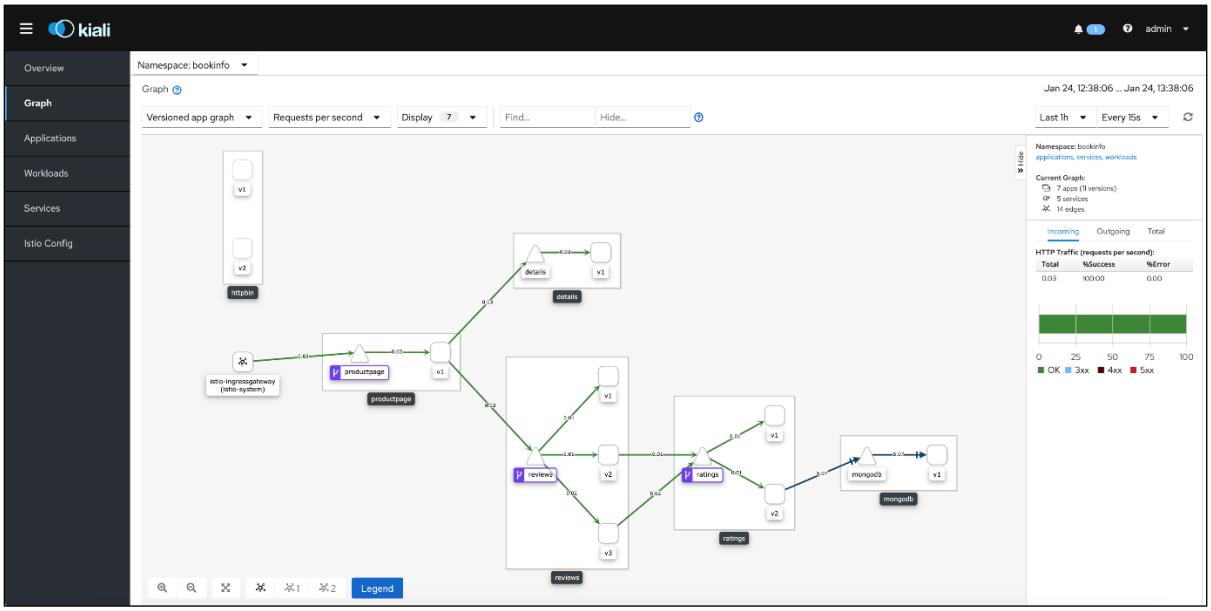

kibana
logstash-2020.01.24
Discover
Visualize
Dashboard
Timelion
Dev Tools
Management
Selected Fields
_source
Available Fields
@timestamp
_id
_index
_score
_type
destination
latency
responseCode
responseSize
severity
source
tag
user
January 24th 2020, 12:52:45.393 - January 24th 2020, 13:07:45.393 — Minute
Count
@timestamp per minute
Time
_source
January 24th 2020, 13:06:53.000
responseCode: 200 responseSize: 5,282 source: istio-ingressgateway user: unknown destination: productpage latency: 32.931521ms severity: info @timestamp: January 24th 2020, 13:06:53.000 tag: newlog.instance.istio-system _id: 2ath2W88KzEgAt2ITsPF _type: fluentd _index: logstash-2020.01.24 _score: -
Table
JSON
View surrounding documents
View single document
January 24th 2020, 13:06:53.000
responseSize: 48 source: reviews user: unknown destination: ratings severity: info latency: 1.760741ms responseCode: 200 @timestamp: January 24th 2020, 13:06:53.000 tag: newlog.instance.istio-system _id: 36th2W88KzEgAt2ITsPF _type: fluentd _index: logstash-2020.01.24 _score: -
January 24th 2020, 13:06:53.000
destination: details latency: 1.664615ms responseCode: 200 responseSize: 178 source: productpage user: unknown severity: info @timestamp: January 24th 2020, 13:06:53.000 tag: newlog.instance.istio-system _id: 5qth2W88KzEgAt2ITsPF _type: fluentd _index: logstash-2020.01.24 _score: -


istio_build
istio_double_request_count
istio_mcp_clients_total
istio_mcp_message_sizes_bytes_bucket
istio_mcp_message_sizes_bytes_count
istio_mcp_message_sizes_bytes_sum
istio_mcp_reconnections
istio_mcp_recv_failures_total
istio_mcp_request_acks_total
istio_request_bytes_bucket
istio_request_bytes_count
istio_request_bytes_sum
istio_request_duration_seconds_bucket
istio_request_duration_seconds_count
istio_request_duration_seconds_sum
istio_requests_total
istio_response_bytes_bucket
istio_response_bytes_count
istio_response_bytes_sum
istio_tcp_connections_closed_total
istio_tcp_connections_opened_total
istio_tcp_received_bytes_total
istio_tcp_sent_bytes_total

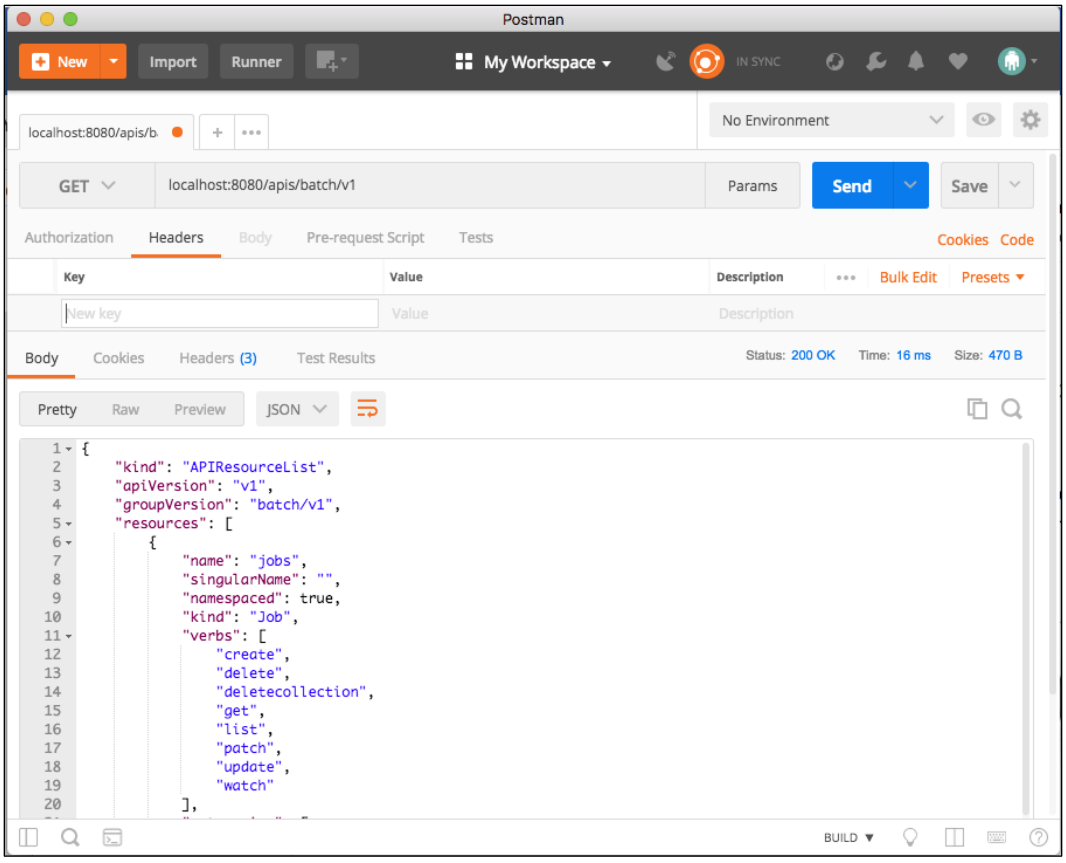

Jaeger UI
Lookup by Trace ID...
Search
Compare
Dependencies
About Jaeger
Search
JSON File
Service (6)
productpage.bookinfo
Operation (6)
all
Tags
http.status_code=200 error=true
Lookback
Last Hour
Min Duration
e.g. 1.2s, 100ms, 500us
Max Duration
e.g. 1.2s, 100ms, 500us
Limit Results
20
Find Traces
Duration
300ms
200ms
100ms
Time
12 Traces
Sort:
Most Recent
Compare traces by selecting result items
istio-ingressgateway: productpage.bookinfo.svc.cluster.local:9080/productpage
61.15ms
8 Spans
details.bookinfo (1)
istio-ingressgateway (1)
productpage.bookinfo (3)
ratings.bookinfo (1)
reviews.bookinfo (2)
Today 4:27:24 pm
a few seconds ago
istio-ingressgateway: productpage.bookinfo.svc.cluster.local:9080/productpage
33.57ms
istio-ingressgateway: productpage.bookinfo.svc.cluster.local:9080/productpage
34.5ms
istio-ingressgateway: productpage.bookinfo.svc.cluster.local:9080/productpage
33.53ms

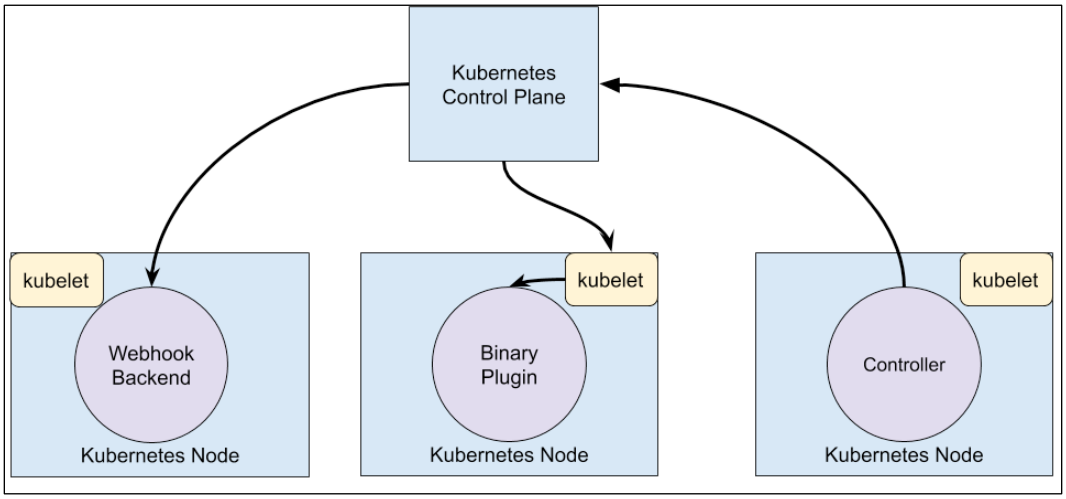

Jaeger UI
Lookup by Trace ID...
Search
Compare
Dependencies
About Jaeger
istio-ingressgateway: productpage.bookinfo.svc.cluster.local:9080/productpage 3d5584c
Find...
Trace Timeline
Trace Start January 24 2020, 16:27:24.809 Duration 61.15ms Services 5 Depth 6 Total Spans 8
0ms
15.29ms
30.57ms
45.86ms
61.15ms
Service & Operation
istio-ingressgateway productpage.bookinfo.svc.cluster.local:9...
productpage.bookinfo productpage.bookinfo.svc.cluster...
productpage.bookinfo details.bookinfo.svc.cluster.lo...
details.bookinfo details.bookinfo.svc.cluster.local:...
3.26ms
2.56ms
details.bookinfo.svc.cluster.local:9080/*
Service: details.bookinfo | Duration: 2.56ms | Start Time: 6.63ms
Tags: node_id = sidecar~172.17.0.27~details-v1-78d78fbddf-sdssb.bookinfo~bookinfo.svc.cluster.local | guid:x-request-id = a336d9be-2ea6-98b7-ad51-e85066c66734 | http.url =...
Process: ip = 172.17.0.27
SpanID: 9c5db0ae97d58d04
productpage.bookinfo reviews.bookinfo.svc.cluster.l...
35.53ms
reviews.bookinfo.svc.cluster.local:9080/*
Service: productpage.bookinfo | Duration: 35.53ms | Start Time: 17.44ms
Tags: node_id = sidecar~172.17.0.24~productpage-v1-596598f447-5pl8n.bookinfo~bookinfo.svc.cluster.local | guid:x-request-id = a336d9be-2ea6-98b7-ad51-e85066c66734 | htt...
Process: ip = 172.17.0.24
SpanID: 1f4b33e6533184ac
reviews.bookinfo reviews.bookinfo.svc.cluster.loc...
reviews.bookinfo ratings.bookinfo.svc.cluster...
ratings.bookinfo ratings.bookinfo.svc.clu...
34.85ms
13.95ms
13.17ms

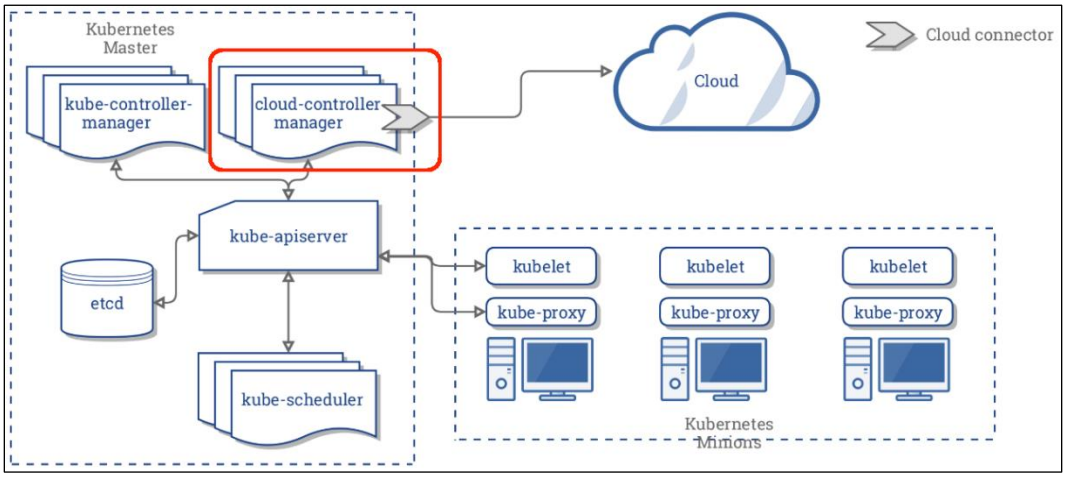

kiali
admin
Overview
Graph
Applications
Workloads
Services
Istio Config
Namespaces
Name
Filter by Name
Name
Show health for
Apps
Compact
Expand
Last 1h
Every 15s
bookinfo
6 Applications
Traffic, last 1h
6
default
0 Applications
N/A
No traffic
istio-system
12 Applications
No traffic
12
logging
3 Applications
No traffic
3

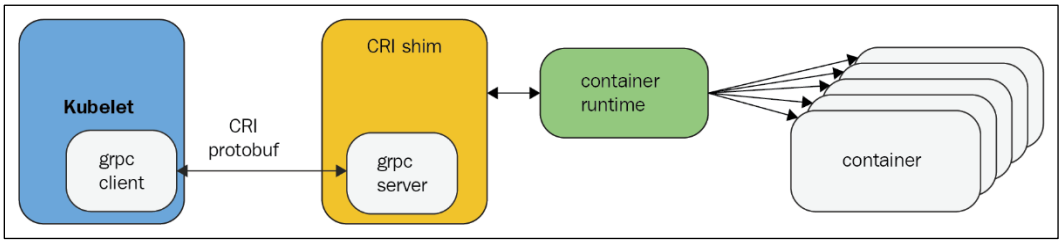

kiali
admin
Overview
Graph
Applications
Workloads
Services
Istio Config
Namespace: bookinfo
Graph
Jan 24, 12:38:06 ... Jan 24, 13:38:06
Versioned app graph
Requests per second
Display 7
Find...
Hide...
Last 1h
Every 15s
Namespace: bookinfo
applications, services, workloads
Current Graph:
7 apps (11 versions)
5 services
14 edges
Incoming
Outgoing
Total
HTTP Traffic (requests per second):
Total
%Success
%Error
0.03
100.00
0.00
0
25
50
75
100
OK
3xx
4xx
5xx
httpbin
v1
v2
istio-ingressgateway
(istio-system)
productpage
details
reviews
ratings
mongodb
Legend

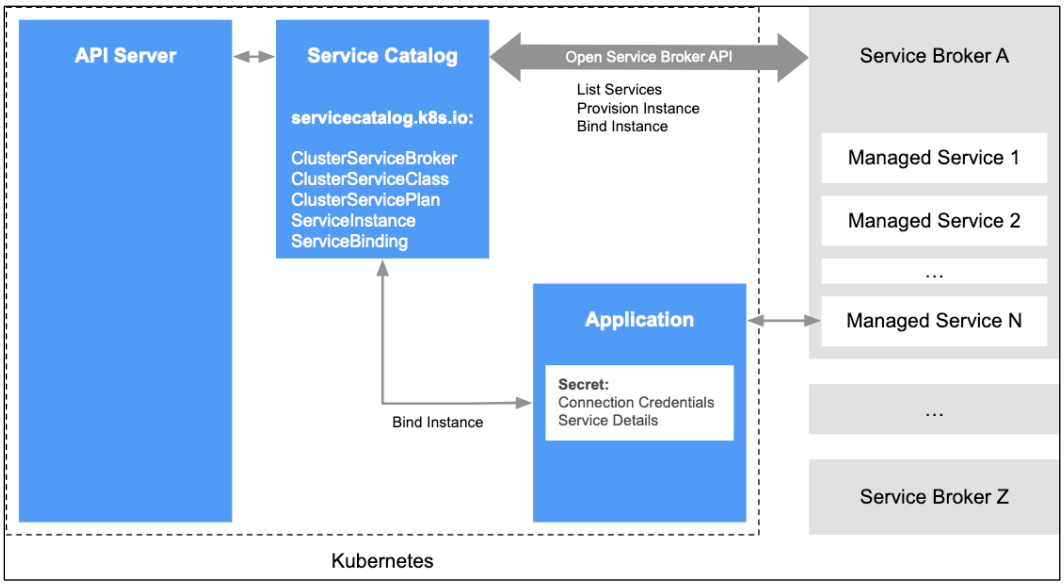
Chapter 15: Extending Kubernetes

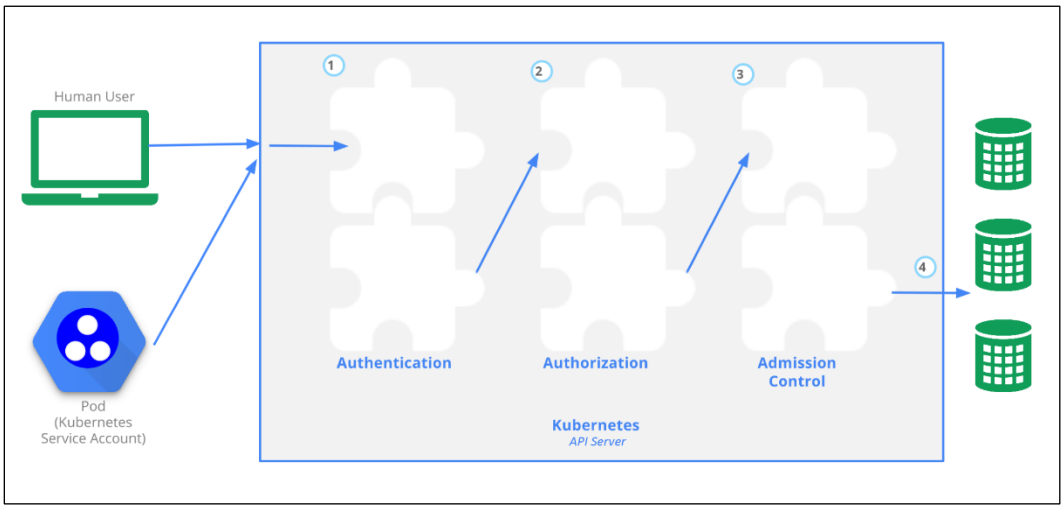



Kubernetes
Control Plane
kubelet
Webhook
Backend
Kubernetes Node
kubelet
Binary
Plugin
Kubernetes Node
kubelet
Controller
Kubernetes Node


Kubernetes
Master
kube-controller-
manager
cloud-controller
manager
Cloud
Cloud connector
kube-apiserver
etcd
kube-scheduler
kubelet
kube-proxy
kubelet
kube-proxy
kubelet
kube-proxy
Kubernetes
Minions


Kubelet
grpc
client
CRI
protobuf
CRI shim
grpc
server
container
runtime
container


API Server
Service Catalog
servicecatalog.k8s.io:
ClusterServiceBroker
ClusterServiceClass
ClusterServicePlan
ServiceInstance
ServiceBinding
Open Service Broker API
List Services
Provision Instance
Bind Instance
Service Broker A
Managed Service 1
Managed Service 2
...
Managed Service N
...
Service Broker Z
Application
Secret:
Connection Credentials
Service Details
Bind Instance
Kubernetes


Human User
Pod
(Kubernetes
Service Account)
1
2
3
4
Authentication
Authorization
Admission
Control
Kubernetes
API Server

Chapter 16: The Future of Kubernetes

